EXAM PACK

EEE475

RF and Microwave Engineering



Exam Pack

Session 2020–21 *Solutions and formula sheet of the Fourth Year First Semester Final Examination 2024*

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Prepared by Nur Mohammad

October 2026

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PAST PAPER SOLUTIONS

EEE475

RF and Microwave Engineering

Question Paper Solutions

Session 2020–21

EEE475 Fourth Year First Semester Final Examination 2024

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

EACH ANSWER

Theory → Answer → Keyword Explanation → Notes and Remarks → Final Concise Answer

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October 2026

BACK OF THE COVER

The paper at a glance

3 h	100
Total time	Total marks

Time = marks × 1.8 min (180 min for 100 marks). Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
PART-A				
Q1	(a) Propagation Constant from the Lumped-Element Circuit	8	14 min	☐
	(b) Shunt Conductance from the Propagation Constant	5	9 min	
	(c) Reflection Coefficient and Input Impedance	12	22 min	
Q2	(a) Constant-r and Constant-x Circles of the Smith Chart	12	22 min	☐
	(b) Single-Stub Matching of $35 - j50\ \Omega$ on a $75\ \Omega$ Line	8	14 min	
	(c) Quarter-Wave Transformer	5	9 min	
Q3	(a) Uniform Plane Wave in a Nonmagnetic Medium	6	11 min	☐
	(b) Telephone Line at 1 kHz	6	11 min	
	(c) Shorted-Stub Matching of an Antenna	6	11 min	
	(d) Propagating Modes of a Filled Rectangular Waveguide	7	13 min	
PART-B				
Q4	(a) Magnetic Field of a Hertzian Dipole	8	14 min	☐
	(b) Electric Field of a Hertzian Dipole	12	22 min	
	(c) Far-Field Average Power Density	5	9 min	
Q5	(a) Mixer Output and Filtering in a Superheterodyne Receiver	6	11 min	☐
	(b) Total Power of an AM Signal	5	9 min	
	(c) EIRP of an Antenna	5	9 min	
	(d) Short Notes: Microwave Tubes and Devices	9	16 min	
Q6	(a) Microstrip Line: Uses, Advantages and Losses	5	9 min	☐
	(b) Jefimenko's Equations and the Need for Accelerating Charges	13	23 min	
	(c) Resonant Frequency of a Filled Cavity	3	5 min	
	(d) Effective Height and Effective Area of an Antenna	4	7 min	

Formulas follow Sadiku, Elements of Electromagnetics, and Pozar, Microwave Engineering; the input impedance in 1(c) keeps the paper's sign convention.

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

B.Sc. (Engg.) Fourth Year, First Semester Final Examination-2024

Course Code: EEE 475 Session: 2020-2021

Course Title: RF and Microwave Engineering

Total Marks: 100 **Credits: 3.0** **Total Time: 3 Hours**

*[Answer any **two** questions from each part. The figure in the right margin indicate full marks.]*

Part-A

1. (a) From the lumped-element equivalent circuit, derive the expression of the propagation constant in terms of series resistance R, series inductance L, shunt conductance G, and shunt capacitance C per unit length. Deduce this expression to low-loss transmission line. 8

   (b) If the propagation constant of a transmission line is $50\angle 89.38°$ m^{-1}, $R = 4\ \Omega$ m^{-1}, $L = 0.5\ \mu$H m^{-1}, $C = 200$ pF m^{-1}, calculate the value of shunt conductance with proper unit. 5

   (c) In Fig. 1, a lossless transmission line is shown where, $V(z) = V_0^+(e^{-j\beta z} + \Gamma e^{j\beta z})$, $I(z) = \frac{V_0^+}{Z_0}(e^{-j\beta z} - \Gamma e^{j\beta z})$, Γ is the reflection coefficient at point $z = 0$, Z_0 is the characteristics impdedance of the line, Z_L is the load impedance, and β is the phase constant. Show that

   (i) $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$

   (ii) $Z_{in}(l) = \left(\frac{Z_L - jZ_0 \tan\beta l}{Z_0 - jZ_L \tan\beta l}\right) Z_0$ 5+7

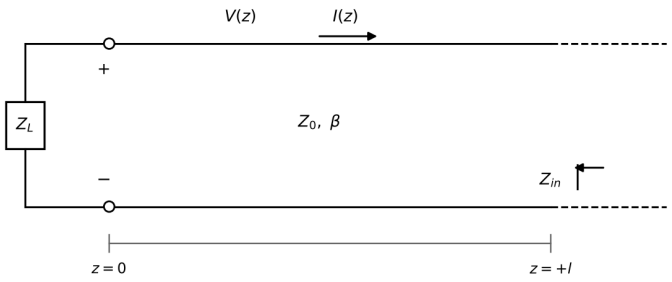


Fig. 1 of the paper (redrawn)

2. (a) Derive the equation of the constant resistance and constant reactance circle of a Smith Chart. You should start form the relation between the reflection coefficient and the normalized load. 12

   (b) A 75 Ω lossless transmission line is connected to a load impedance of $(35 - j50)\ \Omega$. Find the length position, l_p, and the length, l_s of a short-circuited stub required to match the load. You need to show two possible solutions using Smith Chart. 8

   (c) Explain how quarter-wave transformer can be used for impdeance matching. Mention its limitations. 5

3. (a) A plane wave in a nonmagnetic medium has $\vec{E} = 50\sin(10^8 t + 2z)\hat{a}_y$ V/m. Find:

   (i) The direction of wave propagation

   (ii) λ, f, and ϵ_r

   (iii) $\vec{H}$ 6

   (b) A telephone line has $R = 30\ \Omega$ km^{-1}, $L = 100$ mH km^{-1}, $G = 0$, and $C = 20\ \mu$F km^{-1}. At $f = 1$ kHz, obtain:

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(i) The characteristic impedance of the line
(ii) The propagation constant
(iii) The phase velocity 6

(c) An antenna with an impednace of $40 + j30\ \Omega$ is to be matched to a 100 Ω lossless line with a shorted stub. Determine:

(i) The required stub admittance
(ii) The distance between the stub and the antenna
(iii) The stub length
(iv) The standing wave ratio on each segment of the system 6

(d) A rectangular waveguide with dimensions $a = 2.5$ cm, $b = 1$ cm is to operate below 15.1 GHz. How many TE and TM modes can the waveguide transmit if the guide is filled with a medium characterized by $\sigma = 0$, $\epsilon = 4\epsilon_0$, $\mu_r = 1$? Calculate the cutoff frequencies of the modes. 7

Part-B

4. (a) The magnetic vector potential of a Hertzian dipole is given by: $\vec{A} = \frac{\mu_0 I dl}{4\pi r} e^{-j\beta r}\hat{z}$. Show that magnetic field, $H = \frac{Idl}{4\pi}\cos(\theta)\left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r}\hat{\phi}$ [*sic*: should be $\sin(\theta)$]. 8

(b) Using the expression of H found in 4(a), show that the electric field component $E = \frac{\eta Idl}{4\pi}\cos(\theta)\left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}\hat{\theta} + \frac{\eta Idl}{2\pi}\cos(\theta)\left[\frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}\hat{r}$ [*sic*: the $\hat{\theta}$ term should have $\sin(\theta)$] 12

(c) Find the expression of average power density of Hertzian dipole in far field zone. 5

5. (a) An RF signal at $f = 100$ MHz is mixed with local oscillator signal at $f_{LO} = 90$ MHz.

(i) Calculate the IF (intermediate frequency) output.
(ii) Why are active filters used after mixing in a super-heterodyne receiver? 6

(b) For a 1 kHz message signal modulating a 1 MHz carrier with 80% modulation index in AM, calculate the total transmitted power if carrier power is 100 W. 5

(c) For an antenna with gain 10 dB transmitting 10 W power at 2.4 GHz, calculate the effective isotropic radiated power (EIRP) of the antenna. 5

(d) Write short notes (any three).

(i) Klystron
(ii) Magnetron
(iii) Gunn Diode Oscillator
(iv) Travelling Wave Tube (TWT) 3+3+3

6. (a) What are the uses of a microstrip line? Mention its advantages. Describe the type of losses in a microstrip line. 5

(b) Derive Jefeminko's equation and from there, explain the necessity of accelerating charge for electromagnetic radiation. 13

(c) A rectangular cavity resonator has dimensions of $a = 5$ cm, $b = 2$ cm, and $d = 15$ cm. Compute the resonant frequency of the dominant mode for a dielectric-field cavity of $\epsilon_r = 2.56$. 3

(d) What do you mean by effective height and effective area of an antenna? 4

List of the relevant equations:

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$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} a_r & r a_\theta & (r\sin\theta) a_\phi \\ \dfrac{\partial}{\partial r} & \dfrac{\partial}{\partial \theta} & \dfrac{\partial}{\partial \phi} \\ A_r & r A_\theta & (r\sin\theta) A_\phi \end{vmatrix}$$

$$= \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(A_\phi \sin\theta) - \frac{\partial A_\theta}{\partial\phi}\right] a_r + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{\partial}{\partial r}(r A_\phi)\right] a_\theta + \frac{1}{r}\left[\frac{\partial}{\partial r}(r A_\theta) - \frac{\partial A_r}{\partial\theta}\right] a_\phi$$

$$\begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} a_r \\ a_\theta \\ a_\phi \end{bmatrix}$$

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1. (a) From the lumped-element equivalent circuit, derive the expression of the propagation constant in terms of series resistance R, series inductance L, shunt conductance G, and shunt capacitance C per unit length. Deduce this expression to low-loss transmission line. Marks: 8

   (b) If the propagation constant of a transmission line is $50\angle 89.38°$ m^{-1}, $R = 4\ \Omega$ m^{-1}, $L = 0.5\ \mu$H m^{-1}, $C = 200$ pF m^{-1}, calculate the value of shunt conductance with proper unit. Marks: 5

   (c) In Fig. 1, a lossless transmission line is shown where, $V(z) = V_0^+(e^{-j\beta z} + \Gamma e^{j\beta z})$, $I(z) = \frac{V_0^+}{Z_0}(e^{-j\beta z} - \Gamma e^{j\beta z})$, Γ is the reflection coefficient at point $z = 0$, Z_0 is the characteristics impdedance of the line, Z_L is the load impedance, and β is the phase constant. Show that

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   (ii) $Z_{in}(l) = \left(\frac{Z_L - jZ_0 \tan\beta l}{Z_0 - jZ_L \tan\beta l}\right) Z_0$ [Marks: 5+7]

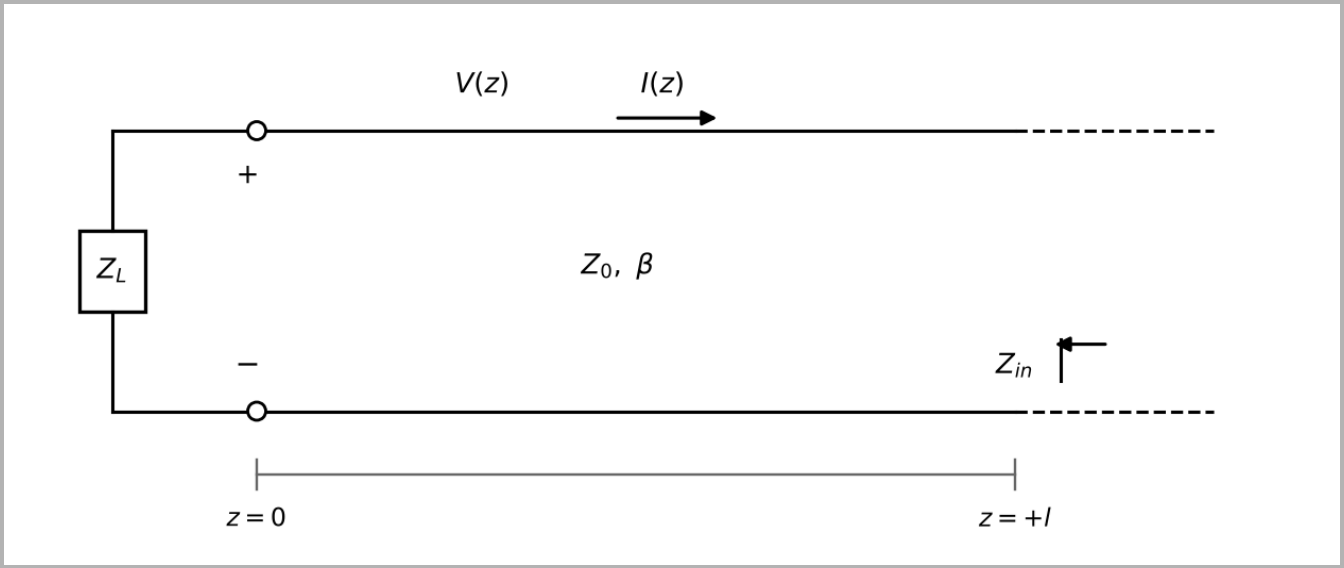


Figure 1: Fig. 1 of the paper (redrawn)

1.(a) From the lumped-element equivalent circuit, derive the expression of the propagation constant in terms of series resistance R, series inductance L, shunt conductance G, and shunt capacitance C per unit length. Deduce this expression to low-loss transmission line. Marks: 8

1. Theory

1. What is a transmission line? (Core Idea)

A **transmission line** is any pair of conductors that carries an electrical signal from one place to another: the two wires of a telephone line, a coaxial TV cable, or a copper track on a circuit board above a ground plane. At low frequency we treat a wire as a simple connection: the voltage at one end appears at the other end at once. That is no longer true at radio frequencies (RF, roughly 3 kHz to 300 GHz) and microwave frequencies (above about 300 MHz).

The reason is **wavelength**. A signal of frequency f travelling at speed v repeats itself every $\lambda = v/f$ metres. At 50 Hz, $\lambda \approx 6000$ km, so a 1 m wire is a tiny fraction of a wavelength and the voltage is the same all along it. At 3 GHz, $\lambda = 10$ cm, so along a 1 m cable the voltage goes up and down ten times. The signal is now a **wave** travelling along the line, and we must study how it travels.

Rule of thumb: once a line is longer than about $\lambda/10$, treat it as a transmission line.

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2. Why can a line be modelled by R, L, G and C?

Every real pair of conductors has four properties, spread evenly along its length. We give them **per unit length** (per metre):

Symbol	Name	Where it comes from	Unit
R	series resistance	the conductors are not perfect; current heats them	Ω/m
L	series inductance	current in the wires makes a magnetic field around them	H/m
G	shunt conductance	the insulation between the wires leaks a little current	S/m
C	shunt capacitance	the two wires are two conductors separated by insulation	F/m

Table 2: The four line constants (per unit length)

R and L are in **series** with the current path because the current flows *through* them. G and C sit **across** (in shunt with) the two wires because they act *between* the wires. A very short piece of line, of length Δz, is therefore the small circuit below: $R\Delta z$ and $L\Delta z$ in series, then $G\Delta z$ and $C\Delta z$ across. A long line is just thousands of these pieces in a chain.

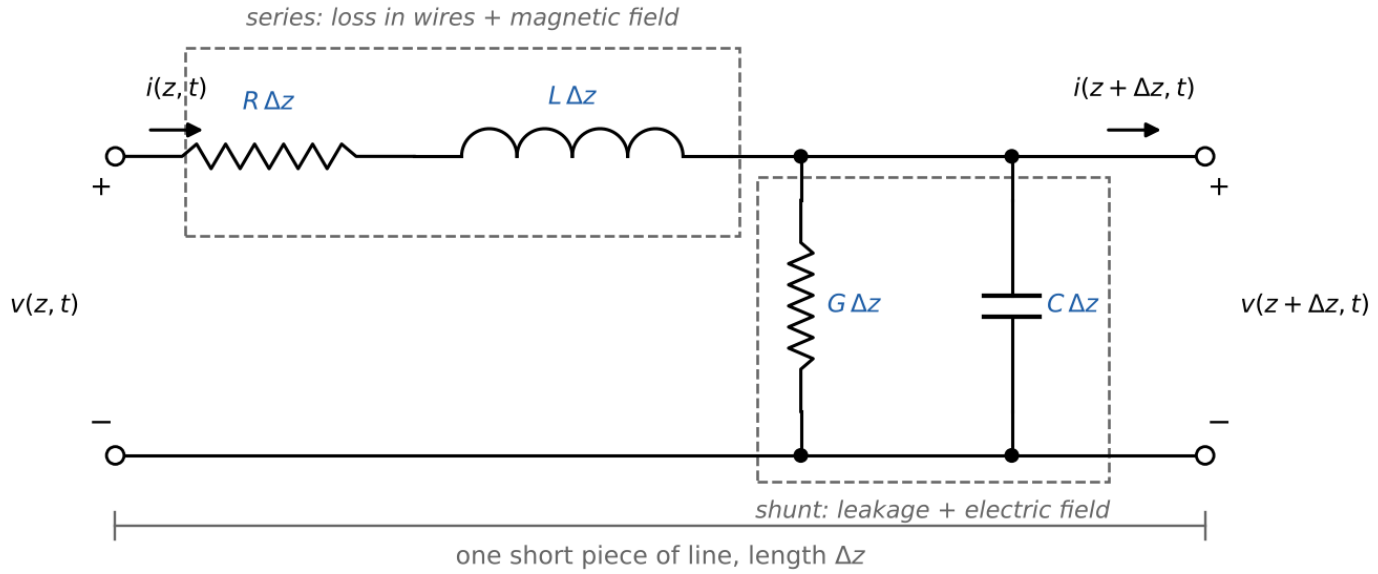


Figure 2: Lumped-element model of a short piece of transmission line

Reading the figure. The voltage $v(z,t)$ and current $i(z,t)$ enter on the left. The series part (top dashed box) drops some voltage, and the shunt part (bottom right box) steals some current, so on the right we get slightly different values $v(z+\Delta z,t)$ and $i(z+\Delta z,t)$. Letting $\Delta z \to 0$ turns these small differences into derivatives.

3. What is the propagation constant?

When a sine wave travels along the line, at each new metre two things happen to it:

- it gets **smaller** (some energy is lost in R and G);
- its **phase shifts** (it takes time to travel).

Both are packed into one complex number, the **propagation constant**

$$\gamma = \alpha + j\beta$$

- α is the **attenuation constant** (Np/m): the amplitude falls as $e^{-\alpha z}$.
- β is the **phase constant** (rad/m): the phase falls by β radians per metre. One full wavelength is 2π radians, so $\lambda = 2\pi/\beta$, and the wave speed is $v_p = \omega/\beta$.

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A travelling voltage wave is then

$$V^+e^{-\gamma z} = V^+e^{-\alpha z}e^{-j\beta z}$$

In the time domain this is $|V^+|e^{-\alpha z}\cos(\omega t - \beta z)$, plotted below.

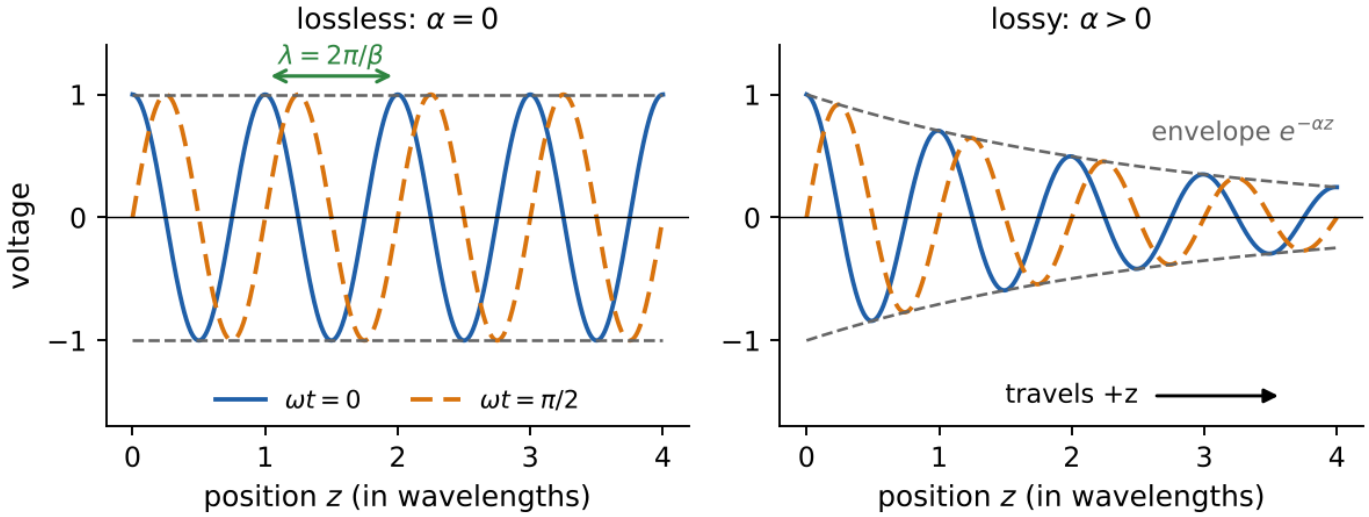


Figure 3: A travelling wave without and with loss

Reading the figure. Each panel shows the voltage along the line at two instants. The dashed orange curve ($\omega t = \pi/2$) is the solid blue curve ($\omega t = 0$) shifted to the right, so the wave moves toward $+z$. On the left ($\alpha = 0$) the height never changes. On the right ($\alpha > 0$) it shrinks inside the thin grey dashed envelope $e^{-\alpha z}$.

4. Phasors in one line

For a sine wave of angular frequency ω we write $v(z,t) = \mathrm{Re}[V(z)e^{j\omega t}]$. The complex number $V(z)$ is the **phasor**. With phasors, $\partial/\partial t$ becomes multiplication by $j\omega$, which turns differential equations in time into simple algebra.

2. Answer

Step 1: Kirchhoff's laws on one small section. For the section of length Δz (figure above), Kirchhoff's voltage law (KVL) around the outer loop gives

$$v(z,t) - R\Delta z\, i(z,t) - L\Delta z\,\frac{\partial i(z,t)}{\partial t} - v(z+\Delta z,t) = 0$$

Kirchhoff's current law (KCL) at the top node of the shunt part gives

$$i(z,t) - G\Delta z\, v(z+\Delta z,t) - C\Delta z\,\frac{\partial v(z+\Delta z,t)}{\partial t} - i(z+\Delta z,t) = 0$$

Step 2: Divide by Δz and let $\Delta z \to 0$. Using $\lim_{\Delta z\to 0}\frac{v(z+\Delta z)-v(z)}{\Delta z} = \frac{\partial v}{\partial z}$:

$$-\frac{\partial v(z,t)}{\partial z} = R\, i(z,t) + L\frac{\partial i(z,t)}{\partial t}$$

$$-\frac{\partial i(z,t)}{\partial z} = G\, v(z,t) + C\frac{\partial v(z,t)}{\partial t}$$

These are the **telegrapher's equations**.

Step 3: Phasor form. For sinusoidal signals, $\partial/\partial t \to j\omega$:

$$\frac{dV(z)}{dz} = -(R + j\omega L)\, I(z) \qquad \frac{dI(z)}{dz} = -(G + j\omega C)\, V(z)$$

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Step 4: Wave equation. Differentiate the first equation with respect to z and substitute the second:

$$\frac{d^2V(z)}{dz^2} = -(R+j\omega L)\frac{dI(z)}{dz} = (R+j\omega L)(G+j\omega C)\,V(z)$$

$$\frac{d^2V(z)}{dz^2} - \gamma^2 V(z) = 0, \qquad \gamma^2 = (R+j\omega L)(G+j\omega C)$$

The same steps give $\frac{d^2I}{dz^2} - \gamma^2 I = 0$. The solutions are waves travelling in the $+z$ and $-z$ directions:

$$V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$$

So the **propagation constant** is

$$\boxed{\gamma = \alpha + j\beta = \sqrt{(R+j\omega L)(G+j\omega C)}}$$

where α is the attenuation constant (Np/m) and β is the phase constant (rad/m). Putting $V(z)$ back into Step 3 also gives the characteristic impedance, $Z_0 = V_0^+/I_0^+ = \sqrt{(R+j\omega L)/(G+j\omega C)}$.

Step 5: Low-loss line. A line is **low-loss** when $R \ll \omega L$ and $G \ll \omega C$. Take $j\omega L$ and $j\omega C$ out of the brackets:

$$\gamma = j\omega\sqrt{LC}\sqrt{\left(1+\frac{R}{j\omega L}\right)\left(1+\frac{G}{j\omega C}\right)} = j\omega\sqrt{LC}\sqrt{1 - j\left(\frac{R}{\omega L}+\frac{G}{\omega C}\right) - \frac{RG}{\omega^2 LC}}$$

The last term is a product of two small numbers, so drop it. Then use $\sqrt{1+x} \approx 1 + x/2$ for small x:

$$\gamma \approx j\omega\sqrt{LC}\left[1 - \frac{j}{2}\left(\frac{R}{\omega L}+\frac{G}{\omega C}\right)\right] = \frac{1}{2}\left(R\sqrt{\frac{C}{L}} + G\sqrt{\frac{L}{C}}\right) + j\omega\sqrt{LC}$$

With $Z_0 \approx \sqrt{L/C}$ for a low-loss line:

$$\boxed{\alpha \approx \frac{R}{2Z_0} + \frac{GZ_0}{2}, \qquad \beta \approx \omega\sqrt{LC}}$$

Interpretation:

- β is the same as for a lossless line, so the wave speed is still $v_p = \omega/\beta = 1/\sqrt{LC}$.
- α has two parts: $R/(2Z_0)$ is the **conductor loss** and $GZ_0/2$ is the **dielectric loss**.
- For a **lossless** line ($R = G = 0$): $\alpha = 0$, $\gamma = j\omega\sqrt{LC}$.

3. Keyword Explanation

1. Lumped-element model

A way to describe something spread out (a line metres long) with ordinary circuit elements, by cutting it into pieces so short that each piece behaves like a small R-L-G-C circuit.

2. Telegrapher's equations

The two first-order equations of Step 2. They link how the voltage changes along the line to the current, and how the current changes to the voltage.

3. Neper (Np)

The unit of α. An amplitude that falls by a factor e has dropped by 1 Np; 1 Np = 8.686 dB.

4. Characteristic impedance Z_0

The ratio V/I for a single wave travelling in one direction. It depends only on the line's R, L, G, C and frequency, not on its length.

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4. Notes and Remarks

- **Why $\sqrt{1+x} \approx 1 + x/2$:** this is the first two terms of the binomial series. It is accurate when $|x| \ll 1$, which is exactly the low-loss condition.
- **Distortionless line:** if $R/L = G/C$, then $\alpha = \sqrt{RG}$ and $\beta = \omega\sqrt{LC}$ exactly, at every frequency. All frequencies are attenuated equally and travel at the same speed, so pulses keep their shape.
- **Choosing the sign of the square root:** take the root with $\alpha \geq 0$. A passive line can only lose energy.

1.(b) If the propagation constant of a transmission line is $50\angle 89.38°$ m^{-1}, $R = 4\ \Omega$ m^{-1}, $L = 0.5\ \mu$H m^{-1}, $C = 200$ pF m^{-1}, calculate the value of shunt conductance with proper unit. Marks: 5

1. Theory

1. What does the given γ tell us?

The propagation constant $\gamma = \alpha + j\beta$ (see the Theory of 1(a), §3) is given in polar form, $50\angle 89.38°$. Converting to rectangular form gives the loss per metre (α, the real part) and the phase shift per metre (β, the imaginary part). The angle is almost 90°, so γ is almost purely imaginary: the line loses only a little per metre compared with how fast its phase turns. That is the definition of a **low-loss line**, so the low-loss formulas of 1(a) apply.

2. How do the losses split?

In a low-loss line, $\alpha = \alpha_c + \alpha_d$:

- $\alpha_c = R/(2Z_0)$ comes from the resistance of the conductors;
- $\alpha_d = GZ_0/2$ comes from leakage through the insulation.

We know α (from γ), R, and Z_0 (from L and C), so the only unknown, G, follows.

2. Answer

Given:

- $\gamma = 50\angle 89.38°$ m^{-1}
- $R = 4\ \Omega$/m
- $L = 0.5\ \mu$H/m $= 0.5 \times 10^{-6}$ H/m
- $C = 200$ pF/m $= 200 \times 10^{-12}$ F/m

Relationships: for a low-loss line (from 1(a))

$$\alpha = \frac{R}{2Z_0} + \frac{GZ_0}{2}, \qquad \beta = \omega\sqrt{LC}, \qquad Z_0 = \sqrt{\frac{L}{C}}$$

Step 1: Split γ into α and β

$$\alpha = 50\cos 89.38° = 50 \times 0.010821 = 0.5410 \text{ Np/m}$$

$$\beta = 50\sin 89.38° = 49.997 \text{ rad/m}$$

Step 2: Characteristic impedance

$$Z_0 = \sqrt{\frac{0.5 \times 10^{-6}}{200 \times 10^{-12}}} = \sqrt{2500} = 50\ \Omega$$

Step 3: Conductor part of the loss

$$\alpha_c = \frac{R}{2Z_0} = \frac{4}{2 \times 50} = 0.0400 \text{ Np/m}$$

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Step 4: Dielectric part and G

$$\alpha_d = \alpha - \alpha_c = 0.5410 - 0.0400 = 0.5010 \text{ Np/m}$$

$$G = \frac{2\alpha_d}{Z_0} = \frac{2 \times 0.5010}{50} = 0.02004 \text{ S/m}$$

$$\boxed{G \approx 0.02 \text{ S/m} = 20.04 \text{ mS/m (or 20 m℧/m)}}$$

Most of the loss on this line is leakage in the insulation ($\alpha_d = 0.501$), not resistance in the wires ($\alpha_c = 0.04$).

Step 5: Check that the line really is low-loss

The operating frequency follows from β:

$$\omega = \frac{\beta}{\sqrt{LC}} = 49.997 \times 10^8 = 5.0 \times 10^9 \text{ rad/s}, \qquad f = \frac{\omega}{2\pi} = 795.7 \text{ MHz}$$

Then $R/(\omega L) = 0.0016$ and $G/(\omega C) = 0.020$. Both are $\ll 1$, so the low-loss formulas were justified.

Final Answer: shunt conductance $G = 0.02004$ S/m ≈ 20 mS/m (the line works at about 796 MHz).

3. Keyword Explanation

1. Siemens (S) and mho (℧)

The unit of conductance, $1/\Omega$. "S/m" means siemens per metre of line. 1 S = 1 ℧.

2. Polar to rectangular

A complex number $M\angle\theta$ equals $M\cos\theta + jM\sin\theta$.

4. Notes and Remarks

- **Exact check (no approximation):** squaring γ and solving $\gamma^2 = (R + j\omega L)(G + j\omega C)$ for both ω and G gives $\omega = 4.9995 \times 10^9$ rad/s and $G = 0.020043$ S/m. That is the same to four figures, so the low-loss method is accurate here.
- **Why the frequency was not given:** $\beta = \omega\sqrt{LC}$ already contains it. One complex number γ gives two real equations, enough for the two unknowns ω and G.

1.(c) In Fig. 1, a lossless transmission line is shown where, $V(z) = V_0^+(e^{-j\beta z} + \Gamma e^{j\beta z})$, $I(z) = \frac{V_0^+}{Z_0}(e^{-j\beta z} - \Gamma e^{j\beta z})$, Γ is the reflection coefficient at point $z = 0$, Z_0 is the characteristics impdedance of the line, Z_L is the load impedance, and β is the phase constant. Show that (i) $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$ (ii) $Z_{in}(l) = \left(\frac{Z_L - jZ_0\tan\beta l}{Z_0 - jZ_L\tan\beta l}\right) Z_0$ [Marks: 5+7]

1. Theory

1. Why does a wave reflect at the load? (Core Idea)

On a line of characteristic impedance Z_0, a single travelling wave always has $V/I = Z_0$ (see 1(a), Keyword 4). At the end of the line sits a load Z_L, and Ohm's law there demands $V/I = Z_L$. If $Z_L \neq Z_0$, one wave alone cannot satisfy both rules. Nature solves this by sending part of the wave back: a **reflected wave**. The incident and reflected waves together give the right V/I at the load.

It is like a rope tied to a heavier rope: part of a pulse goes on and part bounces back. If the two ropes are identical ($Z_L = Z_0$, a **matched** load), nothing bounces back.

2. What is the reflection coefficient?

$$\Gamma = \frac{\text{reflected voltage}}{\text{incident voltage}} \text{ at the load}$$

It is a complex number with $|\Gamma| \leq 1$ for a passive load:

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- $\Gamma = 0$: matched, no reflection.
- $\Gamma = -1$: short circuit ($Z_L = 0$).
- $\Gamma = +1$: open circuit ($Z_L = \infty$).

3. Standing waves

The incident and reflected waves add. Where they are in step the voltage is large (a maximum); half a wavelength later they are out of step and it is small (a minimum). The resulting pattern of $|V|$ along the line stays still and is called a **standing wave**. The **standing wave ratio** is

$$\text{SWR} = \frac{|V|_{max}}{|V|_{min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

It is 1 for a matched load and infinite for a short or open circuit.

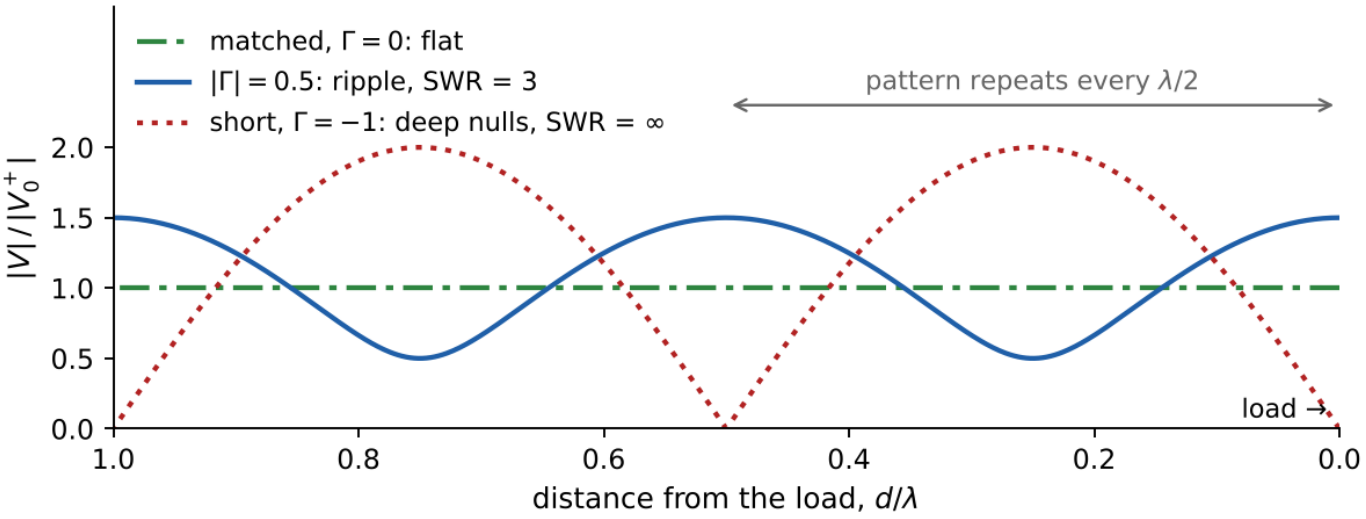


Figure 4: Standing-wave patterns for three loads

Reading the figure. The load is at the right ($d = 0$). With a matched load (green, dash-dot) $|V|$ is flat. With $|\Gamma| = 0.5$ (blue, solid) it ripples between 0.5 and 1.5, giving SWR = 3. With a short (red, dotted) it swings between 0 and 2: no power reaches the load. Every pattern repeats every $\lambda/2$.

4. Why does the input impedance change with length?

At the load end, $V/I = Z_L$. Further back along the line, the incident and reflected waves have travelled different distances, so their phases differ, and the ratio V/I is different. The impedance you "see" looking into the line therefore depends on the length l and on βl: a line **transforms** impedance. This is the key idea behind all matching circuits (Q2, Q3).

2. Answer

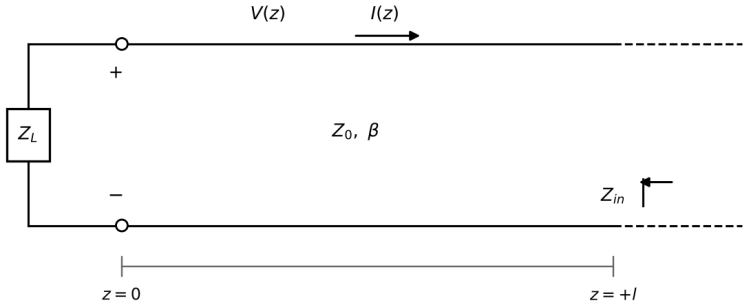


Figure 5: Repeated from the question: Fig. 1 of the paper (redrawn)

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The line is lossless, so $\gamma = j\beta$. With the voltage and current given in the question, the impedance at any point is

$$Z(z) = \frac{V(z)}{I(z)} = Z_0 \frac{e^{-j\beta z} + \Gamma e^{j\beta z}}{e^{-j\beta z} - \Gamma e^{j\beta z}} \qquad (1)$$

(i) Reflection coefficient. At $z = 0$ the line meets the load, so $Z(0) = Z_L$. Put $z = 0$ in (1):

$$Z_L = Z_0 \frac{1+\Gamma}{1-\Gamma}$$

$$Z_L(1-\Gamma) = Z_0(1+\Gamma) \;\Rightarrow\; Z_L - Z_0 = \Gamma(Z_L + Z_0)$$

$$\boxed{\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}}$$

(ii) Input impedance. At $z = l$, $Z_{in}(l) = Z(l)$. Substitute Γ from (i) into (1) and multiply the top and bottom by $(Z_L + Z_0)$:

$$Z_{in}(l) = Z_0 \frac{(Z_L + Z_0)e^{-j\beta l} + (Z_L - Z_0)e^{j\beta l}}{(Z_L + Z_0)e^{-j\beta l} - (Z_L - Z_0)e^{j\beta l}}$$

Group the Z_L and Z_0 terms:

$$\text{numerator} = Z_L(e^{j\beta l} + e^{-j\beta l}) - Z_0(e^{j\beta l} - e^{-j\beta l})$$

$$\text{denominator} = Z_0(e^{j\beta l} + e^{-j\beta l}) - Z_L(e^{j\beta l} - e^{-j\beta l})$$

Use Euler's identities $e^{j\beta l} + e^{-j\beta l} = 2\cos\beta l$ and $e^{j\beta l} - e^{-j\beta l} = 2j\sin\beta l$:

$$Z_{in}(l) = Z_0 \frac{2Z_L\cos\beta l - 2jZ_0\sin\beta l}{2Z_0\cos\beta l - 2jZ_L\sin\beta l}$$

Divide the top and bottom by $2\cos\beta l$:

$$\boxed{Z_{in}(l) = \left(\frac{Z_L - jZ_0\tan\beta l}{Z_0 - jZ_L\tan\beta l}\right) Z_0}$$

which is the required result.

Checks:

- $l = 0$: $Z_{in} = Z_L$ ✔.
- $Z_L = Z_0$: $Z_{in} = Z_0$ for any length ✔ (a matched line looks the same everywhere).

3. Keyword Explanation

1. Matched load

A load with $Z_L = Z_0$. It absorbs the whole incident wave: $\Gamma = 0$, SWR $= 1$.

2. Euler's identities

$e^{\pm j\theta} = \cos\theta \pm j\sin\theta$, so $e^{j\theta} + e^{-j\theta} = 2\cos\theta$ and $e^{j\theta} - e^{-j\theta} = 2j\sin\theta$.

3. Input impedance

The impedance a generator would measure if it were connected at that point of the line, looking toward the load.

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4. Notes and Remarks

Sign convention

Most textbooks (Pozar, Sadiku) put the load at $z = 0$ and the generator on the side $z < 0$, so the incident wave $e^{-j\beta z}$ travels *toward* the load. They write $Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}$ with $+j$. In Fig. 1 the line runs from the load toward $z = +l$, but the question keeps the textbook expressions for $V(z)$ and $I(z)$. Evaluating them at $z = +l$ is the same as putting $l \to -l$, which flips the sign of $\tan \beta l$ and gives the $-j$ form asked for. Both are correct for their own convention. **For numerical work (and on the Smith chart) use the textbook form with $+j$.** That is the one used in Q2(b), Q2(c) and Q3(c).

- **Special cases of the textbook form (useful later):**
  - ➢ Short-circuited line: $Z_{in} = jZ_0 \tan \beta l$.
  - ➢ Open-circuited line: $Z_{in} = -jZ_0 \cot \beta l$.
  - ➢ Quarter-wave line ($\beta l = \pi/2$): $Z_{in} = Z_0^2/Z_L$ (Q2(c)).
  - ➢ Half-wave line ($\beta l = \pi$): $Z_{in} = Z_L$.
- **Γ along the line:** at a distance d from the load, $\Gamma(d) = \Gamma_L e^{-j2\beta d}$. The magnitude stays the same and only the angle turns, by $2\beta d$. On the Smith chart this is a rotation (Q2).

5. Final Concise Answer

(a) From KVL and KCL on a section Δz: $-\frac{dV}{dz} = (R+j\omega L)I$ and $-\frac{dI}{dz} = (G+j\omega C)V$. Then $\frac{d^2V}{dz^2} = \gamma^2 V$, with $\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$. For a low-loss line ($R \ll \omega L$, $G \ll \omega C$): $\alpha \approx \frac{R}{2Z_0} + \frac{GZ_0}{2}$ and $\beta \approx \omega\sqrt{LC}$, with $Z_0 \approx \sqrt{L/C}$.

(b) $\alpha = 0.541$ Np/m, $\beta = 49.997$ rad/m, $Z_0 = 50\ \Omega$, $\alpha_c = 0.04$ Np/m, so $G = 2(\alpha - \alpha_c)/Z_0 = 0.02004$ S/m ≈ 20 mS/m (the line works at about 796 MHz).

(c) (i) At $z = 0$, $Z_L = Z_0(1 + \Gamma)/(1 - \Gamma)$, so $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$. (ii) At $z = l$, using Euler's identities, $Z_{in} = Z_0(Z_L - jZ_0 \tan \beta l)/(Z_0 - jZ_L \tan \beta l)$. With the usual convention (generator on the other side) the signs become $+j$.

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2. (a) Derive the equation of the constant resistance and constant reactance circle of a Smith Chart. You should start form the relation between the reflection coefficient and the normalized load. Marks: 12
   (b) A 75 Ω lossless transmission line is connected to a load impedance of $(35 - j50)$ Ω. Find the length position, l_p, and the length, l_s of a short-circuited stub required to match the load. You need to show two possible solutions using Smith Chart. Marks: 8
   (c) Explain how quarter-wave transformer can be used for impdeance matching. Mention its limitations. Marks: 5

2.(a) Derive the equation of the constant resistance and constant reactance circle of a Smith Chart. You should start form the relation between the reflection coefficient and the normalized load. Marks: 12

1. Theory

1. Why do we need a chart? (Core Idea)

Transmission-line problems keep asking the same thing: "if I move d metres along the line, what impedance do I see?" The formula $Z_{in} = Z_0(Z_L + jZ_0 \tan \beta d)/(Z_0 + jZ_L \tan \beta d)$ (see 1(c)) answers it, but complex division by hand is slow. The **Smith chart**, drawn by Phillip Smith in 1939, does it graphically.

The trick: along a lossless line the **reflection coefficient** only rotates. Its size stays the same and its angle turns by $2\beta d$. A rotation is easy to draw with a compass. So we draw the plane of Γ and, on top of it, the matching impedance values. Then "move along the line" means "turn around the centre", and the impedance is read straight off the grid.

2. Normalized impedance

Divide every impedance by Z_0:

$$z = \frac{Z}{Z_0} = r + jx$$

Here r is the normalized resistance and x the normalized reactance. Then the reflection coefficient of 1(c) becomes

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{z - 1}{z + 1}$$

so one chart works for every line, whatever its Z_0.

3. What the mapping does

Every passive impedance has $r \geq 0$: the whole right half of the z-plane. The formula $\Gamma = (z-1)/(z+1)$ squeezes that infinite half-plane into the inside of a circle of radius 1, $|\Gamma| \leq 1$. Vertical lines (r fixed) become circles, and horizontal lines (x fixed) become arcs of circles.

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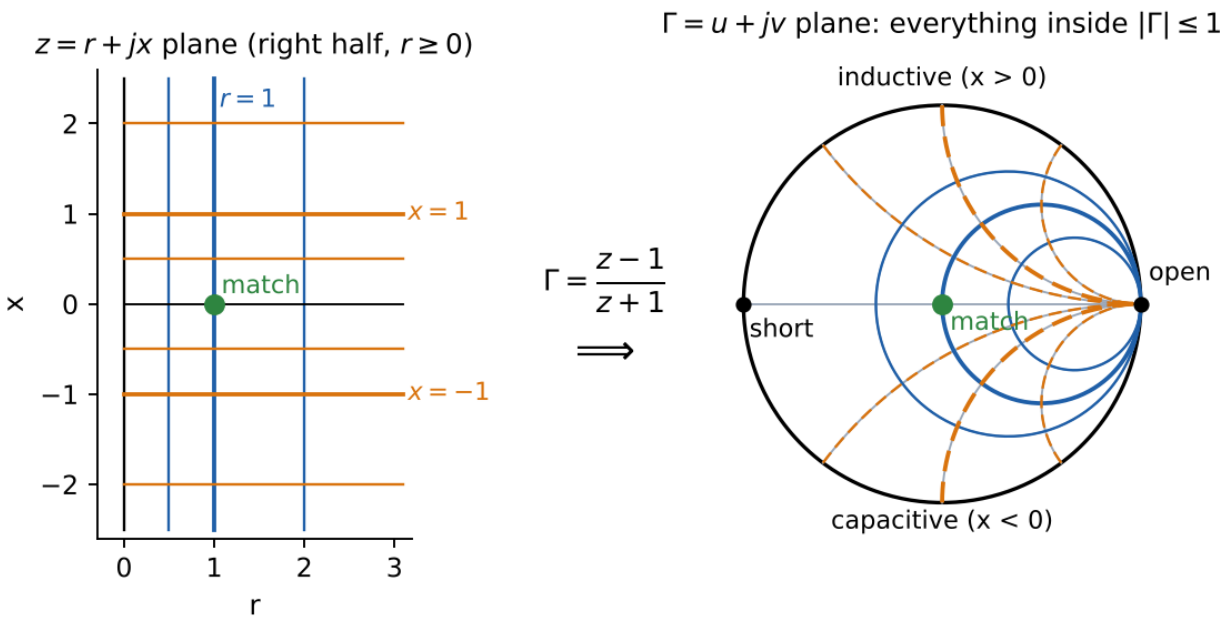


Figure 6: The z-plane is mapped onto the unit circle of the Gamma plane

Reading the figure. Left: the z-plane, with lines of constant r (blue; solid circles on the right) and constant x (orange; dashed arcs on the right). Right: the same lines after the mapping. The match point $z = 1$ (green dot) goes to the centre $\Gamma = 0$, a short ($z = 0$) to the left end, and an open ($z = \infty$) to the right end. Inductive impedances ($x > 0$) fill the top half and capacitive ones ($x < 0$) the bottom half.

2. Answer

Starting relation. With the normalized load $z_L = r + jx$ and $\Gamma = u + jv$ (u = real part, v = imaginary part):

$$\Gamma = \frac{z_L - 1}{z_L + 1} \quad \Longleftrightarrow \quad z_L = \frac{1 + \Gamma}{1 - \Gamma}$$

Step 1: Separate the real and imaginary parts.

$$r + jx = \frac{(1 + u) + jv}{(1 - u) - jv}$$

Multiply the top and bottom by the conjugate of the denominator, $(1 - u) + jv$:

$$r + jx = \frac{[(1 + u) + jv][(1 - u) + jv]}{(1 - u)^2 + v^2} = \frac{(1 - u^2 - v^2) + j2v}{(1 - u)^2 + v^2}$$

Equate the real parts and the imaginary parts:

$$r = \frac{1 - u^2 - v^2}{(1 - u)^2 + v^2} \qquad (1)$$

$$x = \frac{2v}{(1 - u)^2 + v^2} \qquad (2)$$

Step 2: Constant-resistance circles (from (1)). Cross-multiply:

$$r[(1 - u)^2 + v^2] = 1 - u^2 - v^2$$

$$r - 2ru + ru^2 + rv^2 = 1 - u^2 - v^2$$

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$$(1+r)u^2 - 2ru + (1+r)v^2 = 1 - r$$

Divide by $(1+r)$:

$$u^2 - \frac{2r}{1+r}u + v^2 = \frac{1-r}{1+r}$$

Complete the square in u by adding $\left(\frac{r}{1+r}\right)^2$ to both sides:

$$\left(u - \frac{r}{1+r}\right)^2 + v^2 = \frac{1-r}{1+r} + \frac{r^2}{(1+r)^2} = \frac{(1-r)(1+r)+r^2}{(1+r)^2}$$

$$\boxed{\left(u - \frac{r}{1+r}\right)^2 + v^2 = \left(\frac{1}{1+r}\right)^2}$$

This is a circle with **centre** $\left(\frac{r}{1+r}, 0\right)$ and **radius** $\frac{1}{1+r}$.

Step 3: Constant-reactance circles (from (2)). Cross-multiply:

$$x[(1-u)^2 + v^2] = 2v \;\Rightarrow\; (u-1)^2 + v^2 - \frac{2v}{x} = 0$$

Complete the square in v by adding $1/x^2$ to both sides:

$$\boxed{(u-1)^2 + \left(v - \frac{1}{x}\right)^2 = \left(\frac{1}{x}\right)^2}$$

This is a circle with **centre** $\left(1, \frac{1}{x}\right)$ and **radius** $\frac{1}{|x|}$.

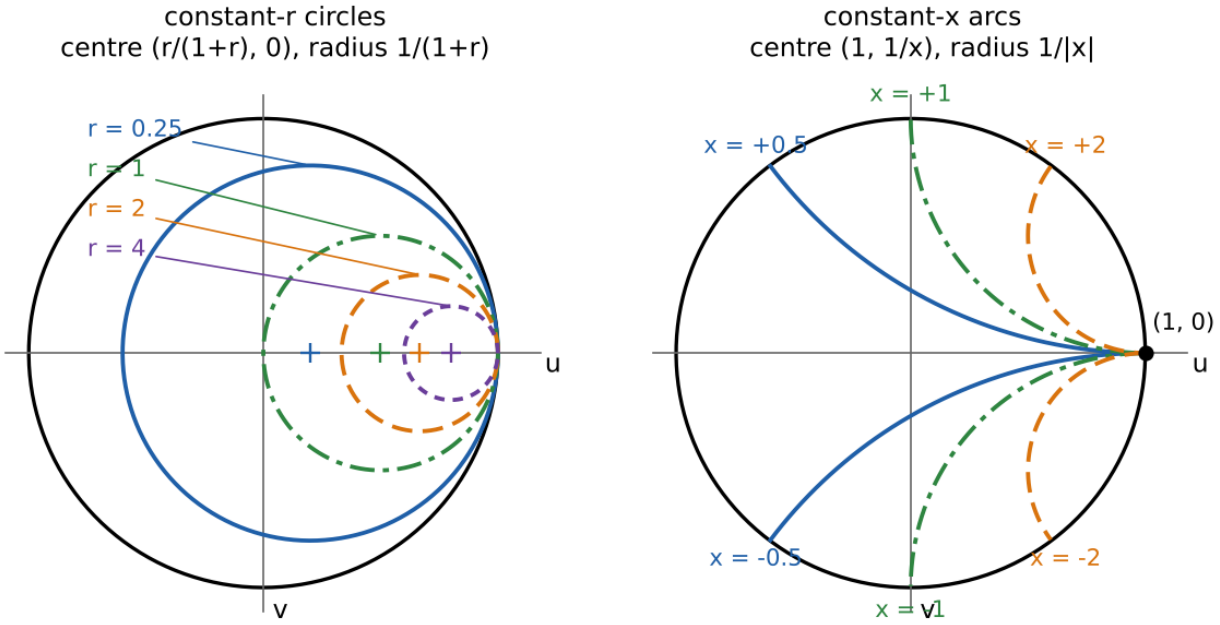


Figure 7: Constant-r circles and constant-x arcs

Reading the figure. Left: every r-circle touches the point $(1, 0)$. As r grows the circle shrinks toward that point. $r = 0$ is the outer rim $|\Gamma| = 1$, and $r = 1$ passes through the centre. Right: every x-arc also passes through $(1, 0)$, and only the part inside the unit circle is used. Positive x lies above the horizontal axis and negative x below.

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Properties of the circles:

Family	Centre	Radius	Special members
constant r	$\left(\frac{r}{1+r}, 0\right)$ on the real axis	$\frac{1}{1+r}$	$r = 0$: the rim; $r = 1$: through the centre; $r \to \infty$: the point $(1, 0)$
constant x	$\left(1, \frac{1}{x}\right)$ on the line $u = 1$	$\frac{1}{\lvert x\rvert}$	$x = 0$: the real axis; $x = \pm 1$: centre $(1, \pm 1)$, radius 1

Table 3: The two families of Smith-chart circles

Using the chart:

- **Special points:** the centre is the match ($z = 1$), the left end is a short ($z = 0$) and the right end is an open ($z = \infty$).
- **Moving along the line:** moving toward the generator by a distance d turns the point **clockwise** by $2\beta d = 4\pi d/\lambda$. One full turn is $\lambda/2$. The outer scale reads "wavelengths toward generator".
- **SWR circle:** a circle around the centre through the load point is the circle of constant $|\Gamma|$. It cuts the right half of the real axis at $r = \text{SWR}$.
- **Admittance:** the point diametrically opposite z is $y = 1/z$, so the same chart can be read as an admittance chart. This is what we do for shunt stubs (Q2(b), Q3(c)).

3. Keyword Explanation

1. Normalized impedance

$z = Z/Z_0$. It has no unit, so one chart serves every line.

2. Completing the square

Rewriting $u^2 - 2au$ as $(u-a)^2 - a^2$. It turns an equation into the standard form of a circle, $(u-u_0)^2 + (v-v_0)^2 = R^2$.

3. Wavelengths toward generator (WTG)

The outer scale of the Smith chart. Moving toward the source is clockwise and moving toward the load is anticlockwise.

4. Notes and Remarks

- **Every circle passes through $(1, 0)$:** that point is $z = \infty$, and every line in the z-plane goes to infinity.
- **Why only the inside of the unit circle:** a passive load has $r \geq 0$, which gives $|\Gamma| \leq 1$. Negative-resistance devices (Gunn diodes, Q5(d)) would plot outside.
- **Common slip:** the centre of an x-circle is at $v = 1/x$, not at $v = x$.

2.(b) A 75 Ω lossless transmission line is connected to a load impedance of $(35 - j50)$ Ω. Find the length position, l_p, and the length, l_s of a short-circuited stub required to match the load. You need to show two possible solutions using Smith Chart. Marks: 8

1. Theory

1. Why match at all? (Core Idea)

If the load is not matched, part of the power is reflected back to the source (see 1(c)). We lose power, the standing wave raises the peak voltage on the line, and transmitters can be damaged. **Matching** means adding something lossless between the line and the load so that the line sees exactly Z_0.

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2. What is a stub?

A **stub** is a short extra piece of the same transmission line, connected in parallel (in **shunt**) across the main line, with its far end shorted or open. A shorted, lossless stub absorbs no power and behaves like a pure reactance. Its input admittance is

$$y_{stub} = -j \cot \beta l_s$$

so by choosing its length l_s we can make it any susceptance we like, positive or negative.

3. The two-step idea

Work with **admittances**, because parallel admittances simply add.

1. **Move along the line** from the load by l_p until the normalized admittance has real part exactly 1: $y = 1 + jb$. On the Smith chart this happens where the SWR circle crosses the $g = 1$ circle. It crosses it twice, which gives the **two solutions**.
2. **Add the stub** with susceptance $-jb$ at that point. The total becomes $1 + jb - jb = 1$: matched.

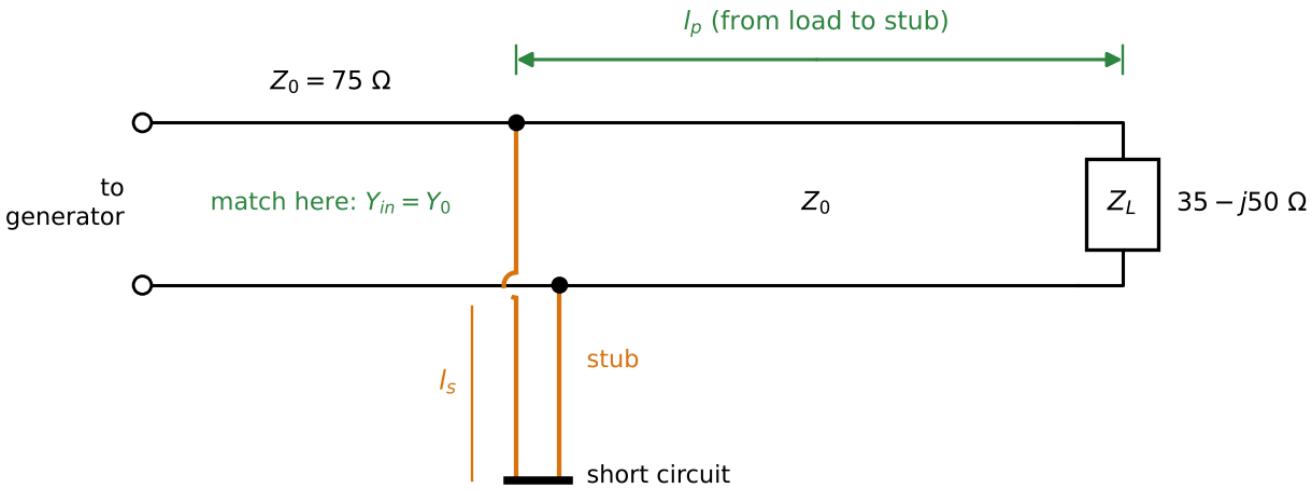


Figure 8: Single shunt stub matching

Reading the figure. The stub (orange, labelled "stub", short-circuited at the bottom) is connected across the main line at a distance l_p from the load. To the left of the stub (toward the generator) the line sees Y_0, so there is no reflection there.

4. Short at the end of a stub on the chart

On an admittance chart a short is $y = \infty$, which is the right-hand end of the real axis. Starting there and turning clockwise along the rim by l_s gives the stub's admittance.

2. Answer

Given: $Z_0 = 75\ \Omega$, $Z_L = 35 - j50\ \Omega$. Shunt, short-circuited stub.

Step 1: Normalize and plot

$$z_L = \frac{35 - j50}{75} = 0.467 - j0.667$$

$$y_L = \frac{1}{z_L} = 0.705 + j1.007$$

On the chart, y_L is the point diametrically opposite z_L. Its reading on the WTG scale is 0.145λ. Its distance from the centre is $|\Gamma_L| = 0.530$, so SWR $= (1 + 0.530)/(1 - 0.530) = 3.25$.

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Step 2: Rotate to the $g = 1$ circle

Draw the SWR circle through y_L and turn **clockwise** (toward the generator) until it meets the $g = 1$ circle:

- **Point A:** $y_A = 1 + j1.25$, at 0.169λ on the WTG scale.
- **Point B:** $y_B = 1 - j1.25$, at 0.331λ.

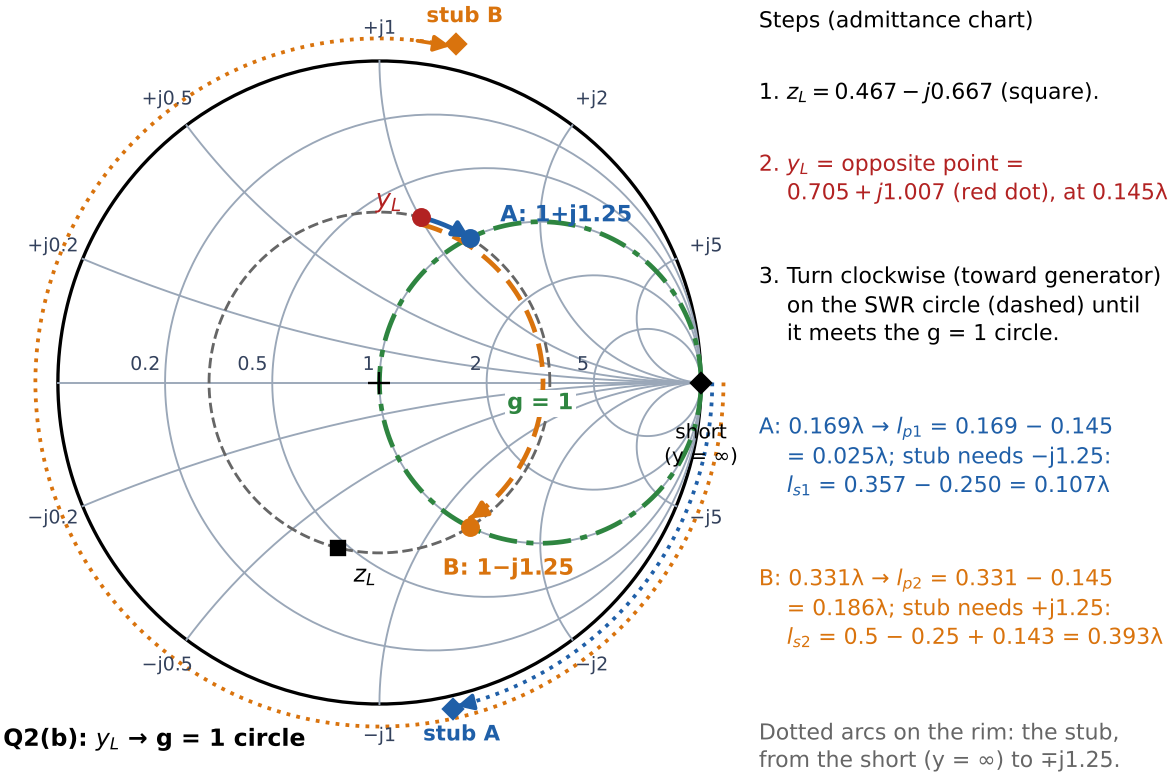


Figure 9: Smith chart solution of Q2(b)

Step 3: Stub position and length

Solution 1 (point A):

$$l_{p1} = 0.169\lambda - 0.145\lambda = 0.025\lambda$$

The stub must cancel $+j1.25$, so it needs $y_s = -j1.25$. On the rim, $-j1.25$ is at 0.357λ, and the short is at 0.25λ:

$$l_{s1} = 0.357\lambda - 0.250\lambda = 0.107\lambda$$

Solution 2 (point B):

$$l_{p2} = 0.331\lambda - 0.145\lambda = 0.186\lambda$$

The stub needs $y_s = +j1.25$, which is at 0.143λ on the rim. Going clockwise from the short at 0.25λ passes 0.5λ $(= 0)$ first:

$$l_{s2} = (0.5 - 0.25)\lambda + 0.143\lambda = 0.393\lambda$$

Step 4: Check with formulas

The input admittance at a distance d, $y(d) = \dfrac{y_L + j\tan\beta d}{1 + jy_L\tan\beta d}$, gives exactly $1.000 + j1.250$ at $d = 0.0247\lambda$ and $1.000 - j1.250$ at $d = 0.1858\lambda$. For the stub, $-j\cot\beta l_s = \mp j1.250$ gives $l_s = 0.1074\lambda$ and 0.3926λ.

Solution	Stub position l_p	Admittance there	Stub susceptance	Stub length l_s
1	0.025λ	$1 + j1.25$	$-j1.25$ (-16.7 mS)	0.107λ
2	0.186λ	$1 - j1.25$	$+j1.25$ ($+16.7$ mS)	0.393λ

Table 4: The two stub-matching solutions of Q2(b)

Final Answer: solution 1 is $l_p = 0.025\lambda$, $l_s = 0.107\lambda$; solution 2 is $l_p = 0.186\lambda$, $l_s = 0.393\lambda$. Solution 1 is usually preferred: the stub is closer to the load and shorter, so the match holds over a wider band.

3. Keyword Explanation

1. Admittance and susceptance

Admittance $Y = 1/Z = G + jB$. G is the conductance and B the susceptance. Normalized: $y = Y/Y_0 = g + jb$.

2. Shunt

Connected in parallel, across the two conductors.

3. $g = 1$ circle

The constant-resistance circle $r = 1$ of 2(a), read as conductance on an admittance chart. It passes through the centre.

4. Notes and Remarks

- **Why two solutions:** the SWR circle cuts the $g = 1$ circle at two points, symmetric about the real axis. Both match; they differ in stub position and length.
- **Adding $\lambda/2$:** any l_p or l_s can be lengthened by $\lambda/2$ with no change, because the chart repeats every half wavelength.
- **Reading accuracy:** chart readings are good to about $\pm 0.002\lambda$. The table uses the calculated values.

2.(c) Explain how quarter-wave transformer can be used for impdeance matching. Mention its limitations. Marks: 5

1. Theory

1. A line as an impedance transformer (Core Idea)

From 1(c), a lossless line of length l and characteristic impedance Z_1, ending in a load Z_L, shows at its input

$$Z_{in} = Z_1 \frac{Z_L + jZ_1 \tan \beta l}{Z_1 + jZ_L \tan \beta l}$$

When the length is exactly a quarter wavelength, $\beta l = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2}$ and $\tan \beta l \to \infty$. Divide the top and bottom by $\tan \beta l$ and let it go to infinity:

$$Z_{in} = Z_1 \frac{jZ_1}{jZ_L} = \frac{Z_1^2}{Z_L}$$

A quarter-wave section therefore **inverts** the load impedance, scaled by Z_1^2. A small load looks large and a large load looks small. If we choose Z_1 well, any resistive load can be made to look like Z_0.

2. Answer

Principle: to match a resistive load R_L to a line Z_0, insert between them a section of line that is $\lambda/4$ long (at the design frequency f_0), with characteristic impedance

$$\boxed{Z_1 = \sqrt{Z_0 R_L}}$$

Then the input impedance of the section is $Z_{in} = Z_1^2/R_L = Z_0 R_L/R_L = Z_0$. The main line sees a perfect match and $\Gamma = 0$ on it.

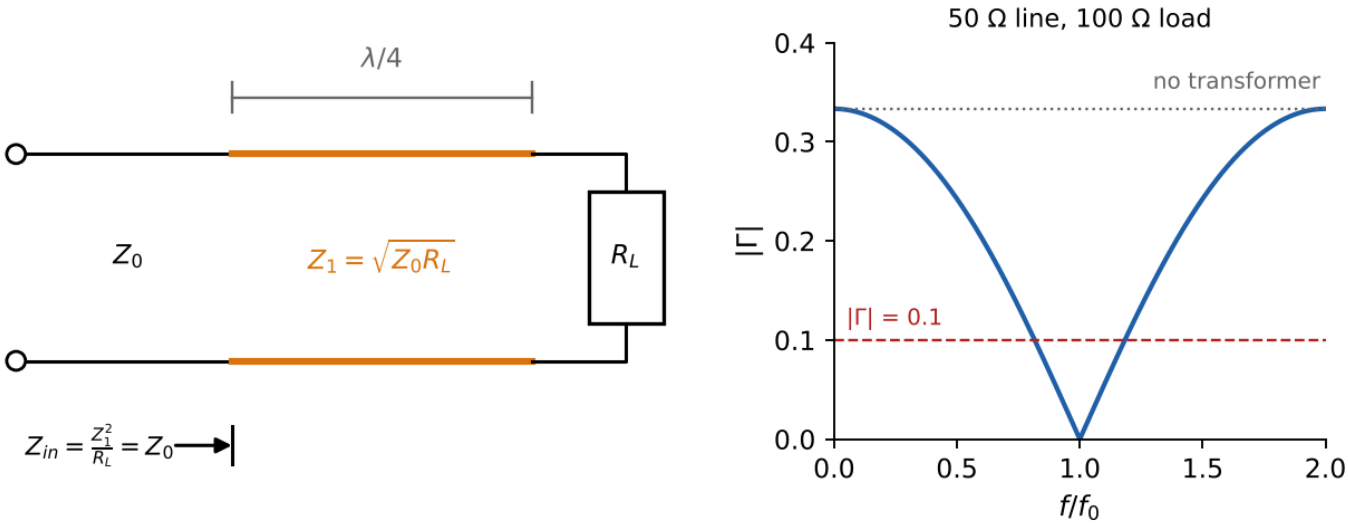


Figure 10: Quarter-wave transformer and its bandwidth

Example (figure)

a 100 Ω load on a 50 Ω line needs $Z_1 = \sqrt{50 \times 100} = 70.7\ \Omega$. At f_0, $|\Gamma| = 0$. At $0.8f_0$ or $1.2f_0$, $|\Gamma| = 0.109$ (SWR 1.24). Without the transformer, $|\Gamma| = 1/3$.

Why it works physically: the wave reflected from the load end and the wave reflected from the input end of the section travel paths that differ by $2 \times \lambda/4 = \lambda/2$, which is 180°. With $Z_1 = \sqrt{Z_0 R_L}$ they are also equal in size, so they cancel.

Limitations:

1. **Real loads only.** The load must be purely resistive. A complex load must first be turned into a real one, either by adding a length of line to reach a voltage maximum or minimum (where $Z = Z_0 \cdot \text{SWR}$ or Z_0/SWR), or by cancelling its reactance with a stub.
2. **Narrow band.** The section is exactly $\lambda/4$ at one frequency only. Away from f_0 the match gets worse (figure). Wider bands need several sections in cascade (binomial or Chebyshev multi-section transformers).
3. **Impractical Z_1.** Z_1 must lie in the range that can actually be built (about 20 to 150 Ω for microstrip). Very large or very small ratios R_L/Z_0 give impractical values.
4. **Physical size.** At low frequencies $\lambda/4$ is long (75 cm at 100 MHz), so the section is bulky.
5. **Fixed design.** It cannot be adjusted (tuned) after it is made, unlike a stub.

3. Keyword Explanation

1. Bandwidth (of a match)

The range of frequencies over which the reflection stays below a chosen limit, for example $|\Gamma| \leq 0.1$.

2. Impedance inverter

Another name for a $\lambda/4$ line, because $Z_{in} \propto 1/Z_L$.

4. Notes and Remarks

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- **Odd multiples:** $3\lambda/4$, $5\lambda/4$, … also work at f_0, but they are even more narrowband.
- **Fractional bandwidth:** for the 50 Ω to 100 Ω example, $|\Gamma| \le 0.1$ holds over about 37 % of f_0. The closer R_L is to Z_0, the wider the band.

5. Final Concise Answer

(a) From $z = \frac{1+\Gamma}{1-\Gamma}$ with $\Gamma = u+jv$: $r = \frac{1-u^2-v^2}{(1-u)^2+v^2}$ and $x = \frac{2v}{(1-u)^2+v^2}$. These rearrange to $\left(u - \frac{r}{1+r}\right)^2 + v^2 = \left(\frac{1}{1+r}\right)^2$ (constant-r circles: centre $(\frac{r}{1+r}, 0)$, radius $\frac{1}{1+r}$) and $(u-1)^2 + (v-\frac{1}{x})^2 = \left(\frac{1}{x}\right)^2$ (constant-x circles: centre $(1, \frac{1}{x})$, radius $\frac{1}{|x|}$).

(b) $z_L = 0.467 - j0.667$, $y_L = 0.705 + j1.007$, SWR $= 3.25$. Solution 1: $l_p = 0.025\lambda$ ($y = 1 + j1.25$), stub $-j1.25$, $l_s = 0.107\lambda$. Solution 2: $l_p = 0.186\lambda$ ($y = 1 - j1.25$), stub $+j1.25$, $l_s = 0.393\lambda$.

(c) A $\lambda/4$ section of impedance $Z_1 = \sqrt{Z_0 R_L}$ makes $Z_{in} = Z_1^2/R_L = Z_0$. Limitations: resistive loads only, narrow band (exact at one frequency), Z_1 must be realizable, bulky at low frequency, not adjustable.

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3. (a) A plane wave in a nonmagnetic medium has $\vec{E} = 50\sin(10^8 t + 2z)\hat{a}_y$ V/m. Find:

   (i) The direction of wave propagation
   (ii) λ, f, and ϵ_r
   (iii) $\vec{H}$ Marks: 6

   (b) A telephone line has $R = 30\ \Omega\ \text{km}^{-1}$, $L = 100$ mH km^{-1}, $G = 0$, and $C = 20\ \mu$F km^{-1}. At $f = 1$ kHz, obtain:

   (i) The characteristic impedance of the line
   (ii) The propagation constant
   (iii) The phase velocity Marks: 6

   (c) An antenna with an impednace of $40 + j30\ \Omega$ is to be matched to a $100\ \Omega$ lossless line with a shorted stub. Determine:

   (i) The required stub admittance
   (ii) The distance between the stub and the antenna
   (iii) The stub length
   (iv) The standing wave ratio on each segment of the system Marks: 6

   (d) A rectangular waveguide with dimensions $a = 2.5$ cm, $b = 1$ cm is to operate below 15.1 GHz. How many TE and TM modes can the waveguide transmit if the guide is filled with a medium characterized by $\sigma = 0$, $\epsilon = 4\epsilon_0$, $\mu_r = 1$? Calculate the cutoff frequencies of the modes. Marks: 7

3.(a) A plane wave in a nonmagnetic medium has $\vec{E} = 50\sin(10^8 t + 2z)\hat{a}_y$ V/m. Find: (i) The direction of wave propagation (ii) λ, f, and ϵ_r (iii) $\vec{H}$ Marks: 6

1. Theory

1. What is an electromagnetic wave? (Core Idea)

An **electromagnetic (EM) wave** is a travelling disturbance of electric field **E** and magnetic field **H**. A changing **E** creates **H** and a changing **H** creates **E** (Maxwell's equations), so the pair keeps itself going through space with no wires at all. Radio, Wi-Fi, microwaves and light are all EM waves; only the frequency differs.

2. What is a uniform plane wave?

Far from the antenna, the wave fronts are so large that a small piece of them looks flat: a **plane**. On such a plane **E** and **H** are the same everywhere (**uniform**). For a uniform plane wave:

- **E**, **H** and the direction of travel are all **perpendicular** to each other (a TEM wave);
- the direction of travel is the direction of $\mathbf{E} \times \mathbf{H}$;
- $|\mathbf{E}|/|\mathbf{H}| = \eta$, the **intrinsic impedance** of the medium ($\eta_0 = 120\pi \approx 377\ \Omega$ in free space).

3. How to read the wave equation

A wave written as $E_0 \sin(\omega t \mp \beta z)$ has:

- angular frequency ω (rad/s), so $f = \omega/2\pi$;
- phase constant β (rad/m), so $\lambda = 2\pi/\beta$;
- speed $u = \omega/\beta$.

Direction rule: follow a crest, where the phase $\omega t \mp \beta z$ is constant. With $\omega t - \beta z$, as t grows z must grow: the wave travels toward $+z$. With $\omega t + \beta z$, z must shrink: it travels toward $-z$.

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4. Speed in a material

In a medium with relative permittivity ϵ_r and relative permeability μ_r:

$$u = \frac{c}{\sqrt{\mu_r \epsilon_r}}, \qquad \eta = \eta_0 \sqrt{\frac{\mu_r}{\epsilon_r}}$$

"Nonmagnetic" means $\mu_r = 1$, so only ϵ_r slows the wave.

2. Answer

Given: $\vec{E} = 50\sin(10^8 t + 2z)\hat{a}_y$ V/m, so $E_0 = 50$ V/m, $\omega = 10^8$ rad/s, $\beta = 2$ rad/m, and $\mu_r = 1$.

(i) Direction of propagation. The phase is $\omega t + \beta z$ (plus sign), so the wave travels in the $-z$ **direction** $(-\hat{a}_z)$.

(ii) λ, f and ϵ_r.

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{2} = \pi = 3.142 \text{ m}$$

$$f = \frac{\omega}{2\pi} = \frac{10^8}{2\pi} = 15.92 \text{ MHz}$$

$$u = \frac{\omega}{\beta} = \frac{10^8}{2} = 5 \times 10^7 \text{ m/s}$$

Since $u = c/\sqrt{\epsilon_r}$ (with $\mu_r = 1$):

$$\epsilon_r = \left(\frac{c}{u}\right)^2 = \left(\frac{3 \times 10^8}{5 \times 10^7}\right)^2 = 6^2 = 36$$

$$\boxed{\lambda = 3.142 \text{ m}, \quad f = 15.92 \text{ MHz}, \quad \epsilon_r = 36}$$

(iii) Magnetic field $\vec{H}$. The intrinsic impedance is

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{120\pi}{6} = 20\pi = 62.83\ \Omega$$

so $H_0 = E_0/\eta = 50/(20\pi) = 0.796$ A/m.

For the direction, $\hat{a}_E \times \hat{a}_H$ must equal the direction of travel, $-\hat{a}_z$. With $\hat{a}_E = \hat{a}_y$: since $\hat{a}_y \times \hat{a}_x = -\hat{a}_z$, we need $\hat{a}_H = +\hat{a}_x$. Therefore

$$\boxed{\vec{H} = 0.796\sin\left(10^8 t + 2z\right)\hat{a}_x \text{ A/m} = 796\sin\left(10^8 t + 2z\right)\hat{a}_x \text{ mA/m}}$$

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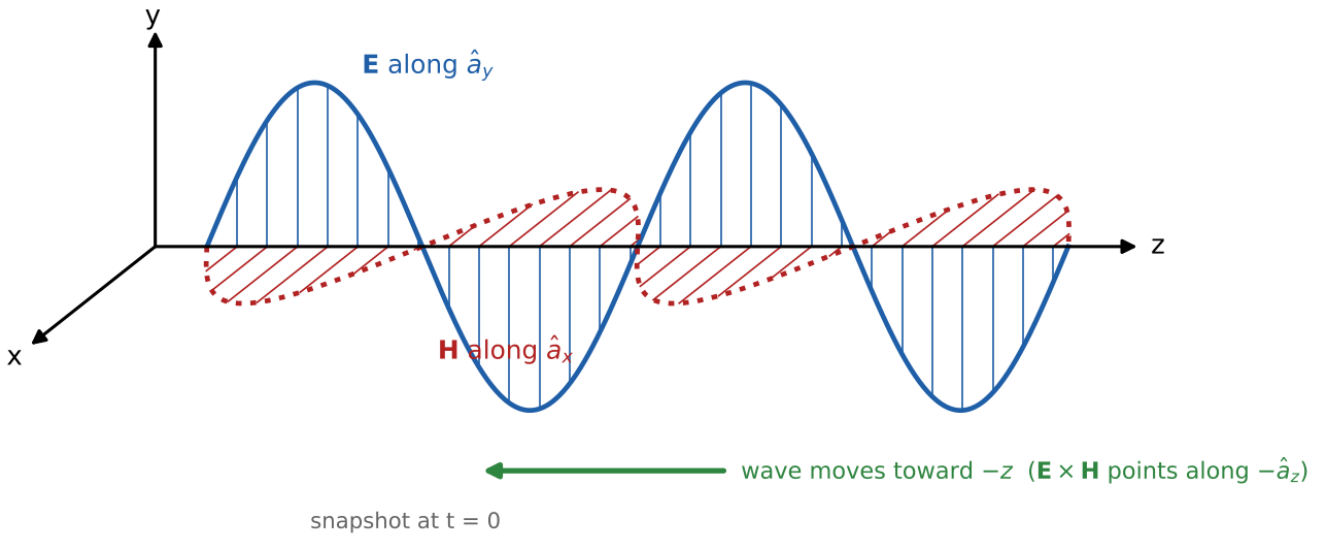


Figure 11: The plane wave of Q3(a)

Reading the figure. At $t = 0$, **E** (blue, solid outline, vertical hatching) swings up and down along y and **H** (red, dotted outline, slanted hatching) swings along x (drawn slanting into the page). They are in phase and at right angles, and the pattern slides toward $-z$ (the green arrow at the bottom).

3. Keyword Explanation

1. Intrinsic impedance η

The ratio E/H for a plane wave in a medium. It plays the same role for waves in space that Z_0 plays for waves on a line.

2. Nonmagnetic medium

A material with $\mu = \mu_0$ ($\mu_r = 1$): most dielectrics, water, glass, plastics.

3. Lossless medium

No conduction ($\sigma = 0$), so **E** and **H** stay in phase and the amplitude does not decay.

4. Notes and Remarks

- **Check:** $\beta = \omega\sqrt{\mu\epsilon} = \omega\sqrt{\epsilon_r}/c = 10^8 \times 6/(3 \times 10^8) = 2$ rad/m ✔.
- **Common slip:** taking the direction from the sign of ωt. Only the sign in front of βz matters, relative to ωt.

3.(b) A telephone line has $R = 30\ \Omega\ \mathrm{km}^{-1}$, $L = 100$ mH km^{-1}, $G = 0$, and $C = 20\ \mu$F km^{-1}. At $f = 1$ kHz, obtain: (i) The characteristic impedance of the line (ii) The propagation constant (iii) The phase velocity Marks: 6

1. Theory

1. A real line with losses (Core Idea)

A telephone line is a transmission line (see the Theory of 1(a)) with a large series resistance, because the copper wires are thin and kilometres long. At 1 kHz its R is not small compared with ωL, so we cannot use the low-loss shortcut. We use the full formulas:

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}, \qquad \gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}, \qquad u_p = \frac{\omega}{\beta}$$

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2. Complex square roots in polar form

To take $\sqrt{M\angle\theta}$, take the root of the magnitude and *half* the angle: $\sqrt{M\angle\theta} = \sqrt{M}\angle(\theta/2)$. Products and quotients work the same way: multiply (divide) the magnitudes and add (subtract) the angles.

3. Units "per km"

All four line constants are given per km. If we keep them per km, γ comes out per km and u_p in km/s; Z_0 (a ratio) has no length unit.

2. Answer

Given: $R = 30\ \Omega$/km, $L = 100$ mH/km $= 0.1$ H/km, $G = 0$, $C = 20\ \mu$F/km $= 20 \times 10^{-6}$ F/km, $f = 1$ kHz.

$$\omega = 2\pi f = 2\pi \times 10^3 = 6283.2 \text{ rad/s}$$

Series impedance and shunt admittance per km:

$$R + j\omega L = 30 + j(6283.2)(0.1) = 30 + j628.32 = 629.03\angle 87.27^\circ\ \Omega/\text{km}$$

$$G + j\omega C = 0 + j(6283.2)(20 \times 10^{-6}) = j0.12566 = 0.12566\angle 90^\circ\ \text{S/km}$$

(i) Characteristic impedance.

$$Z_0 = \sqrt{\frac{629.03\angle 87.27^\circ}{0.12566\angle 90^\circ}} = \sqrt{5005.7\angle -2.73^\circ} = 70.75\angle -1.367^\circ\ \Omega$$

$$\boxed{Z_0 = 70.75\angle -1.367^\circ\ \Omega = (70.73 - j1.69)\ \Omega}$$

(ii) Propagation constant.

$$\gamma = \sqrt{(629.03\angle 87.27^\circ)(0.12566\angle 90^\circ)} = \sqrt{79.05\angle 177.27^\circ} = 8.891\angle 88.63^\circ\ /\text{km}$$

$$\boxed{\gamma = 0.2121 + j8.888\ /\text{km}}$$

so $\alpha = 0.2121$ Np/km and $\beta = 8.888$ rad/km.

(iii) Phase velocity.

$$u_p = \frac{\omega}{\beta} = \frac{6283.2}{8.888} = 706.9 \text{ km/s}$$

$$\boxed{u_p = 7.069 \times 10^5 \text{ m/s}}$$

That is only about 0.24 % of the speed of light, because this line has a very large L and C.

Final Answer: $Z_0 = 70.75\angle -1.37^\circ\ \Omega$; $\gamma = 0.212 + j8.888$ /km; $u_p = 7.07 \times 10^5$ m/s.

3. Keyword Explanation

1. Phase velocity

The speed at which a point of constant phase (a crest) moves along the line, $u_p = \omega/\beta$.

2. Loaded telephone line

Old long-distance telephone lines had extra coils added (loading coils) to raise L. This brought the line closer to the distortionless condition $R/L = G/C$ (see 1(a), Notes).

4. Notes and Remarks

- **Lossless comparison:** $\sqrt{L/C} = 70.71\ \Omega$ and $1/\sqrt{LC} = 707.1$ km/s. These are very close to the exact answers, because $R/(\omega L) = 0.048$ is already fairly small at 1 kHz.

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- **Wavelength:** $\lambda = 2\pi/\beta = 0.707$ km, so even a few kilometres of telephone line is several wavelengths long at 1 kHz.

3.(c) An antenna with an impednace of $40 + j30\ \Omega$ is to be matched to a $100\ \Omega$ lossless line with a shorted stub. Determine: (i) The required stub admittance (ii) The distance between the stub and the antenna (iii) The stub length (iv) The standing wave ratio on each segment of the system Marks: 6

1. Theory

1. The idea (summary of 2(b))

An antenna is a load at the end of a feed line. If its impedance differs from Z_0, some power bounces back (see the Theory of 1(c)). A **shunt short-circuited stub** fixes this in two moves (see the Theory of 2(b) for the full explanation and figure):

1. Move from the antenna toward the generator by a distance d until the normalized admittance becomes $y = 1 \pm jb$ (the SWR circle meets the $g = 1$ circle).
2. Connect a shorted stub there whose admittance is $\mp jb$. The total becomes 1, and the line is matched from that point back to the generator.

2. What SWR to expect on each piece

- **Between the antenna and the stub:** the reflected wave from the antenna is still there, so the SWR is that of the antenna itself, $(1 + |\Gamma_L|)/(1 - |\Gamma_L|)$.
- **Between the stub and the generator:** matched, so the SWR is 1.
- **On the stub:** the shorted end reflects everything ($|\Gamma| = 1$), so the SWR is infinite.

2. Answer

Given: $Z_L = 40 + j30\ \Omega$, $Z_0 = 100\ \Omega$ (so $Y_0 = 10$ mS). Shunt shorted stub.

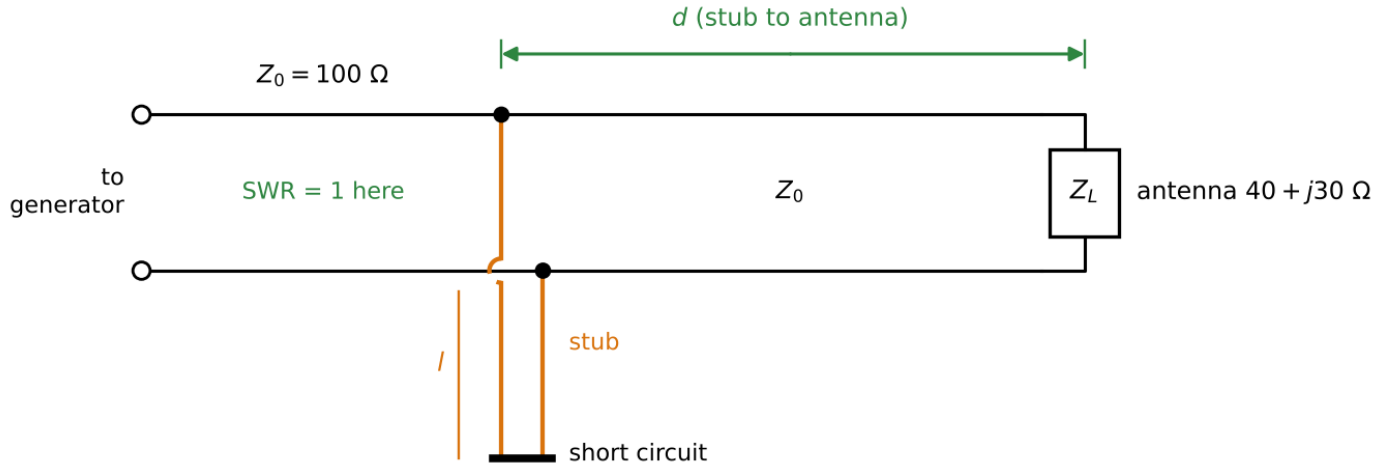


Figure 12: Stub matching of the antenna

Step 1: Normalize

$$z_L = \frac{40 + j30}{100} = 0.4 + j0.3, \qquad y_L = \frac{1}{0.4 + j0.3} = \frac{0.4 - j0.3}{0.25} = 1.6 - j1.2$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{-60 + j30}{140 + j30} = 0.4685\angle 141.34^\circ$$

y_L reads 0.304λ on the WTG scale of the chart.

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Step 2: Move to the $g = 1$ circle

Turning clockwise on the SWR circle ($|\Gamma| = 0.4685$), the $g = 1$ circle is met at:

- **A:** $y = 1 - j1.06$ at 0.336λ, so $d_1 = 0.336\lambda - 0.304\lambda = 0.0325\lambda$;
- **B:** $y = 1 + j1.06$ at 0.164λ, so $d_2 = (0.5 - 0.304 + 0.164)\lambda = 0.360\lambda$.

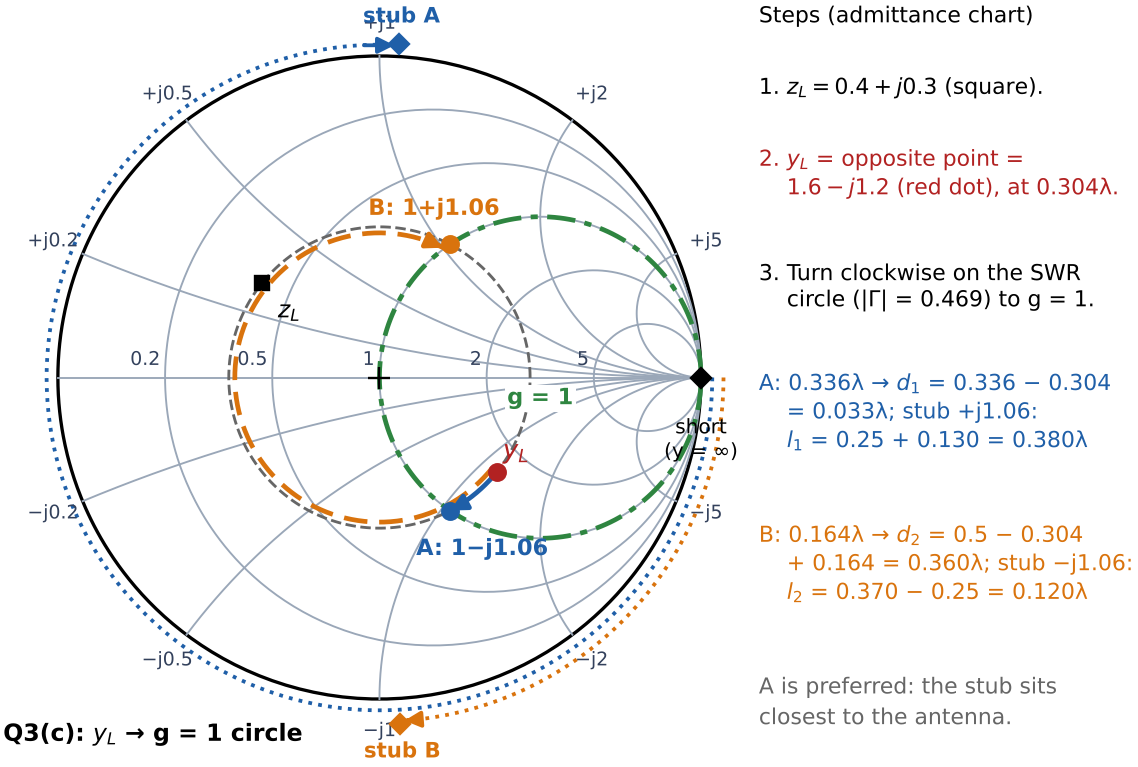


Figure 13: Smith chart solution of Q3(c)

(i) Required stub admittance. The stub must cancel the susceptance:

- for A, $y_s = +j1.06$, so $Y_s = +j1.06 \times 10$ mS $= +j10.6$ mS;
- for B, $y_s = -j1.06$, so $Y_s = -j10.6$ mS.

$$\boxed{Y_s = \pm j10.6 \text{ mS } (y_s = \pm j1.06)}$$

(ii) Distance between the stub and the antenna.

$$\boxed{d = 0.0325\lambda \text{ (solution A)} \quad \text{or} \quad d = 0.360\lambda \text{ (solution B)}}$$

(iii) Stub length. A short is at 0.25λ on the admittance chart. Turning clockwise:

- for A, to $+j1.06$ (at 0.130λ): $l = (0.5 - 0.25 + 0.130)\lambda = 0.380\lambda$;
- for B, to $-j1.06$ (at 0.370λ): $l = (0.370 - 0.25)\lambda = 0.120\lambda$.

Check with $y_s = -j \cot \beta l$: $\cot \beta l = \mp 1.0607$ gives $l = 0.3797\lambda$ and 0.1203λ ✔.

$$\boxed{l = 0.380\lambda \text{ (A)} \quad \text{or} \quad l = 0.120\lambda \text{ (B)}}$$

(iv) SWR on each segment.

$$s_{antenna\text{-}stub} = \frac{1 + 0.4685}{1 - 0.4685} = 2.76$$

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Segment	SWR	Why
antenna to stub	2.76	the antenna's reflection is still present
stub to generator	1	matched
stub itself	∞	short circuit, total reflection

Table 5: SWR on each segment

Final Answer (solution A, the usual choice): $Y_s = +j10.6$ mS, $d = 0.0325\lambda$, $l = 0.380\lambda$, SWR $= 2.76$, 1 and ∞. (Solution B: $-j10.6$ mS, $d = 0.360\lambda$, $l = 0.120\lambda$, with the same SWRs.)

3. Keyword Explanation

1. Segment

One piece of the system between two junctions: here antenna to stub, stub to generator, and the stub itself.

4. Notes and Remarks

- **Which solution:** A puts the stub only 0.0325λ from the antenna, so the length of line carrying the high SWR is short. That means less loss and a wider bandwidth. Textbooks (Sadiku, Example on single-stub matching) give A as the answer.
- **Signs:** the stub admittance is opposite in sign to the susceptance b at the stub position.

3.(d) A rectangular waveguide with dimensions $a = 2.5$ cm, $b = 1$ cm is to operate below 15.1 GHz. How many TE and TM modes can the waveguide transmit if the guide is filled with a medium characterized by $\sigma = 0$, $\epsilon = 4\epsilon_0$, $\mu_r = 1$? Calculate the cutoff frequencies of the modes. Marks: 7

1. Theory

1. What is a waveguide? (Core Idea)

At microwave frequencies, two-wire lines and coaxial cables lose a lot of power. A **waveguide** is a hollow metal pipe, here with a rectangular cross-section $a \times b$ ($a > b$). The wave travels inside the pipe by bouncing between the walls. There is no centre conductor, so there is less loss and it can carry high power.

2. What is a mode?

The metal walls force the electric field parallel to them to be zero at the walls. Across the width a the field can therefore only form a whole number m of half-waves, and across the height b a whole number n of half-waves, like a guitar string fixed at both ends. Each allowed pattern (m, n) is a **mode**.

- $\mathbf{TE}_{mn}$ (transverse electric): **E** has no component along the guide. It needs m or $n \neq 0$, but not both zero.
- $\mathbf{TM}_{mn}$ (transverse magnetic): **H** has no component along the guide. It needs *both* $m \geq 1$ and $n \geq 1$.

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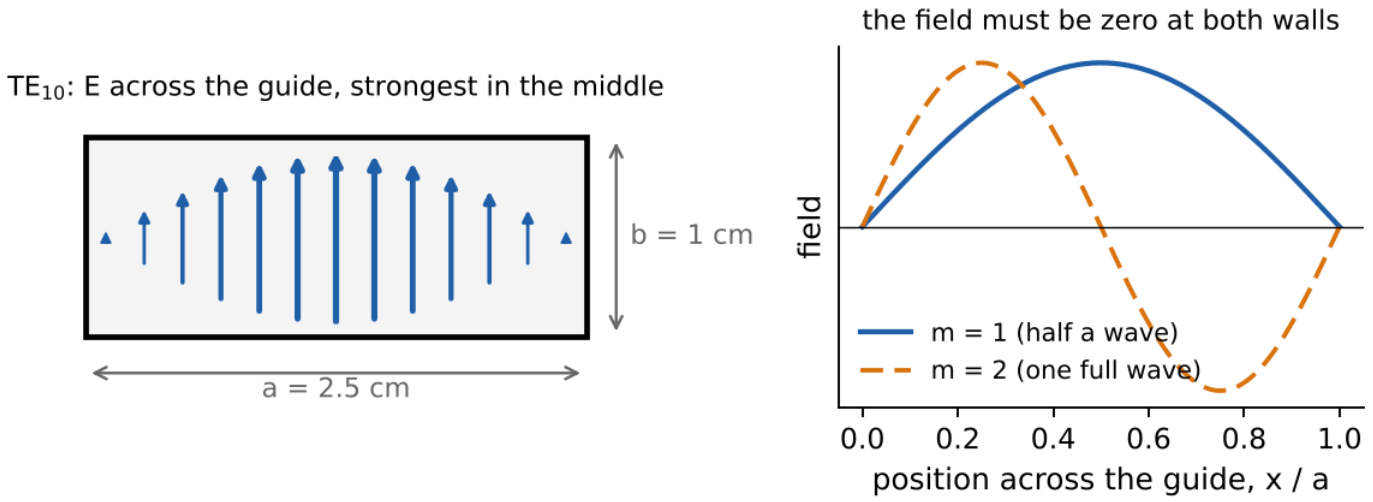


Figure 14: TE10 field across the guide

Reading the figure. Left: in the TE_{10} mode the electric field (arrows) points across the short side, is strongest in the middle and falls to zero at the side walls (one half-wave across a, none across b). Right: $m = 1$ is half a wave across the guide and $m = 2$ is a full wave. Both are zero at both walls.

3. Cutoff frequency

A mode can fit in the guide only if half its wavelength is small enough to fit the pattern. Below a certain frequency, the **cutoff frequency**, the mode cannot travel and dies out within a short distance:

$$f_{c,mn} = \frac{u'}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}, \qquad u' = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{\mu_r \epsilon_r}}$$

A mode **propagates** only when the operating frequency is above its cutoff, $f > f_c$. The mode with the lowest cutoff, TE_{10} when $a > b$, is the **dominant mode**.

4. Effect of filling the guide

A dielectric slows the wave ($u' < c$), so every cutoff frequency drops by $\sqrt{\epsilon_r}$. More modes then fit below a given frequency.

2. Answer

Given: $a = 2.5$ cm $= 0.025$ m, $b = 1$ cm $= 0.01$ m, $\epsilon_r = 4$, $\mu_r = 1$, $\sigma = 0$, operating frequency $f < 15.1$ GHz.

Wave speed in the filling:

$$u' = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{2} = 1.5 \times 10^8 \text{ m/s}$$

Cutoff formula with numbers:

$$f_{c,mn} = \frac{1.5 \times 10^8}{2}\sqrt{\left(\frac{m}{0.025}\right)^2 + \left(\frac{n}{0.01}\right)^2} = 0.75 \times 10^8 \sqrt{1600m^2 + 10^4 n^2} \text{ Hz}$$

$$f_{c,mn} = 3\sqrt{m^2 + 6.25n^2} \text{ GHz}$$

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Every mode with $f_c < 15.1$ GHz:

Mode	f_c (GHz)	Mode	f_c (GHz)
TE_{10}	3.00	TE_{21}, TM_{21}	9.60
TE_{20}	6.00	TE_{31}, TM_{31}	11.72
TE_{01}	7.50	TE_{40}	12.00
TE_{11}, TM_{11}	8.08	TE_{41}, TM_{41}	14.15
TE_{30}	9.00	TE_{02}, TE_{50}	15.00

Table 6: Modes that propagate below 15.1 GHz

The next modes are cut off: TE_{12}/TM_{12} at 15.30 GHz, TE_{22}/TM_{22} at 16.16 GHz and TE_{51}/TM_{51} at 16.77 GHz.

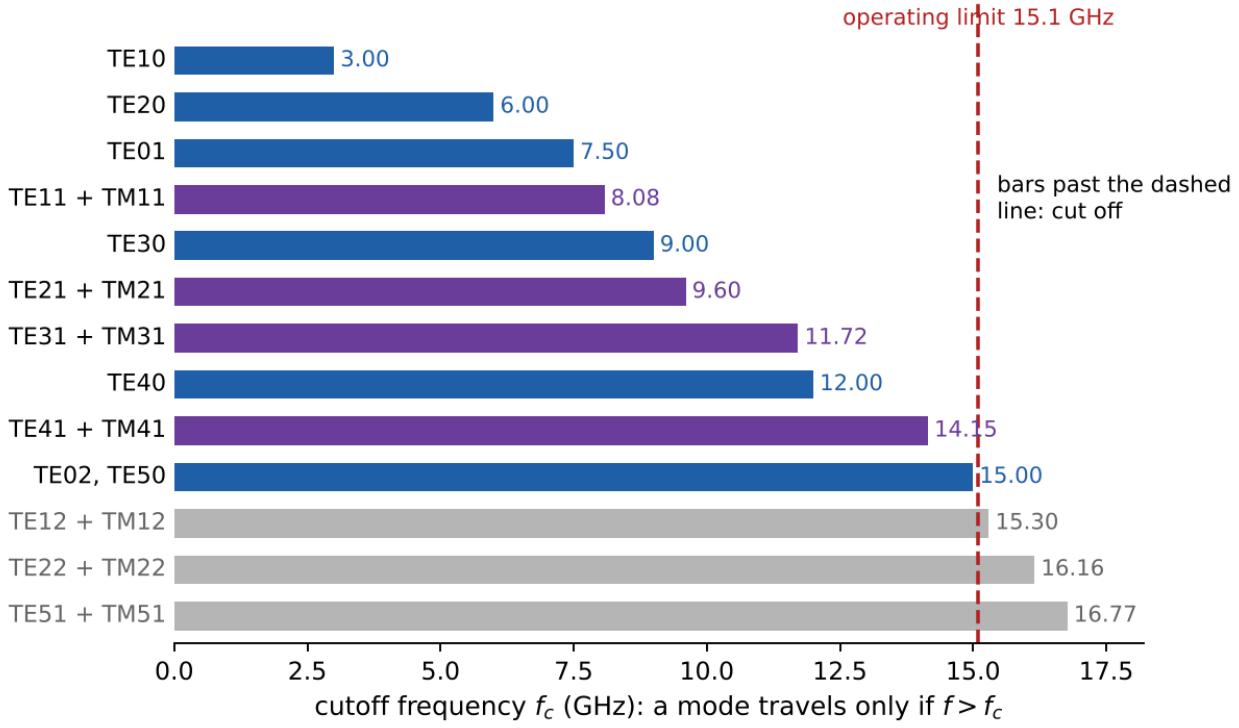


Figure 15: Cutoff frequencies of the filled guide

Count:

- TE modes: TE_{10}, TE_{20}, TE_{01}, TE_{11}, TE_{30}, TE_{21}, TE_{31}, TE_{40}, TE_{41}, TE_{02}, TE_{50}: **11 modes**.
- TM modes: TM_{11}, TM_{21}, TM_{31}, TM_{41}: **4 modes**.

$$\boxed{11 \text{ TE modes} + 4 \text{ TM modes} = 15 \text{ modes}}$$

Final Answer: the guide can carry 11 TE and 4 TM modes (15 in total) below 15.1 GHz. Their cutoff frequencies are listed above; the dominant TE_{10} mode has $f_c = 3$ GHz.

3. Keyword Explanation

1. Dominant mode

The mode with the lowest cutoff frequency. Waveguides are normally used between the first and second cutoffs, so that only this one mode travels (here 3 to 6 GHz).

2. Evanescent mode

A mode below its cutoff. It does not travel but decays exponentially along the guide.

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4. Notes and Remarks

- **Why TM_{m0} and TM_{0n} do not exist:** the TM field contains $\sin(m\pi x/a)\sin(n\pi y/b)$, which is zero everywhere if m or n is 0.
- **TE_{02} and TE_{50} at exactly 15 GHz:** they are below 15.1 GHz, so they count. This is why the paper says 15.1 GHz and not 15 GHz.
- **Air-filled comparison:** with $\epsilon_r = 1$ every cutoff doubles (TE_{10} at 6 GHz), and only TE_{10}, TE_{20} and TE_{01} would propagate below 15.1 GHz.

5. Final Concise Answer

(a) (i) The wave travels in $-\hat{a}_z$. (ii) $\lambda = \pi = 3.142$ m, $f = 15.92$ MHz, $\epsilon_r = 36$. (iii) $\eta = 20\pi\ \Omega$, so $\vec{H} = 0.796\sin(10^8 t + 2z)\hat{a}_x$ A/m.

(b) (i) $Z_0 = 70.75\angle -1.367^\circ\ \Omega$. (ii) $\gamma = 0.2121 + j8.888$ /km. (iii) $u_p = \omega/\beta = 7.069 \times 10^5$ m/s.

(c) $y_L = 1.6 - j1.2$, SWR $= 2.76$. (i) $Y_s = \pm j10.6$ mS. (ii) $d = 0.0325\lambda$ (or 0.360λ). (iii) $l = 0.380\lambda$ (or 0.120λ). (iv) SWR $= 2.76$ between the antenna and the stub, 1 between the stub and the generator, ∞ on the stub.

(d) $u' = 1.5 \times 10^8$ m/s and $f_c = 3\sqrt{m^2 + 6.25n^2}$ GHz. Below 15.1 GHz: 11 TE modes (TE_{10} 3, TE_{20} 6, TE_{01} 7.5, TE_{11} 8.08, TE_{30} 9, TE_{21} 9.60, TE_{31} 11.72, TE_{40} 12, TE_{41} 14.15, TE_{02} 15, TE_{50} 15 GHz) and 4 TM modes (TM_{11}, TM_{21}, TM_{31}, TM_{41}), 15 in total.

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4. (a) The magnetic vector potential of a Hertzian dipole is given by: $\vec{A} = \frac{\mu_0 I dl}{4\pi r} e^{-j\beta r}\hat{z}$. Show that magnetic field, $H = \frac{Idl}{4\pi}\cos(\theta)\left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r}\hat{\phi}$ [*sic*: should be $\sin(\theta)$]. Marks: 8

   (b) Using the expression of H found in 4(a), show that the electric field component $E = \frac{\eta Idl}{4\pi}\cos(\theta)\left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}\hat{\theta} + \frac{\eta Idl}{2\pi}\cos(\theta)\left[\frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}\hat{r}$ [*sic*: the $\hat{\theta}$ term should have $\sin(\theta)$] Marks: 12

   (c) Find the expression of average power density of Hertzian dipole in far field zone. Marks: 5

4.(a) The magnetic vector potential of a Hertzian dipole is given by: $\vec{A} = \frac{\mu_0 I dl}{4\pi r} e^{-j\beta r}\hat{z}$. Show that magnetic field, $H = \frac{Idl}{4\pi}\cos(\theta)\left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r}\hat{\phi}$ [*sic*: should be $\sin(\theta)$]. Marks: 8

1. Theory

1. What is an antenna, and what is a Hertzian dipole? (Core Idea)

An **antenna** turns a current in a wire into an EM wave in space (transmitting), or the other way round (receiving). The simplest antenna to analyse is the **Hertzian dipole**: a very short piece of wire, of length $dl \ll \lambda$, carrying a current I that is the same along its whole length and varies as $I\cos\omega t$. Any real antenna can be thought of as many Hertzian dipoles end to end, so its fields are the building block of antenna theory.

2. Why use a vector potential?

Finding **E** and **H** straight from a current is hard. It is easier to first find a helper field, the **magnetic vector potential A**, which:

- points in the same direction as the current ($\hat{z}$ for a wire along z);
- comes from adding up $\mu_0 I\, dl/(4\pi r)$ from each bit of current, delayed by the travel time r/c, which gives the factor $e^{-j\beta r}$.

Then the fields follow by differentiation:

$$\mathbf{B} = \mu_0\mathbf{H} = \nabla \times \mathbf{A}, \qquad \mathbf{E} = \frac{1}{j\omega\epsilon}\nabla \times \mathbf{H} \text{ (away from the source)}$$

3. Spherical coordinates

Around a dipole, spherical coordinates are natural:

- r: distance from the dipole;
- θ: angle down from the dipole axis (z-axis);
- ϕ: angle around the axis.

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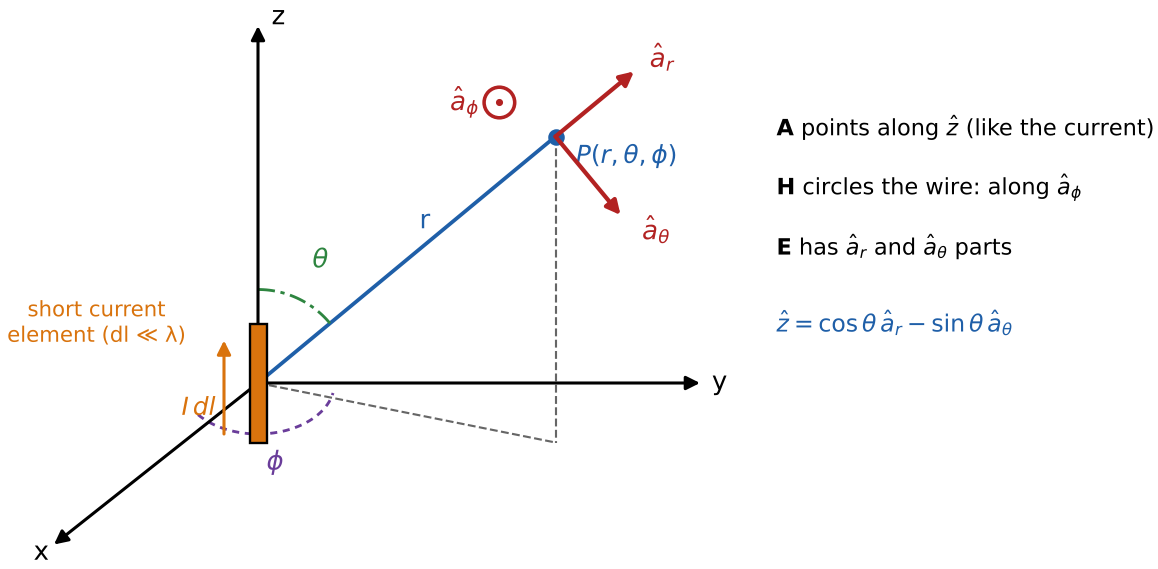


Figure 16: Hertzian dipole and spherical coordinates

Reading the figure. The dipole (the short orange bar labelled $I\,dl$) sits at the origin along z. A point P is at distance r and angle θ from the axis. At P the unit vectors are: $\hat{a}_r$ pointing away from the dipole, $\hat{a}_\theta$ pointing "south" (increasing θ), and $\hat{a}_\phi$ circling the axis (out of the page here). From the triangle, $\hat{z}$ has a part $\cos\theta$ along $\hat{a}_r$ and a part $-\sin\theta$ along $\hat{a}_\theta$.

2. Answer

Note on the paper

The printed answer has $\cos(\theta)$. Starting from the given $\vec{A}$, the curl gives $\sin\theta$ (as below, and as in every textbook). The $\cos\theta$ is a printing error, and the correct result is derived here.

Step 1: Write A in spherical components. Using $\hat{z} = \cos\theta\,\hat{a}_r - \sin\theta\,\hat{a}_\theta$:

$$A_r = \frac{\mu_0 I dl}{4\pi r} e^{-j\beta r}\cos\theta, \qquad A_\theta = -\frac{\mu_0 I dl}{4\pi r} e^{-j\beta r}\sin\theta, \qquad A_\phi = 0$$

Step 2: Curl in spherical coordinates. None of the components depends on ϕ (the dipole looks the same from every side) and $A_\phi = 0$. So the r and θ components of the curl are zero, and only the ϕ component remains:

$$(\nabla \times \mathbf{A})_\phi = \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial \theta}\right]$$

Step 3: The two derivatives.

$$rA_\theta = -\frac{\mu_0 I dl \sin\theta}{4\pi} e^{-j\beta r} \;\Rightarrow\; \frac{\partial}{\partial r}(rA_\theta) = \frac{j\beta\,\mu_0 I dl \sin\theta}{4\pi} e^{-j\beta r}$$

$$\frac{\partial A_r}{\partial \theta} = -\frac{\mu_0 I dl \sin\theta}{4\pi r} e^{-j\beta r}$$

Step 4: Combine.

$$(\nabla \times \mathbf{A})_\phi = \frac{1}{r}\left[\frac{j\beta\,\mu_0 I dl \sin\theta}{4\pi} + \frac{\mu_0 I dl \sin\theta}{4\pi r}\right] e^{-j\beta r} = \frac{\mu_0 I dl \sin\theta}{4\pi}\left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r}$$

Step 5: Divide by μ_0. Since $\mathbf{H} = \frac{1}{\mu_0}\nabla \times \mathbf{A}$:

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$$\boxed{\mathbf{H} = \frac{Idl}{4\pi}\sin\theta\left[\frac{j\beta}{r} + \frac{1}{r^2}\right]e^{-j\beta r}\,\hat{a}_\phi}$$

So $H_r = H_\theta = 0$ and only H_ϕ exists: the magnetic field lines are circles around the dipole axis, as for any current in a straight wire.

Interpretation:

- **Near the dipole** ($\beta r \ll 1$) the $1/r^2$ term wins: $H_\phi \approx Idl\sin\theta/(4\pi r^2)$, which is the **Biot–Savart** field of a current element.
- **Far away** ($\beta r \gg 1$) the $j\beta/r$ term wins: this is the **radiation** field (see 4(c)).

3. Keyword Explanation

1. Magnetic vector potential A

A field whose curl gives **B**. For currents it is simpler to compute than **B** itself, because it points along the current.

2. Curl

$\nabla \times \mathbf{A}$ measures how much a field "circulates" around a point.

3. $e^{-j\beta r}$ (retardation)

The phase delay of a wave that has travelled a distance r. The effect of the current reaches distance r only after a time r/c.

4. Notes and Remarks

- **Why sin θ and not cos θ:** at $\theta = 0$ (on the axis) the field must vanish, because a current element produces no magnetic field along its own line (Biot–Savart has $d\mathbf{l} \times \hat{a}_r = 0$ there). $\sin 0 = 0$ fits; $\cos 0 = 1$ would not.
- **Units:** $I\,dl$ is in A · m and $1/r^2$ in $1/\text{m}^2$, so H is in A/m ✔.

4.(b) Using the expression of H found in 4(a), show that the electric field component $E = \frac{\eta Idl}{4\pi}\cos(\theta)\left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right]e^{-j\beta r}\hat{\theta} + \frac{\eta Idl}{2\pi}\cos(\theta)\left[\frac{1}{r^2} - \frac{j}{\beta r^3}\right]e^{-j\beta r}\hat{r}$ [*sic*: the $\hat{\theta}$ term should have $\sin(\theta)$]

Marks: 12

1. Theory

1. Getting E from H (Core Idea)

Outside the wire there is no current, so Maxwell's (Ampère's) law is $\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E}$. Therefore

$$\mathbf{E} = \frac{1}{j\omega\epsilon}\nabla \times \mathbf{H}$$

Once **H** is known (from 4(a)), one more curl gives **E**.

2. A useful identity

With $\eta = \sqrt{\mu/\epsilon}$ and $\beta = \omega\sqrt{\mu\epsilon}$, we get $\omega\epsilon = \beta/\eta$. Hence

$$\frac{1}{j\omega\epsilon} = \frac{\eta}{j\beta}$$

This is how η enters the answer.

3. Three kinds of terms

The answer contains $1/r$, $1/r^2$ and $1/r^3$ terms:

- $1/r^3$: the **electrostatic** (dipole) field of the charges piling up at the ends of the wire;

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- $1/r^2$: the **induction** field (energy that moves out and back each cycle);
- $1/r$: the **radiation** field (energy that leaves for good).

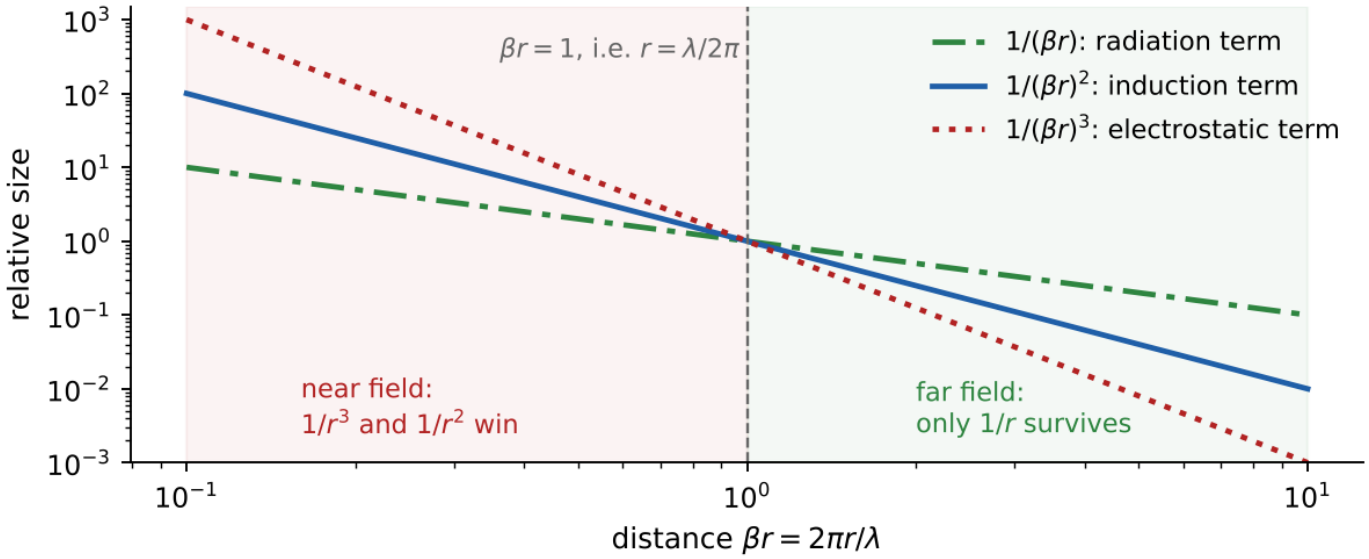


Figure 17: How the three kinds of terms compare with distance

Reading the figure. All three terms are equal at $\beta r = 1$ ($r = \lambda/2\pi$). Closer in (left), $1/r^3$ dominates; farther out (right), only $1/r$ matters. This is why the far field (4(c)) keeps only the $1/r$ terms.

2. Answer

Note on the paper

As in 4(a), the $\hat{\theta}$ term should contain $\sin\theta$, not $\cos\theta$. The $\hat{r}$ term correctly has $\cos\theta$. The derivation below gives the correct result.

From 4(a), $\mathbf{H} = H_\phi \hat{a}_\phi$ with

$$H_\phi = \frac{Idl\sin\theta}{4\pi}\left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r}$$

Step 1: Curl of a field with only a ϕ component (no ϕ-dependence):

$$\nabla \times \mathbf{H} = \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(H_\phi \sin\theta)\,\hat{a}_r - \frac{1}{r}\frac{\partial}{\partial r}(rH_\phi)\,\hat{a}_\theta$$

Step 2: Radial component E_r.

$$H_\phi \sin\theta = \frac{Idl\sin^2\theta}{4\pi}\left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r} \;\Rightarrow\; \frac{\partial}{\partial\theta}(H_\phi\sin\theta) = \frac{Idl\,2\sin\theta\cos\theta}{4\pi}\left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r}$$

$$E_r = \frac{\eta}{j\beta}\cdot\frac{1}{r\sin\theta}\cdot\frac{2Idl\sin\theta\cos\theta}{4\pi}\left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r} = \frac{\eta Idl\cos\theta}{2\pi}\cdot\frac{1}{j\beta}\left[\frac{j\beta}{r^2} + \frac{1}{r^3}\right] e^{-j\beta r}$$

Using $\frac{1}{j\beta}\cdot j\beta = 1$ and $\frac{1}{j\beta} = -\frac{j}{\beta}$:

$$\boxed{E_r = \frac{\eta Idl}{2\pi}\cos\theta\left[\frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}}$$

Step 3: Polar component E_θ.

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$$rH_\phi = \frac{Idl\sin\theta}{4\pi}\left[j\beta + \frac{1}{r}\right]e^{-j\beta r}$$

Differentiate with the product rule ($\frac{d}{dr}e^{-j\beta r} = -j\beta e^{-j\beta r}$):

$$\frac{\partial}{\partial r}(rH_\phi) = \frac{Idl\sin\theta}{4\pi}\left[-\frac{1}{r^2} + \left(j\beta + \frac{1}{r}\right)(-j\beta)\right]e^{-j\beta r} = \frac{Idl\sin\theta}{4\pi}\left[-\frac{1}{r^2} + \beta^2 - \frac{j\beta}{r}\right]e^{-j\beta r}$$

Then

$$E_\theta = \frac{\eta}{j\beta}\left(-\frac{1}{r}\right)\frac{\partial}{\partial r}(rH_\phi) = \frac{\eta Idl\sin\theta}{4\pi}\cdot\frac{1}{j\beta}\left[\frac{1}{r^3} - \frac{\beta^2}{r} + \frac{j\beta}{r^2}\right]e^{-j\beta r}$$

Divide each term by $j\beta$: $\frac{1}{j\beta r^3} = -\frac{j}{\beta r^3}$, $-\frac{\beta^2}{j\beta r} = \frac{j\beta}{r}$ and $\frac{j\beta}{j\beta r^2} = \frac{1}{r^2}$. So

$$\boxed{E_\theta = \frac{\eta Idl}{4\pi}\sin\theta\left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right]e^{-j\beta r}}$$

Result:

$$\mathbf{E} = \frac{\eta Idl}{4\pi}\sin\theta\left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right]e^{-j\beta r}\hat{a}_\theta + \frac{\eta Idl}{2\pi}\cos\theta\left[\frac{1}{r^2} - \frac{j}{\beta r^3}\right]e^{-j\beta r}\hat{a}_r$$

This is the paper's expression, with $\sin\theta$ in the $\hat{\theta}$ term. $E_\phi = 0$.

Interpretation:

- **Near field** ($\beta r \ll 1$): the $1/r^3$ terms dominate and give $E_r \propto 2\cos\theta/r^3$, $E_\theta \propto \sin\theta/r^3$. This is the field of an electric dipole of charges $\pm q$ at the ends of the wire.
- **Far field** ($\beta r \gg 1$): only $E_\theta = j\eta\beta Idl\sin\theta\, e^{-j\beta r}/(4\pi r)$ survives. E_r falls as $1/r^2$ and disappears, so the far field is transverse, like a plane wave.

3. Keyword Explanation

1. Intrinsic impedance η

$\eta = \sqrt{\mu/\epsilon}$, which is $120\pi \approx 377\ \Omega$ in free space (see 3(a), Keyword 1).

2. Source-free region

A region with no charges or currents, where $\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E}$ holds without the $\mathbf{J}$ term.

4. Notes and Remarks

- **Checking the far field:** $E_\theta/H_\phi = \eta\,(j\beta/r)/(j\beta/r) = \eta$. The far field is a local plane wave, with $E \perp H$ and both perpendicular to $\hat{a}_r$.
- **Why the r component has $\cos\theta$:** it comes from $\partial(\sin^2\theta)/\partial\theta = 2\sin\theta\cos\theta$, and one $\sin\theta$ cancels.

4.(c) Find the expression of average power density of Hertzian dipole in far field zone. Marks: 5

1. Theory

1. Power carried by a wave (Core Idea)

An EM wave carries energy. The power flowing through 1 m^2 is given by the **Poynting vector**. For phasors, the time-average value is

$$\mathbf{P}_{avg} = \frac{1}{2}\text{Re}\,(\mathbf{E}\times\mathbf{H}^*)\ \ \text{W/m}^2$$

where * means the complex conjugate. It points in the direction the energy flows.

2. Far-field zone

Far from the antenna ($r \gg \lambda/2\pi$, in practice $r > 2D^2/\lambda$ for an antenna of size D), the $1/r^2$ and $1/r^3$ terms are negligible (see the figure in 4(b)). Only the $1/r$ terms remain. These are the fields that carry power away for good: their power density falls as $1/r^2$, while the area of a sphere grows as r^2, so the total power through any large sphere stays the same.

2. Answer

Far-field components (keep only the $1/r$ terms of 4(a) and 4(b)):

$$H_\phi = \frac{j\beta I dl \sin\theta}{4\pi r} e^{-j\beta r}, \qquad E_\theta = \eta H_\phi = \frac{j\eta\beta I dl \sin\theta}{4\pi r} e^{-j\beta r}, \qquad E_r \approx 0$$

Poynting vector. Since $\hat{a}_\theta \times \hat{a}_\phi = \hat{a}_r$:

$$\mathbf{P}_{avg} = \frac{1}{2}\mathrm{Re}(E_\theta H_\phi^*)\,\hat{a}_r = \frac{1}{2}\eta |H_\phi|^2\,\hat{a}_r$$

$$\boxed{\mathbf{P}_{avg} = \frac{\eta\beta^2 I_0^2 dl^2 \sin^2\theta}{32\pi^2 r^2}\,\hat{a}_r \ \mathrm{W/m^2}}$$

where I_0 is the peak current. In free space, with $\eta = 120\pi$ and $\beta = 2\pi/\lambda$:

$$\boxed{\mathbf{P}_{avg} = \frac{15\pi I_0^2}{r^2}\left(\frac{dl}{\lambda}\right)^2 \sin^2\theta\,\hat{a}_r \ \mathrm{W/m^2}}$$

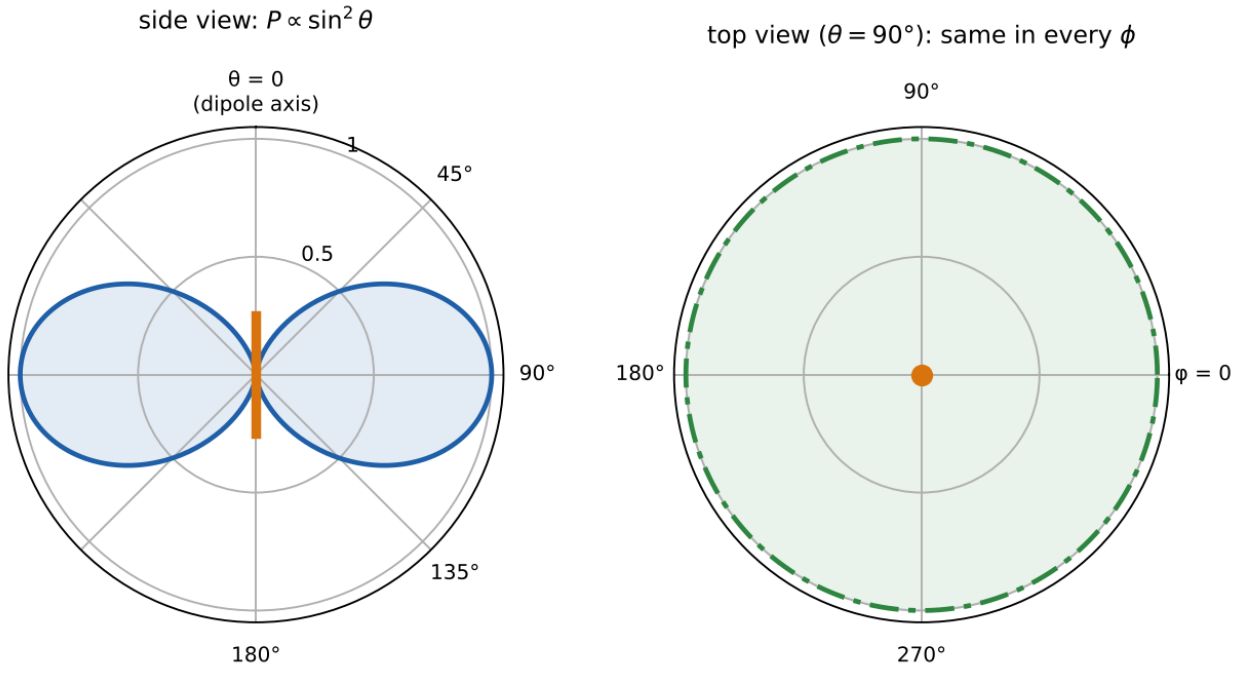


Figure 18: Radiation pattern of a Hertzian dipole

Reading the figure. Side view (left): the power density goes as $\sin^2\theta$. It is largest at right angles to the dipole ($\theta = 90°$) and zero along its axis, giving a doughnut shape. Top view (right): it is the same in every horizontal direction (ϕ).

Total radiated power and radiation resistance (a useful addition). Integrate over a sphere:

$$P_{rad} = \int_0^{2\pi}\int_0^{\pi} P_{avg}\, r^2 \sin\theta\, d\theta\, d\phi = 40\pi^2 \left(\frac{dl}{\lambda}\right)^2 I_0^2$$

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$$R_{rad} = \frac{2P_{rad}}{I_0^2} = 80\pi^2 \left(\frac{dl}{\lambda}\right)^2 \ \Omega$$

Interpretation:

- The power density falls as $1/r^2$ (inverse-square law).
- It grows with $(dl/\lambda)^2$, so a very short dipole radiates poorly. For $dl = \lambda/50$, R_{rad} is only 0.32 Ω.

3. Keyword Explanation

1. Poynting vector

$\mathbf{E} \times \mathbf{H}$: the power per unit area carried by the fields, and its direction.

2. Radiation pattern

A plot of how strongly an antenna radiates in each direction.

3. Radiation resistance

The resistance that would absorb the same power as the antenna radiates when carrying the same current.

4. Notes and Remarks

- **rms current:** if I is an rms value, drop the factor $\frac{1}{2}$ in $\mathbf{P}_{avg}$. The result is then $\eta\beta^2 I_{rms}^2 dl^2 \sin^2\theta/(16\pi^2 r^2)$.
- **Where $\int_0^\pi \sin^3\theta\, d\theta = 4/3$ comes in:** this integral is what gives the $40\pi^2$ in P_{rad}.

5. Final Concise Answer

(a) $\hat{z} = \cos\theta\,\hat{a}_r - \sin\theta\,\hat{a}_\theta$, so $A_r = A_z\cos\theta$ and $A_\theta = -A_z\sin\theta$. Only $(\nabla\times\mathbf{A})_\phi = \frac{1}{r}\left[\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial\theta}\right]$ survives, giving $\mathbf{H} = \frac{Idl}{4\pi}\sin\theta\left[\frac{j\beta}{r} + \frac{1}{r^2}\right]e^{-j\beta r}\hat{a}_\phi$ (the printed $\cos\theta$ should be $\sin\theta$).

(b) $\mathbf{E} = \frac{\eta}{j\beta}\nabla\times\mathbf{H}$ gives $E_r = \frac{\eta Idl}{2\pi}\cos\theta\left[\frac{1}{r^2} - \frac{j}{\beta r^3}\right]e^{-j\beta r}$ and $E_\theta = \frac{\eta Idl}{4\pi}\sin\theta\left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right]e^{-j\beta r}$.

(c) Far field: $E_\theta = \eta H_\phi = \frac{j\eta\beta Idl\sin\theta}{4\pi r}e^{-j\beta r}$, so $\mathbf{P}_{avg} = \frac{\eta\beta^2 I_0^2 dl^2\sin^2\theta}{32\pi^2 r^2}\hat{a}_r = \frac{15\pi I_0^2}{r^2}\left(\frac{dl}{\lambda}\right)^2\sin^2\theta\,\hat{a}_r$ W/m^2. Total $P_{rad} = 40\pi^2(dl/\lambda)^2 I_0^2$.

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5. (a) An RF signal at $f = 100$ MHz is mixed with local oscillator signal at $f_{LO} = 90$ MHz.
   (i) Calculate the IF (intermediate frequency) output.
   (ii) Why are active filters used after mixing in a super-heterodyne receiver? Marks: 6
   (b) For a 1 kHz message signal modulating a 1 MHz carrier with 80% modulation index in AM, calculate the total transmitted power if carrier power is 100 W. Marks: 5
   (c) For an antenna with gain 10 dB transmitting 10 W power at 2.4 GHz, calculate the effective isotropic radiated power (EIRP) of the antenna. Marks: 5
   (d) Write short notes (any three).
   (i) Klystron
   (ii) Magnetron
   (iii) Gunn Diode Oscillator
   (iv) Travelling Wave Tube (TWT) [Marks: 3+3+3]

5.(a) An RF signal at $f = 100$ MHz is mixed with local oscillator signal at $f_{LO} = 90$ MHz. (i) Calculate the IF (intermediate frequency) output. (ii) Why are active filters used after mixing in a super-heterodyne receiver? Marks: 6

1. Theory

1. The problem a radio receiver must solve (Core Idea)

The air is full of signals at many frequencies. A receiver must pick one station, amplify it a great deal, and reject its neighbours. Building a very sharp, high-gain filter that can also be **tuned** across a whole band is hard. The **superheterodyne** idea (Armstrong, 1918) avoids it: shift every wanted station down to **one fixed frequency**, the **intermediate frequency (IF)**, and do all the sharp filtering and most of the amplification there, with circuits that never need retuning.

2. How a mixer shifts frequency

A **mixer** multiplies the incoming RF signal by a **local oscillator (LO)** signal. The product of two cosines is

$$\cos\omega_{RF}t \cdot \cos\omega_{LO}t = \frac{1}{2}\cos(\omega_{RF} - \omega_{LO})t + \frac{1}{2}\cos(\omega_{RF} + \omega_{LO})t$$

so the output contains the **difference** $|f_{RF} - f_{LO}|$ and the **sum** $f_{RF} + f_{LO}$. Real mixers also leak the original RF and LO, plus their harmonics. The **difference** is normally chosen as the IF. Turning the tuning knob moves f_{LO}, so that each station in turn lands on the same IF.

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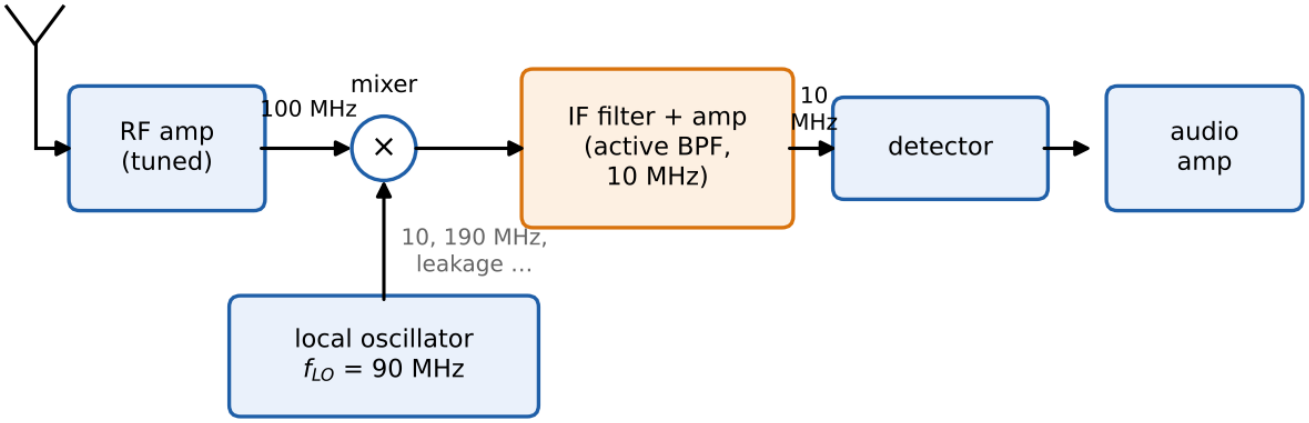


Figure 19: Superheterodyne receiver

Reading the figure. The antenna signal is amplified (RF amp) and multiplied by the LO in the mixer. The mixer output holds many frequencies; the IF filter and amplifier keep only the IF. The detector then recovers the message, and the audio amplifier drives the speaker.

3. The image frequency

A second RF frequency also gives the same difference: $f_{image} = f_{LO} - f_{IF}$ (when the LO is below the RF), which is $2f_{IF}$ away from the wanted signal. It must be removed by the tuned RF amplifier *before* the mixer, because after mixing it cannot be told apart from the wanted signal.

2. Answer

(i) IF output.

$$f_{IF} = f_{RF} - f_{LO} = 100 - 90 = 10 \text{ MHz}$$

$$\boxed{f_{IF} = 10 \text{ MHz}}$$

The mixer also produces the sum, $100 + 90 = 190$ MHz, which is filtered out. (The image frequency for this setting is $f_{LO} - f_{IF} = 80$ MHz.)

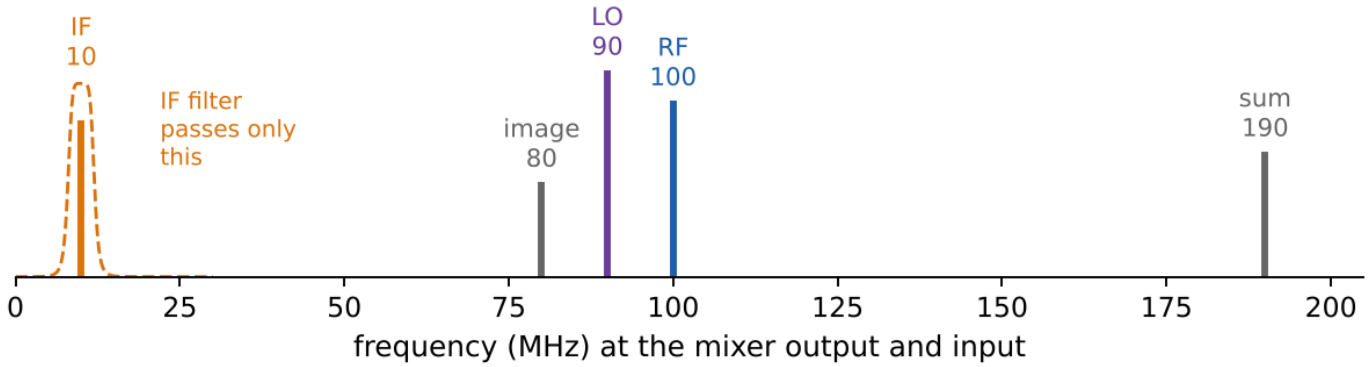


Figure 20: Frequencies around the mixer

(ii) Why active filters are used after the mixer. The mixer output contains the wanted 10 MHz IF together with the 190 MHz sum, LO and RF leakage, harmonics, and the IF products of nearby stations. The filter after the mixer must keep only the IF band. **Active filters** (amplifiers with frequency-selective feedback) are used because they:

1. **Select the IF band sharply** (good selectivity), rejecting the sum frequency, leakage and adja-

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cent channels.

2. **Provide gain at the same time.** The signal after the mixer is still very weak, and most of the receiver's gain is placed in the IF stage.
3. **Need no inductors.** Op-amp and RC (or integrated gm-C) filters can be built on a chip; bulky, lossy coils are avoided.
4. **Buffer the stages.** High input and low output impedance, so the mixer and the detector do not load each other.
5. **Are easy to set precisely.** The centre frequency and bandwidth are fixed by resistor and capacitor values, and the IF never changes.

3. Keyword Explanation

1. Intermediate frequency (IF)

The fixed frequency to which every received station is shifted before detection. Common values are 455 kHz (AM radio) and 10.7 MHz (FM radio).

2. Local oscillator

The receiver's own oscillator, tuned together with the RF stage so that $|f_{RF} - f_{LO}| = f_{IF}$ for the wanted station.

3. Selectivity

The ability to accept the wanted signal and reject signals at nearby frequencies.

4. Notes and Remarks

- **Low-side vs high-side LO:** here $f_{LO} < f_{RF}$ (low-side injection). With high-side injection, $f_{LO} = f_{RF} + f_{IF} = 110$ MHz, and the image would be at 120 MHz.
- **Practical note:** at 10.7 MHz, ceramic and crystal filters are also widely used, often together with IF amplifier chips. The question asks about the active-filter approach.

5.(b) For a 1 kHz message signal modulating a 1 MHz carrier with 80% modulation index in AM, calculate the total transmitted power if carrier power is 100 W. Marks: 5

1. Theory

1. What is AM? (Core Idea)

In **amplitude modulation (AM)** the height (amplitude) of a high-frequency carrier is made to follow the message. With a single-tone message $\mu \cos \omega_m t$:

$$s(t) = A_c[1 + \mu \cos \omega_m t] \cos \omega_c t$$

μ is the **modulation index**. It must be at most 1, or the envelope can no longer follow the message.

2. Where the power goes

Expanding the product with $\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$:

$$s(t) = A_c \cos \omega_c t + \frac{\mu A_c}{2} \cos(\omega_c - \omega_m)t + \frac{\mu A_c}{2} \cos(\omega_c + \omega_m)t$$

That is a carrier plus a **lower sideband (LSB)** and an **upper sideband (USB)**. Power is proportional to amplitude squared, so each sideband has $(\mu/2)^2$ of the carrier power:

$$P_t = P_c + 2 \cdot \frac{\mu^2}{4} P_c = P_c \left(1 + \frac{\mu^2}{2}\right)$$

2. Answer

Given: $P_c = 100$ W, $\mu = 0.8$, $f_m = 1$ kHz, $f_c = 1$ MHz.

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$$P_t = P_c\left(1+\frac{\mu^2}{2}\right) = 100\left(1+\frac{0.8^2}{2}\right) = 100(1+0.32) = 132 \text{ W}$$

$$\boxed{P_t = 132 \text{ W}}$$

Breakdown:

- The two sidebands carry $132 - 100 = 32$ W, which is 16 W each, at 999 kHz and 1001 kHz.
- Only the sidebands carry the message, so the efficiency is $32/132 = 24.2$ %.

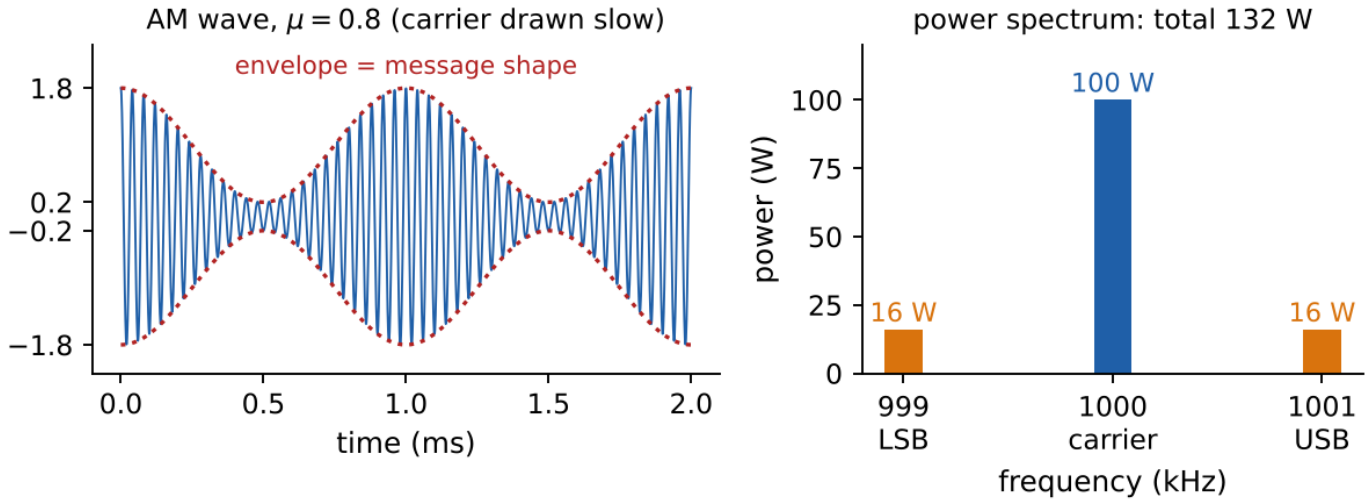


Figure 21: The AM wave and its power spectrum

3. Keyword Explanation

1. Modulation index μ

The ratio of the message amplitude to the carrier amplitude. 80 % means $\mu = 0.8$.

2. Sideband

The new frequency components created by modulation, at $f_c \pm f_m$.

4. Notes and Remarks

- **Frequencies not needed:** P_t depends only on P_c and μ. The 1 kHz and 1 MHz tell us where the sidebands sit (999 and 1001 kHz) and that the bandwidth is $2f_m = 2$ kHz.
- **At $\mu = 1$:** $P_t = 1.5P_c$, so at best only one third of the power carries information.

5.(c) For an antenna with gain 10 dB transmitting 10 W power at 2.4 GHz, calculate the effective isotropic radiated power (EIRP) of the antenna. Marks: 5

1. Theory

1. Isotropic antenna and gain (Core Idea)

An **isotropic antenna** is an imaginary antenna that radiates equally in all directions. A real antenna concentrates its power in some directions. Its **gain** G says how many times stronger the radiation is in its best direction than an isotropic antenna fed with the same power would give. The gain does not create power; it only redistributes it, like a torch reflector focusing the light of the same bulb.

2. dB to ratio

$$G_{dB} = 10\log_{10} G \quad \Longleftrightarrow \quad G = 10^{G_{dB}/10}$$

so 10 dB is a factor of 10.

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3. What EIRP means

Effective isotropic radiated power is the power an isotropic antenna would need to give the same signal strength as our antenna gives in its best direction:

$$\text{EIRP} = P_t G_t$$

2. Answer

Given: $G = 10$ dB, $P_t = 10$ W, $f = 2.4$ GHz.

$$G = 10^{10/10} = 10$$

$$\text{EIRP} = P_t G = 10 \times 10 = 100 \text{ W}$$

In dB units:

$$\text{EIRP} = 10\log_{10}(10) + 10 = 10 + 10 = 20 \text{ dBW} = 50 \text{ dBm}$$

$$\boxed{\text{EIRP} = 100 \text{ W} = 20 \text{ dBW} = 50 \text{ dBm}}$$

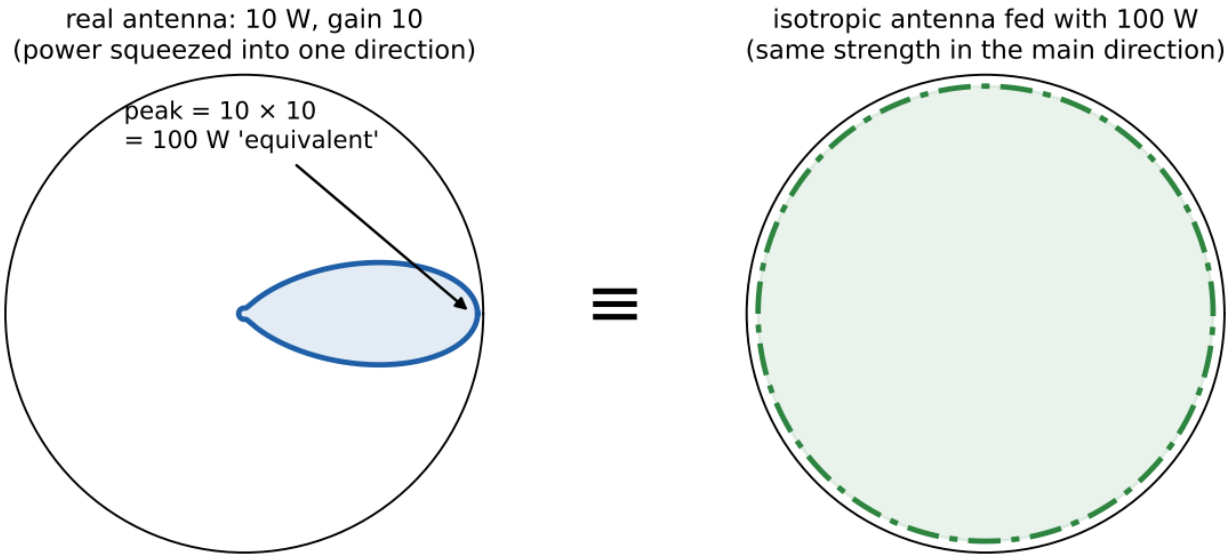


Figure 22: EIRP: a directive antenna and its isotropic equivalent

Reading the figure. Left: the real antenna radiates 10 W, but squeezes it into a narrow beam ten times stronger than isotropic. Right: an isotropic antenna would need 100 W to be equally strong in that direction.

3. Keyword Explanation

1. dBW and dBm

Power in decibels relative to 1 W (dBW) or 1 mW (dBm). 1 W = 0 dBW = 30 dBm.

4. Notes and Remarks

- **Frequency not needed:** EIRP does not depend on frequency. The 2.4 GHz matters for path loss and antenna size: at 2.4 GHz, $\lambda = 12.5$ cm, and the antenna's effective area is $A_e = G\lambda^2/4\pi = 124$ cm^2 (see 6(d)).
- **Use:** regulators limit EIRP. For example, Wi-Fi at 2.4 GHz is commonly limited to about 36 dBm EIRP.

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5.(d) Write short notes (any three). (i) Klystron (ii) Magnetron (iii) Gunn Diode Oscillator (iv) Travelling Wave Tube (TWT) [Marks: 3+3+3]

1. Theory

1. Why ordinary transistors and tubes fail at microwaves (Core Idea)

At microwave frequencies the period of one cycle is under a nanosecond. In an ordinary vacuum tube or transistor, the electrons take longer than that to cross from one electrode to the other (**transit time**), and lead inductances and stray capacitances short out the signal. Microwave tubes turn this problem into the solution. They let a fast **electron beam** travel a long way and hand its DC energy to an RF field *while it travels*.

2. The two key ideas

- **Velocity modulation and bunching** (klystron): speed some electrons up and slow others down. After a drift distance the fast ones catch the slow ones, and the beam forms **bunches** that can drive a cavity in step.
- **Slow-wave interaction** (TWT, magnetron): make the RF wave travel at about the same speed as the electrons, so they interact continuously and the electrons slowly give up energy to the wave.

3. Solid-state alternative

A **Gunn diode** produces microwaves with no beam at all. It uses a property of GaAs: above a threshold field, electrons get *slower* as the field increases (**negative differential resistance**).

4. Resonant cavity

A closed metal box that rings at a microwave frequency, like a bell. It is the microwave version of an LC tank circuit (see 6(c)).

2. Answer

(i) Klystron

A **klystron** is a linear-beam microwave tube used as an **amplifier** (two-cavity or multi-cavity) or as an **oscillator** (reflex klystron).

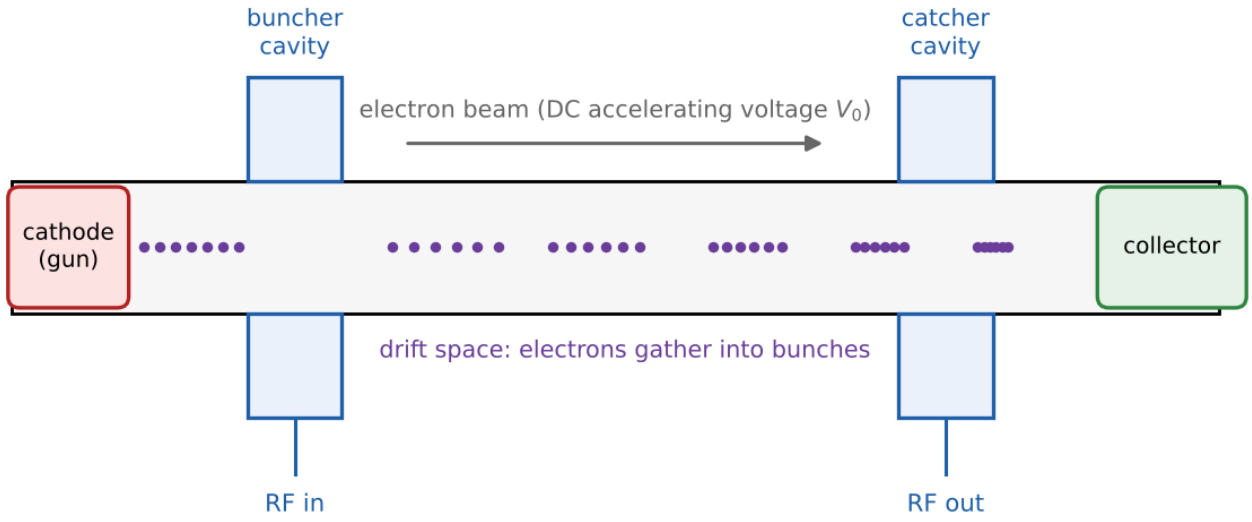


Figure 23: Two-cavity klystron

Construction: an electron gun (cathode) sends a beam through two resonant cavities, the **buncher** and the **catcher**, separated by a field-free **drift space**, to a **collector**.

Working:

1. **Velocity modulation:** the weak RF input sets up an alternating voltage across the buncher gap. Electrons passing during the positive half-cycle are sped up, and those during the negative half-cycle are slowed down.

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2. **Bunching:** in the drift space the fast electrons catch up with the slow ones that left earlier, so the beam gathers into bunches, once per RF cycle (Applegate diagram below).
3. **Energy transfer:** the bunches pass the catcher gap at the moment its RF field opposes them. They are slowed and give energy to the cavity, which then holds a much larger RF signal: the amplified output.

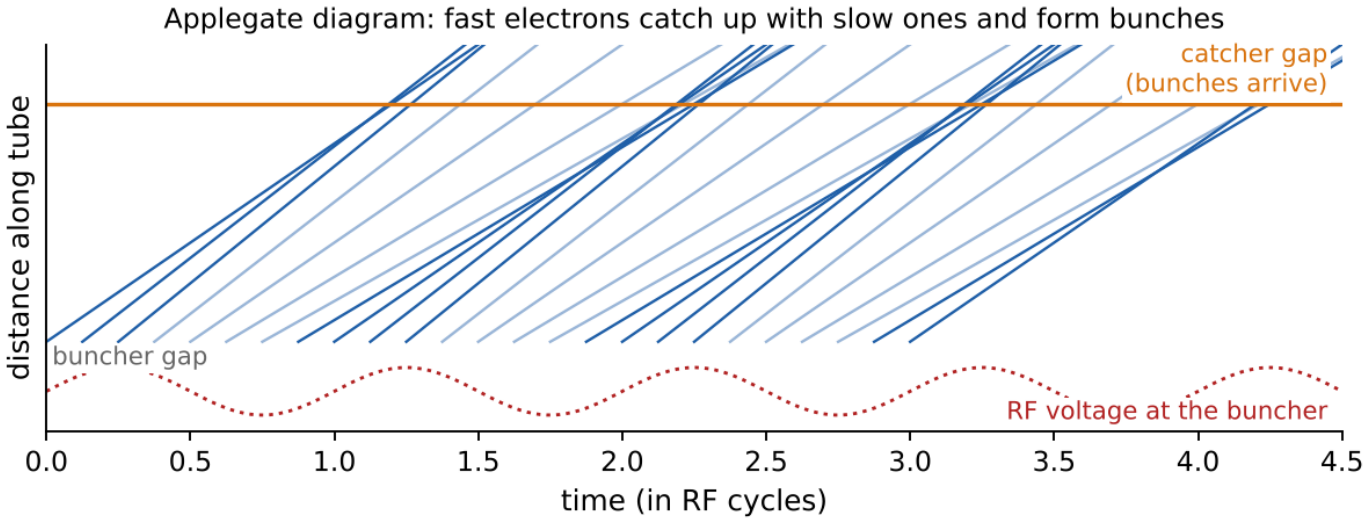


Figure 24: Applegate diagram

Typical performance: gain of 30 to 70 dB, output from watts up to megawatts (pulsed), 0.5 to 100 GHz, but narrow bandwidth (because of the cavities).

Uses: UHF TV transmitters, radar transmitters, satellite uplinks, particle accelerators. The reflex klystron (a single cavity, with a repeller electrode sending the beam back) was a common low-power local oscillator.

(ii) Magnetron

A **magnetron** is a crossed-field microwave **oscillator**: the electric field is radial, and a magnetic field is applied along the axis, at right angles to it.

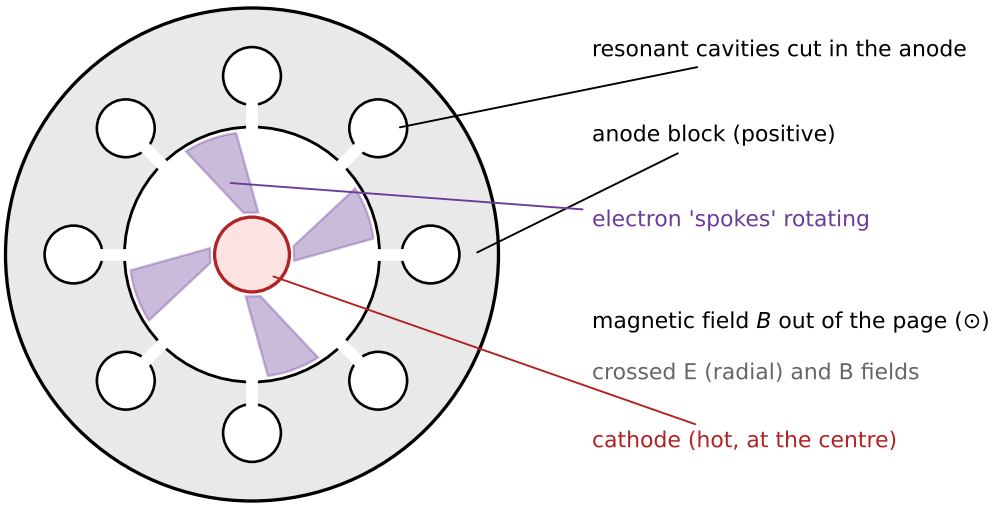


Figure 25: Cavity magnetron cross-section

Construction: a cylindrical cathode at the centre, surrounded by a copper anode block with several resonant cavities cut into it (often 8). Strong magnets apply B along the axis.

Working:

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1. Electrons leave the hot cathode toward the positive anode, but the magnetic field bends their paths into curves that circle the cathode.
2. Noise excites the cavities, and the RF fields at the cavity slots act on the circling electrons. Electrons in step with the RF field give it energy and are pulled toward the anode; electrons out of step take energy and are sent back to the cathode. This sorting forms rotating **spokes** of electron charge.
3. The spokes rotate in step with the RF field (in the **π-mode**, adjacent cavities are 180° apart) and keep feeding it. Power is taken out through a coupling loop in one cavity.

Typical performance: high efficiency (60 to 80 %), high power (kW continuous, MW pulsed), 1 to 40 GHz.
Uses: microwave ovens (2.45 GHz, about 1 kW), pulsed radar transmitters, industrial heating.

(iii) Gunn Diode Oscillator

A **Gunn diode** is a two-terminal solid-state device made of n-type GaAs (or InP), with no p–n junction. It works by the **transferred-electron effect** (Gunn effect, 1963).

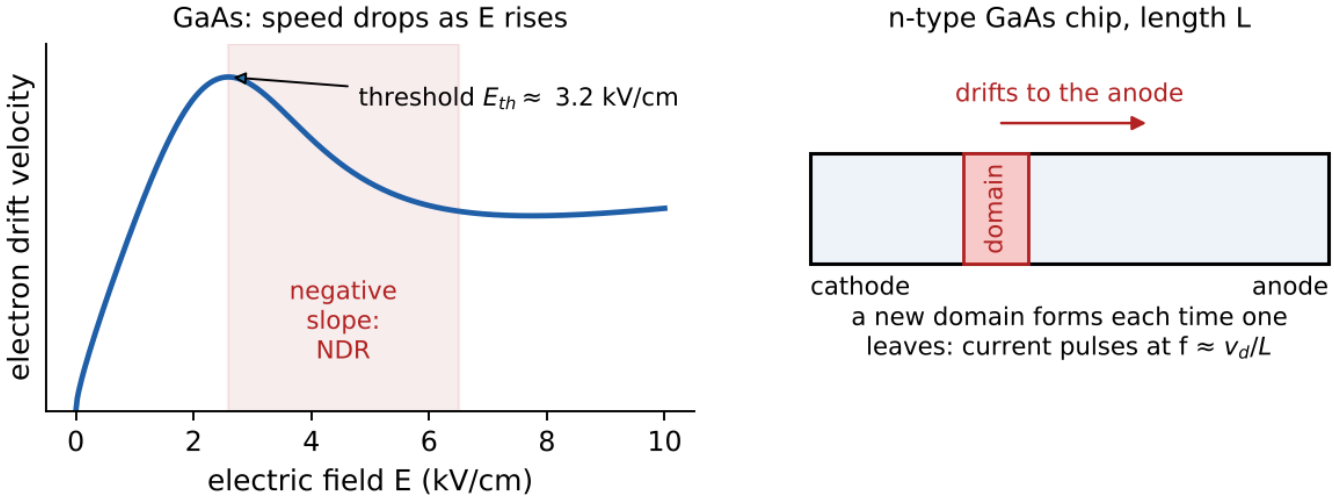


Figure 26: Gunn effect

Principle: the conduction band of GaAs has a low valley, where electrons are light and fast, and an upper valley 0.36 eV higher, where they are heavy and slow. Above a **threshold field** of about 3.2 kV/cm, electrons gain enough energy to jump into the upper valley and slow down. So the drift velocity *falls* as the field rises: **negative differential resistance (NDR)** (left of the figure).
Working as an oscillator:

1. With a DC bias above threshold, a small region of high field forms near the cathode: a **domain**, where electrons are slow and pile up.
2. The domain drifts to the anode at about 10^5 m/s. While it travels, the current stays low.
3. When it reaches the anode, the current jumps up, and a new domain forms at the cathode.

The current is therefore a series of pulses at $f \approx v_d/L$, where L is the device length (10 µm gives about 10 GHz). Mounted in a resonant cavity, the diode gives a stable sine-wave output, and the frequency can be tuned by the cavity.
Typical performance: 1 to 100 GHz, mW to about 1 W, low noise, low voltage, small and cheap.
Uses: police radar guns, door-opening sensors, local oscillators, microwave test sources.

(iv) Travelling Wave Tube (TWT)

A **TWT** is a linear-beam **broadband amplifier**.

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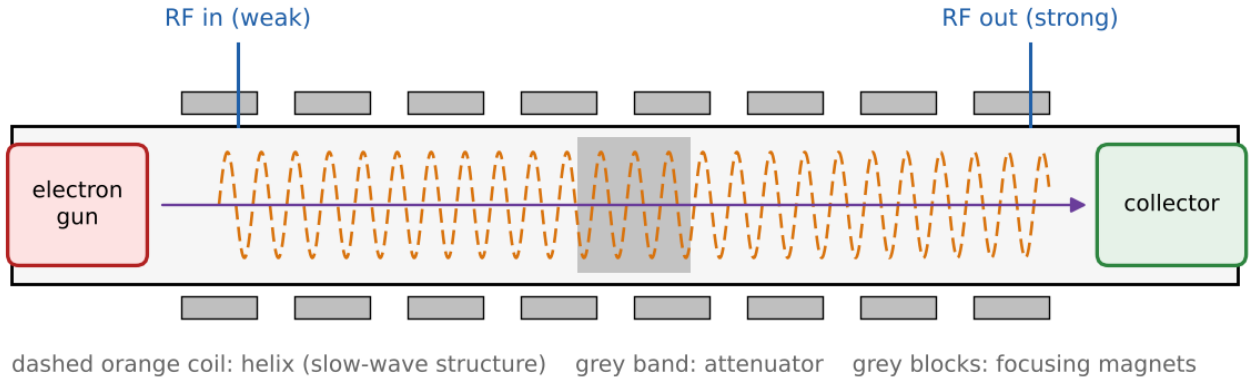


Figure 27: Helix travelling wave tube

Construction: an electron gun, a long **helix** (a coiled wire, the slow-wave structure) with the electron beam running along its axis, a collector, focusing magnets along the tube, and an attenuator near the middle of the helix.
Working:

1. The RF signal enters at the gun end and travels along the helix wire at nearly the speed of light. Because it must go round and round the coil, its speed **along the axis** is only about $c \times$ (pitch/circumference), roughly $0.1c$.
2. The beam is accelerated to a speed slightly **above** this axial wave speed. The two travel together, so the RF field continuously bunches the electrons, and the bunches give energy to the wave all along the tube.
3. The wave grows exponentially and leaves at the far end, amplified. The **attenuator** absorbs any wave reflected backward, which prevents oscillation.

Typical performance: very wide bandwidth (an octave or more, because there are no resonant cavities), gain of 30 to 60 dB, low noise.
Uses: satellite transponders and ground stations, radar, electronic warfare, broadband microwave links.

Device	Type	Main feature	Typical use
Klystron	linear-beam amplifier or oscillator	high gain and power, narrow band	radar, TV transmitters
Magnetron	crossed-field oscillator	high efficiency and power	microwave ovens, radar
Gunn diode	solid-state oscillator	small, low voltage	radar guns, sensors
TWT	linear-beam amplifier	very wide band	satellites, EW

Table 7: The four microwave sources compared

3. Keyword Explanation

1. Velocity modulation

Changing the speed of the electrons in a beam with an RF voltage, instead of changing their number.

2. Slow-wave structure

A structure (helix, coupled cavities) that makes an RF wave travel along an axis much slower than light, so electrons can keep pace with it.

3. Negative differential resistance

A region where increasing the voltage *reduces* the current. It can cancel circuit losses and sustain oscillation.

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4. Notes and Remarks

- **Any three:** all four notes are written so that you can choose. Each is about one page, sized for 3 marks; in the exam, write the construction, the working in three or four steps, and the uses, with the sketch.
- **Common confusion:** the klystron and the TWT both use a linear beam. The klystron interacts at a few cavity gaps (narrow band); the TWT interacts along its whole length (broadband).

5. Final Concise Answer

(a) (i) $f_{IF} = f_{RF} - f_{LO} = 100 - 90 = 10$ MHz (the sum, 190 MHz, is rejected; the image is at 80 MHz). (ii) Active filters after the mixer pick out the IF band and reject the sum frequency, leakage and adjacent channels, while also giving gain. They need no inductors, isolate the stages and are easy to set precisely.

(b) $P_t = P_c(1 + \mu^2/2) = 100(1 + 0.32) = 132$ W (16 W in each sideband).

(c) $G = 10^{10/10} = 10$, so EIRP $= P_t G = 100$ W $= 20$ dBW $= 50$ dBm.

(d) Klystron: velocity modulation in the buncher cavity, bunching in the drift space, energy given to the catcher cavity; a high-gain, narrowband amplifier. Magnetron: crossed E and B fields and rotating electron spokes driving anode cavities; an efficient high-power oscillator (microwave ovens, radar). Gunn diode: GaAs with negative differential mobility above 3.2 kV/cm; domains transit the device, giving $f \approx v_d/L$; a small solid-state oscillator. TWT: a beam travelling with a wave slowed by a helix, giving continuous interaction; a broadband amplifier (satellites).

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6. (a) What are the uses of a microstrip line? Mention its advantages. Describe the type of losses in a microstrip line. Marks: 5
   (b) Derive Jefeminko's equation and from there, explain the necessity of accelerating charge for electromagnetic radiation. Marks: 13
   (c) A rectangular cavity resonator has dimensions of $a = 5$ cm, $b = 2$ cm, and $d = 15$ cm. Compute the resonant frequency of the dominant mode for a dielectric-field cavity of $\epsilon_r = 2.56$. Marks: 3
   (d) What do you mean by effective height and effective area of an antenna? Marks: 4

6.(a) What are the uses of a microstrip line? Mention its advantages. Describe the type of losses in a microstrip line. Marks: 5

1. Theory

1. What is a microstrip line? (Core Idea)

A **microstrip** is a transmission line printed on a circuit board: a narrow copper strip of width W on top of a dielectric substrate (thickness h, permittivity ϵ_r), with a solid copper **ground plane** underneath. It is the two-conductor line of 1(a) in flat, printable form.

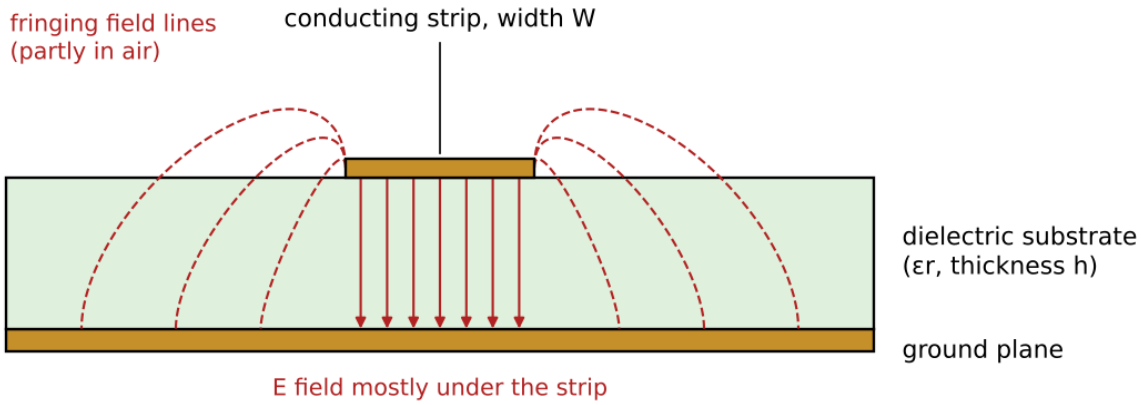


Figure 28: Microstrip cross-section and field

Reading the figure. Most of the electric field (the red arrows) runs straight down from the strip to the ground through the substrate. Near the edges some lines **fringe** out through the air above. Because the field is partly in the dielectric and partly in air, the wave travels at a speed set by an **effective permittivity**, between 1 and ϵ_r.

2. Characteristic impedance

Z_0 depends on W/h and ϵ_r: a wider strip gives a lower Z_0. Typical values are 20 to 150 Ω, and 50 Ω is the usual design value.

2. Answer

Uses of microstrip lines:

- Interconnections between components in microwave and RF circuits (mobile phones, Wi-Fi, GPS, radar boards).
- Printed passive circuits: filters, couplers, power dividers, impedance-matching networks (stubs, quarter-wave transformers, see 2(b) and 2(c)).
- Patch antennas and antenna arrays.

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- Microwave integrated circuits (MICs and MMICs), and the biasing and matching of transistor amplifiers and oscillators.

Advantages:

1. Small, light and cheap, because it is made by ordinary PCB etching.
2. Planar and open, so components (transistors, chips) are easily soldered onto the strip.
3. Easy to integrate with other circuits on the same board, and well suited to mass production.
4. Z_0 is set simply by the strip width, so lines of different impedance are easy to make.
5. Wide bandwidth (a quasi-TEM line with no cutoff frequency).

Types of losses:

1. **Conductor loss:** I^2R heating in the strip and the ground plane. At microwave frequencies the current flows in a thin skin layer, so the resistance rises as $\sqrt{f}$. It is made worse by surface roughness.
2. **Dielectric loss:** the substrate is not a perfect insulator, so its molecules absorb energy from the alternating field. It is set by the loss tangent $\tan\delta$ and grows with f.
3. **Radiation loss:** because the line is open, discontinuities (bends, steps, open ends) and the fringing fields radiate some power into space. It is worse for thick, low-ϵ_r substrates and at high frequency.
4. **Surface-wave loss:** at high frequency some energy is trapped in waves travelling inside the substrate, away from the line.

3. Keyword Explanation

1. Quasi-TEM

Almost a TEM wave (E and H both transverse). It is not exactly TEM because the field sits in two media (air and dielectric).

2. Skin effect

At high frequency, current crowds into a thin layer at the conductor surface, with skin depth $\delta_s = 1/\sqrt{\pi f \mu \sigma}$.

3. Loss tangent

$\tan\delta = \sigma/(\omega\epsilon)$ for a dielectric: the ratio of the lossy (conduction) current to the lossless (displacement) current.

4. Notes and Remarks

- **Disadvantages (for contrast):** higher loss than waveguides, lower power handling, radiation, and dispersion (ϵ_{eff} changes with frequency).

6.(b) Derive Jefeminko's equation and from there, explain the necessity of accelerating charge for electromagnetic radiation. Marks: 13

1. Theory

1. The question behind the question (Core Idea)

A charge at rest has an electric field (Coulomb's law). A steady current has a magnetic field (Biot–Savart). Neither of these sends energy away: the fields just sit there. Yet antennas *do* send energy away. So what exactly must the charges do to radiate? **Jefimenko's equations** answer this. They give **E** and **B** at any point and time directly from the charges and currents, and show clearly which terms radiate.

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2. Signals take time to travel (retarded time)

Nothing travels faster than light. If a charge at point $\mathbf{r}'$ moves now, a point $\mathbf{r}$ a distance $R = |\mathbf{r} - \mathbf{r}'|$ away does not feel it until a time R/c later. So the fields at $(\mathbf{r}, t)$ depend on what the source was doing at the earlier **retarded time**

$$t_r = t - \frac{R}{c}$$

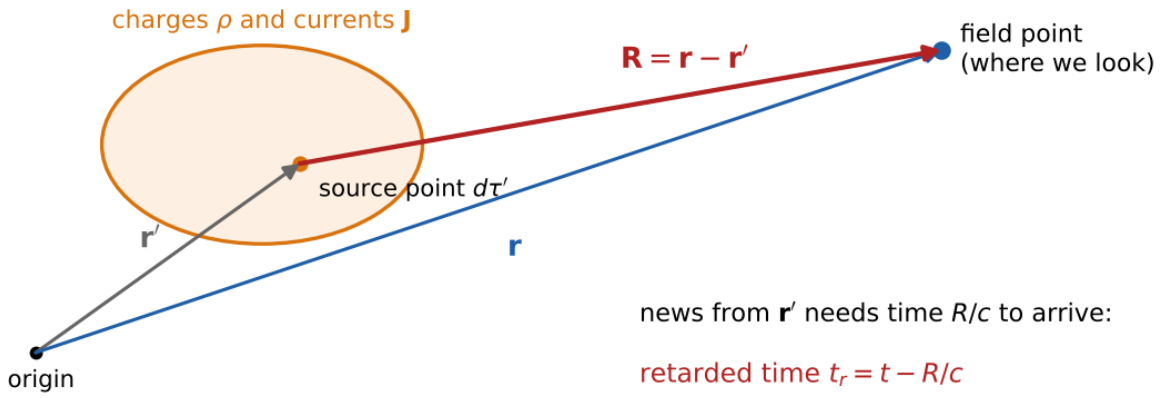


Figure 29: Source point, field point and retarded time

Reading the figure. $\mathbf{r}'$ locates a small piece $d\tau'$ of the source (shaded orange region) and $\mathbf{r}$ the point where we want the field. The separation vector is $\mathbf{R} = \mathbf{r} - \mathbf{r}'$, with length R and unit vector $\hat{\mathbf{R}}$.

3. Retarded potentials (the starting point)

In electrostatics, $V = \frac{1}{4\pi\epsilon_0}\int \frac{\rho}{R} d\tau'$ and $\mathbf{A} = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}}{R} d\tau'$. For time-varying sources the same formulas hold, provided each source is taken at its retarded time:

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\mathbf{r}', t_r)}{R}\, d\tau', \qquad \mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}(\mathbf{r}', t_r)}{R}\, d\tau'$$

These solve Maxwell's equations (in the Lorenz gauge). The fields follow from

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}, \qquad \mathbf{B} = \nabla \times \mathbf{A}$$

4. Two derivative facts we need

The source quantities depend on $\mathbf{r}$ (the field point) only through t_r, and $\nabla R = \hat{\mathbf{R}}$. So

$$\nabla t_r = -\frac{1}{c}\nabla R = -\frac{1}{c}\hat{\mathbf{R}}, \qquad \nabla \rho(\mathbf{r}', t_r) = \dot{\rho}\, \nabla t_r = -\frac{\dot{\rho}}{c}\hat{\mathbf{R}}$$

where a dot means $\partial/\partial t$ taken at t_r. Also $\nabla\left(\frac{1}{R}\right) = -\frac{\hat{\mathbf{R}}}{R^2}$.

2. Answer

Step 1: Gradient of the retarded scalar potential

$$\nabla V = \frac{1}{4\pi\epsilon_0}\int \left[\frac{\nabla\rho}{R} + \rho\,\nabla\frac{1}{R}\right] d\tau' = \frac{1}{4\pi\epsilon_0}\int \left[-\frac{\dot{\rho}}{c}\frac{\hat{\mathbf{R}}}{R} - \rho\frac{\hat{\mathbf{R}}}{R^2}\right] d\tau'$$

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Step 2: Time derivative of the retarded vector potential

$$\frac{\partial \mathbf{A}}{\partial t} = \frac{\mu_0}{4\pi}\int \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{R}\, d\tau'$$

Step 3: The electric field

$\mathbf{E} = -\nabla V - \partial\mathbf{A}/\partial t$. Using $\mu_0 = 1/(\epsilon_0 c^2)$:

$$\boxed{\mathbf{E}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0}\int \left[\frac{\rho(\mathbf{r}', t_r)}{R^2}\hat{\mathbf{R}} + \frac{\dot{\rho}(\mathbf{r}', t_r)}{c\,R}\hat{\mathbf{R}} - \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{c^2 R}\right] d\tau'}$$

Step 4: Curl of the retarded vector potential

Use the product rule $\nabla \times (f\mathbf{J}) = f\,\nabla \times \mathbf{J} + \nabla f \times \mathbf{J}$ with $f = 1/R$:

$$\mathbf{B} = \frac{\mu_0}{4\pi}\int \left[\frac{1}{R}\nabla \times \mathbf{J} + \nabla\frac{1}{R} \times \mathbf{J}\right] d\tau'$$

Since $\mathbf{J}$ depends on $\mathbf{r}$ only through t_r:

$$\nabla \times \mathbf{J} = \nabla t_r \times \dot{\mathbf{J}} = -\frac{1}{c}\hat{\mathbf{R}} \times \dot{\mathbf{J}} = \frac{1}{c}\dot{\mathbf{J}} \times \hat{\mathbf{R}}$$

and $\nabla\frac{1}{R} \times \mathbf{J} = -\frac{\hat{\mathbf{R}}}{R^2} \times \mathbf{J} = \frac{\mathbf{J}\times\hat{\mathbf{R}}}{R^2}$. So

$$\boxed{\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi}\int \left[\frac{\mathbf{J}(\mathbf{r}', t_r)}{R^2} + \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{c\,R}\right] \times \hat{\mathbf{R}}\, d\tau'}$$

The two boxed results are **Jefimenko's equations**.

Step 5: What the terms mean

Term	Falls as	Present when	Name
$\rho\hat{\mathbf{R}}/R^2$	$1/R^2$	any charge	Coulomb field (retarded)
$\mathbf{J} \times \hat{\mathbf{R}}/R^2$	$1/R^2$	any current	Biot–Savart field (retarded)
$\dot{\rho}$, $\dot{\mathbf{J}}$ terms	$1/R$	**changing** charge or current	radiation fields

Table 8: The terms of Jefimenko's equations

Static limit (check): if $\dot{\rho} = 0$ and $\dot{\mathbf{J}} = 0$, the equations reduce exactly to Coulomb's law and the Biot–Savart law. These fields fall as $1/R^2$.

Step 6: Why radiation needs accelerating charges

1. **Radiation means power escaping to infinity.** The power through a large sphere of radius r_0 is (Poynting vector, $\propto EB$) $\times$ (area, $\propto r_0^2$). If E and B fall as $1/r_0^2$, then $EB \propto 1/r_0^4$ and the power $\propto 1/r_0^2 \to 0$: nothing escapes. Only fields falling as $1/r_0$ give $EB \times r_0^2 =$ constant, a finite power that reaches infinity. So **only the $1/R$ terms radiate**.
2. **The $1/R$ terms contain only $\dot{\rho}$ and $\dot{\mathbf{J}}$.** Static charges (ρ constant) and steady currents ($\mathbf{J}$ constant) give no $1/R$ term and **do not radiate**.
3. **A changing current means accelerating charges.** In a conductor $\mathbf{J} = nq\mathbf{v}$, so $\dot{\mathbf{J}} = nq\,\dot{\mathbf{v}} = nq\,\mathbf{a}$. The radiation term $\dot{\mathbf{J}}/(c^2R)$ is driven directly by the **acceleration a** of the charges.
4. **Uniform motion does not radiate.** For a single point charge in uniform motion ($\mathbf{a} = 0$), the $\dot{\rho}$ and $\dot{\mathbf{J}}$ contributions (which arise only because the charge moves from place to place) cancel in the $1/R$ part. What remains is a field falling as $1/R^2$ (the Liénard–Wiechert velocity field). This must be so: in a frame moving with the charge it is at rest, and a static charge cannot radiate.

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5. **The radiated power grows with acceleration.** Carrying the analysis through for a point charge gives the Larmor formula, $P = \dfrac{q^2 a^2}{6\pi\epsilon_0 c^3}$: zero when $a = 0$.

Conclusion: electromagnetic radiation requires **time-varying currents**, that is, **accelerating (or decelerating) charges**. An antenna driven by an alternating current works precisely because the electrons in it accelerate back and forth every cycle. The Hertzian dipole fields of Q4 are an example: their $1/r$ parts are proportional to $j\omega I = dI/dt$.

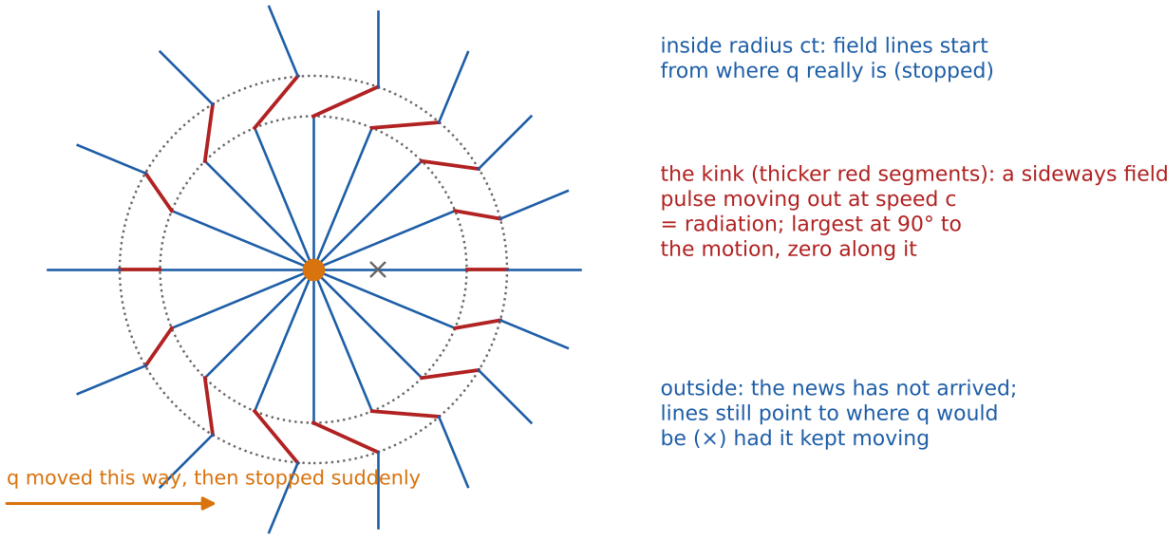


Figure 30: Why a stopping charge radiates

Reading the figure. A charge was moving to the right and suddenly stopped. Information about the stop spreads out at speed c. Inside the inner circle the field lines already point from where the charge really is; outside the outer circle they still point to where it *would* be (×). The lines must join across the thin shell, so they run **sideways** there (the thicker red segments). This sideways field pulse travels outward at speed c: it is the radiation. It exists only because the velocity *changed*. It is strongest at 90° to the motion and zero along it, just like the $\sin\theta$ pattern of the dipole in Q4.

3. Keyword Explanation

1. Retarded time

$t_r = t - R/c$: the moment at which the source must have acted for its effect to reach the field point at time t.

2. Retarded potentials

The potentials V and $\mathbf{A}$ computed from the sources at the retarded time. They are the general solutions of Maxwell's equations for given ρ and $\mathbf{J}$.

3. Larmor formula

The power radiated by a slowly moving point charge with acceleration a: $P = q^2a^2/(6\pi\epsilon_0 c^3)$.

4. Notes and Remarks

- **Spelling:** the paper prints "Jefeminko"; the name is Oleg **Jefimenko** (1966).
- **Why not just retard the fields?** It is tempting to write Coulomb's law with $\rho(t_r)$ and stop. Jefimenko's equations show that this misses the $\dot\rho$ and $\dot{\mathbf{J}}$ terms, and those are exactly the ones that radiate.
- **Where this is used:** it explains why antennas need alternating currents, why X-rays come from electrons stopping in a metal (bremsstrahlung), and why synchrotrons (circular motion, so constant acceleration) radiate.

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6.(c) A rectangular cavity resonator has dimensions of $a = 5$ cm, $b = 2$ cm, and $d = 15$ cm. Compute the resonant frequency of the dominant mode for a dielectric-field cavity of $\epsilon_r = 2.56$. Marks: 3

1. Theory

1. What is a cavity resonator? (Core Idea)

Take a piece of rectangular waveguide (see the Theory of 3(d)) and close both ends with metal plates. The wave now bounces back and forth between the ends, and only frequencies for which a whole number of half-waves fits in *all three* directions can survive. The box **rings** at these resonant frequencies, like an organ pipe for sound. It is the microwave equivalent of an LC tank circuit, with a very high quality factor Q.

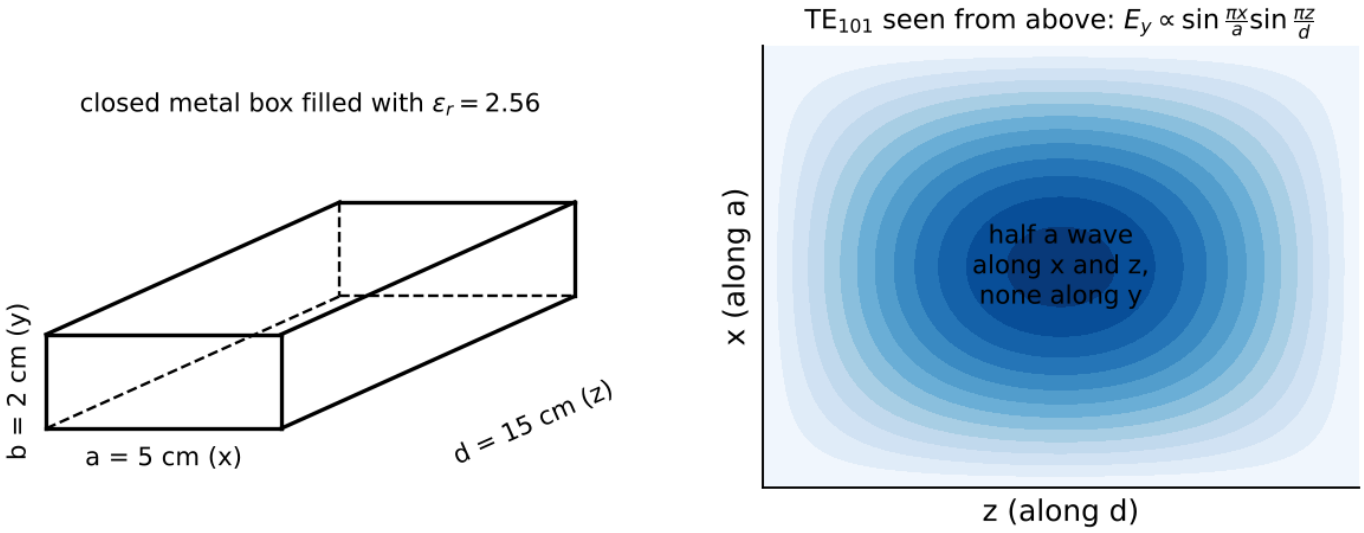


Figure 31: Cavity dimensions and the dominant mode

Reading the figure. Left: the box, $a = 5$ cm wide (x), $b = 2$ cm high (y) and $d = 15$ cm long (z). Right: in the TE_{101} mode, seen from above, the electric field (pointing along y) is strongest in the middle and zero at the four side walls: half a wave across a, half a wave along d, and no variation along b.

2. Resonant frequency formula

$$f_{r,mnp} = \frac{u'}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}, \qquad u' = \frac{c}{\sqrt{\epsilon_r}}$$

- TM_{mnp} needs $m, n \geq 1$ and $p \geq 0$.
- TE_{mnp} needs $p \geq 1$, with m and n not both 0.

3. Dominant mode

The **dominant mode** is the one with the lowest resonant frequency. To make the square root smallest, use 1 for the two **largest** dimensions and 0 for the smallest. Here $d > a > b$, so the dominant mode is $\mathbf{TE}_{101}$ ($m = 1$, $n = 0$, $p = 1$).

2. Answer

Given: $a = 0.05$ m, $b = 0.02$ m, $d = 0.15$ m, $\epsilon_r = 2.56$ ($\mu_r = 1$).

$$u' = \frac{3 \times 10^8}{\sqrt{2.56}} = \frac{3 \times 10^8}{1.6} = 1.875 \times 10^8 \text{ m/s}$$

Dominant mode ($d > a > b$): TE_{101}.

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$$f_{r,101} = \frac{1.875 \times 10^8}{2}\sqrt{\left(\frac{1}{0.05}\right)^2 + \left(\frac{1}{0.15}\right)^2} = 0.9375 \times 10^8\sqrt{400 + 44.44}$$

$$f_{r,101} = 0.9375 \times 10^8 \times 21.08 = 1.976 \times 10^9 \text{ Hz}$$

$$\boxed{f_{r,101} = 1.976 \text{ GHz}}$$

Check that it is the lowest: the next modes are TE_{102} at 2.25 GHz and TE_{103} at 2.65 GHz. The lowest TM mode, TM_{110}, is at 5.05 GHz. TE_{101} is indeed lowest.

3. Keyword Explanation

1. Quality factor Q

Stored energy divided by the energy lost per cycle (times 2π). Cavities reach Q of 10^3 to 10^4, much higher than LC circuits.

2. Dielectric-filled cavity

A cavity filled with a dielectric. The paper prints "dielectric-field"; "dielectric-filled" is meant. Filling lowers every resonant frequency by $\sqrt{\epsilon_r}$.

4. Notes and Remarks

- **Air-filled comparison:** the same box empty resonates at $1.976 \times 1.6 = 3.162$ GHz. Filling a cavity is a way to make it smaller for a given frequency.

6.(d) What do you mean by effective height and effective area of an antenna? Marks: 4

1. Theory

1. Describing a receiving antenna simply (Core Idea)

A receiving antenna sits in an incoming wave and delivers a voltage and a power to its receiver. Its shape and its current distribution can be complicated. Two simple numbers summarise what it does:

- **effective height** (or effective length) h_e: how much **voltage** it gets from a given field;
- **effective area** (or effective aperture) A_e: how much **power** it collects from a given power density.

2. Why "height"?

A straight wire of length h in a uniform field E parallel to it would ideally have a voltage Eh across it. A real antenna's current is not uniform along its length (it falls to zero at the tips), so it behaves like a *shorter* ideal wire. That shorter length is h_e.

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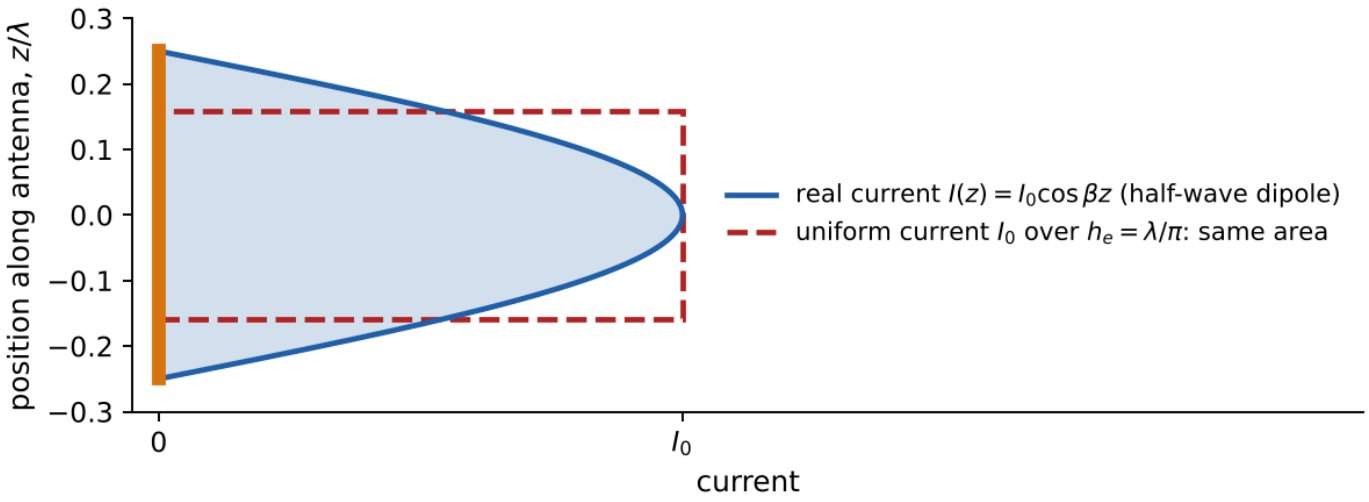


Figure 32: Effective height of a half-wave dipole

Reading the figure. The real current on a half-wave dipole (solid blue curve) is largest at the centre and zero at the ends. A uniform current I_0 over a shorter length $h_e = \lambda/\pi$ (red dashed rectangle) has the same area under it, which means the same radiation and reception.

2. Answer

Effective height h_e. For a receiving antenna it is the ratio of the open-circuit voltage at its terminals to the incident electric field (polarized along the antenna):

$$\boxed{h_e = \frac{V_{oc}}{E} \text{ (m)}}$$

Equivalently, for a transmitting antenna it is the length of an imaginary antenna with a **uniform** current equal to the terminal current I_0, having the same current-length product (the same far field):

$$h_e = \frac{1}{I_0} \int I(z)\, dz$$

Examples: Hertzian dipole $h_e = dl$ (its current is uniform); half-wave dipole $h_e = \lambda/\pi = 0.318\lambda$ (less than its physical length 0.5λ).

Effective area A_e. It is the area that, multiplied by the incident power density P_{inc} (W/m^2), gives the power delivered to a matched load:

$$\boxed{A_e = \frac{P_{received}}{P_{inc}} \text{ (m}^2\text{)}}$$

It is linked to gain by

$$A_e = \frac{\lambda^2}{4\pi} G$$

so a high-gain antenna collects more power. For an antenna with a radiation resistance R_{rad} (matched, lossless):

$$A_e = \frac{\eta\, h_e^2}{4 R_{rad}}$$

Examples: Hertzian dipole $A_e = 3\lambda^2/8\pi = 0.119\lambda^2$; half-wave dipole $A_e = 0.13\lambda^2$. A dish antenna has an A_e of about 55 to 70 % of its physical aperture.

Interpretation: h_e tells you the **voltage** you get, $V = h_e E$; A_e tells you the **power** you get, $P = A_e P_{inc}$.

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3. Keyword Explanation

1. Open-circuit voltage

The voltage across the antenna terminals with nothing connected.

2. Power density

Power per unit area of a wave, in W/m^2 (the Poynting vector magnitude, see 4(c)).

4. Notes and Remarks

- **Check for the Hertzian dipole:** $A_e = \eta h_e^2/(4R_{rad}) = 120\pi\, dl^2/[4 \times 80\pi^2(dl/\lambda)^2] = 3\lambda^2/8\pi$ ✔, the same as $\lambda^2 G/4\pi$ with $G = 1.5$.
- **Not the physical size:** a thin wire dipole has almost no physical area but an A_e of $0.13\lambda^2$. It collects power from the field around it, not only from what hits the wire.

5. Final Concise Answer

(a) Uses: RF/microwave PCB interconnects, printed filters, couplers, dividers and matching networks, patch antennas, MICs and MMICs. Advantages: small, light, cheap, planar, easy to mount components and integrate, Z_0 set by the width. Losses: conductor (skin effect), dielectric ($\tan\delta$), radiation, surface waves.

(b) From the retarded potentials $V = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(t_r)}{R} d\tau'$ and $\mathbf{A} = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}(t_r)}{R} d\tau'$, with $\nabla t_r = -\hat{\mathbf{R}}/c$: $\mathbf{E} = \frac{1}{4\pi\epsilon_0}\int \left[\frac{\rho}{R^2}\hat{\mathbf{R}} + \frac{\dot{\rho}}{cR}\hat{\mathbf{R}} - \frac{\dot{\mathbf{J}}}{c^2 R}\right] d\tau'$ and $\mathbf{B} = \frac{\mu_0}{4\pi}\int \left[\frac{\mathbf{J}}{R^2} + \frac{\dot{\mathbf{J}}}{cR}\right] \times \hat{\mathbf{R}}\, d\tau'$. Only the $1/R$ terms carry power to infinity, and they need $\dot{\mathbf{J}} = nq\mathbf{a} \neq 0$: radiation needs accelerating charges (static and uniformly moving charges do not radiate).

(c) $u' = 1.875 \times 10^8$ m/s; dominant mode TE_{101} ($d > a > b$); $f_r = \frac{u'}{2}\sqrt{(1/a)^2 + (1/d)^2} = 1.976$ GHz.

(d) Effective height $h_e = V_{oc}/E$: the length of an equivalent uniform-current antenna (λ/π for a half-wave dipole). Effective area $A_e = P_{received}/P_{inc} = G\lambda^2/4\pi$: the area that collects the received power ($0.13\lambda^2$ for a half-wave dipole).

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FORMULA SHEET

EEE475

RF and Microwave Engineering

Formula Sheet

Session 2020–21 paper

Every formula behind the Fourth Year First Semester Final Examination 2024

TOPICS

1 Transmission lines — 8 formulas

2 Reflection and matching — 9 formulas

3 Plane waves — 3 formulas

4 Waveguides and cavities — 2 formulas

5 Hertzian dipole and antennas — 14 formulas

6 Receivers and AM — 5 formulas

7 Radiation from moving charges — 4 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

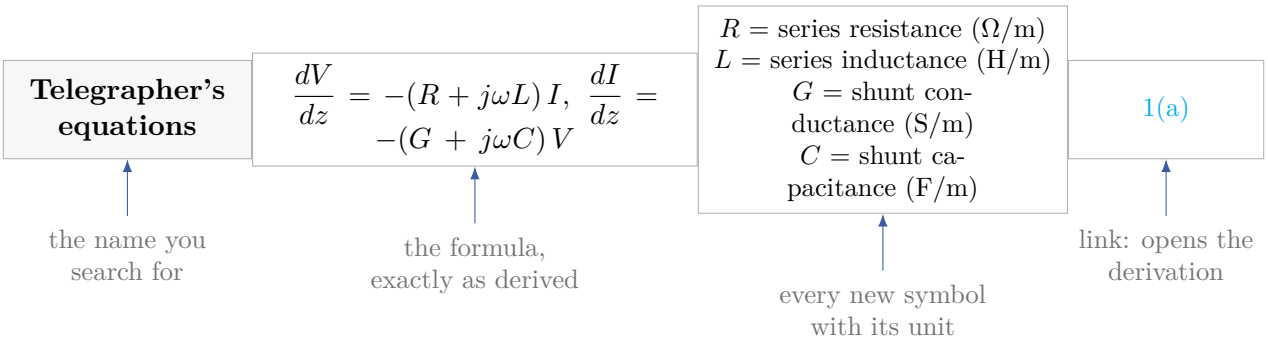
Prepared by Nur Mohammad

October 2026

BACK OF THE COVER

Before you use the sheet

How to read a row



In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Transmission lines **4**

2 Reflection and matching **4**

3 Plane waves **5**

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5 Hertzian dipole and antennas **6**

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7 Radiation from moving charges **7**

8 Key formulas **8**

9 Symbol index **10**

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Conventions: phasors with $e^{j\omega t}$; R, L, G, C are per unit length; z, y are impedance and admittance normalized to Z_0; $\Gamma = u + jv$ on the Smith chart; $\eta_0 = 120\pi\ \Omega$ and $c = 3 \times 10^8$ m/s; $u' = c/\sqrt{\mu_r \epsilon_r}$ is the speed in the filling of a guide or cavity; Idl is the current moment of a Hertzian dipole and I_0 its peak current. In 1(c) the paper's choice of line coordinate gives $-j$ in Z_{in}; with the usual convention (generator on the other side) the signs become $+j$. References point to the solutions of the 2020–21 paper.

Columns: Title · Equation · Notation · Ref. (reference).

1. Transmission lines:

Title	Equation	Notation	Ref.
Telegrapher's equations	$\frac{dV}{dz} = -(R + j\omega L)\, I,\ \frac{dI}{dz} = -(G + j\omega C)\, V$	R = series resistance (Ω/m) L = series inductance (H/m) G = shunt conductance (S/m) C = shunt capacitance (F/m)	1(a)
Line voltage	$V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$	V_0^+, V_0^- = forward and backward amplitudes (V)	1(a)
★ **Propagation constant**	$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$	α = attenuation constant (Np/m) β = phase constant (rad/m)	1(a)
★ **Characteristic impedance**	$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$	Z_0 = characteristic impedance (Ω)	3(b)
★ **Low-loss line**	$\alpha \approx \frac{R}{2Z_0} + \frac{GZ_0}{2},\ \beta \approx \omega\sqrt{LC},\ Z_0 \approx \sqrt{\frac{L}{C}}$	valid for $R \ll \omega L$, $G \ll \omega C$	1(a)
Conductor loss	$\alpha_c = \frac{R}{2Z_0},\ \alpha_d = \alpha - \alpha_c$	α_c, α_d = conductor and dielectric parts of α (Np/m)	1(b)
Shunt conductance	$G = \frac{2(\alpha - \alpha_c)}{Z_0}$	from the low-loss α	1(b)
Phase velocity	$u_p = \frac{\omega}{\beta}$	u_p = phase velocity (m/s)	3(b)

Table 1: Transmission lines

2. Reflection and matching:

Title	Equation	Notation	Ref.
★ **Reflection coefficient**	$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$	Γ = reflection coefficient at the load Z_L = load impedance (Ω)	1(c)
Load from Γ	$Z_L = Z_0 \frac{1 + \Gamma}{1 - \Gamma}$	the inverse of the Γ formula	1(c)
★ **Standing wave ratio**	$\mathrm{SWR} = \frac{\lvert V\rvert_{max}}{\lvert V\rvert_{min}} = \frac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert}$	1 for a matched load	1(c)
★ **Input impedance**	$Z_{in}(l) = \left(\frac{Z_L - jZ_0 \tan\beta l}{Z_0 - jZ_L \tan\beta l}\right) Z_0$	l = distance from the load (m); $+j$ in the usual convention	1(c)

Title	Equation	Notation	Ref.
Normalized impedance	$z = \frac{Z}{Z_0} = r + jx,\ \Gamma = \frac{z-1}{z+1}$	r, x = normalized resistance and reactance	2(a)
★ **Constant-r circle**	$\left(u - \frac{r}{1+r}\right)^2 + v^2 = \left(\frac{1}{1+r}\right)^2$	u, v = real and imaginary parts of Γ	2(a)
★ **Constant-x circle**	$(u-1)^2 + \left(v - \frac{1}{x}\right)^2 = \left(\frac{1}{x}\right)^2$	centre $(1, 1/x)$, radius $1/\lvert x\rvert$	2(a)
Short-circuited stub	$y_{stub} = -j \cot \beta l_s$	l_s = stub length (m)	2(b)
★ **Quarter-wave transformer**	$Z_{in} = \frac{Z_1^2}{Z_L},\ Z_1 = \sqrt{Z_0 R_L}$	Z_1 = impedance of the $\lambda/4$ section (Ω) R_L = resistive load (Ω)	2(c)

Table 2: Reflection and matching

3. Plane waves:

Title	Equation	Notation	Ref.
★ **Wave speed**	$u = \frac{c}{\sqrt{\mu_r \epsilon_r}}$	μ_r, ϵ_r = relative permeability and permittivity	3(a)
★ **Intrinsic impedance**	$\eta = \eta_0 \sqrt{\frac{\mu_r}{\epsilon_r}}$	η = ratio of E to H (Ω)	3(a)
Wavelength	$\lambda = \frac{2\pi}{\beta}$	λ = wavelength (m)	3(a)

Table 3: Plane waves

4. Waveguides and cavities:

Title	Equation	Notation	Ref.
★ **Cutoff frequency**	$f_{c,mn} = \frac{u'}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$	a, b = guide width and height (m) m, n = mode indices; TM needs $m, n \geq 1$	3(d)
★ **Cavity resonance**	$f_{r,mnp} = \frac{u'}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$	d = cavity length (m) p = third mode index	6(c)

Table 4: Waveguides and cavities

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5. Hertzian dipole and antennas:

Title	Equation	Notation	Ref.
Vector potential	$A_r = \frac{\mu_0 I dl}{4\pi r} e^{-j\beta r} \cos\theta,\ A_\theta = -\frac{\mu_0 I dl}{4\pi r} e^{-j\beta r} \sin\theta$	r, θ = spherical coordinates (m, rad)	4(a)
★ **Magnetic field**	$\mathbf{H} = \frac{Idl}{4\pi} \sin\theta \left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r}\, \hat{a}_\phi$	$\mathbf{H}$ = magnetic field (A/m)	4(a)
Field from $\mathbf{H}$	$\mathbf{E} = \frac{1}{j\omega\epsilon} \nabla \times \mathbf{H},\ \frac{1}{j\omega\epsilon} = \frac{\eta}{j\beta}$	away from the source	4(b)
★ **Radial E field**	$E_r = \frac{\eta I dl}{2\pi} \cos\theta \left[\frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}$	dies out in the far field	4(b)
★ **Polar E field**	$E_\theta = \frac{\eta I dl}{4\pi} \sin\theta \left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}$	far field: $E_\theta = \eta H_\phi$	4(b)
Average Poynting vector	$\mathbf{P}_{avg} = \frac{1}{2}\mathrm{Re}\left(\mathbf{E} \times \mathbf{H}^*\right)$	$\mathbf{P}_{avg}$ = power density (W/m^2)	4(c)
★ **Far-field power density**	$\mathbf{P}_{avg} = \frac{15\pi I_0^2}{r^2} \left(\frac{dl}{\lambda}\right)^2 \sin^2\theta\, \hat{a}_r$	in free space; $= \frac{\eta\beta^2 I_0^2 dl^2 \sin^2\theta}{32\pi^2 r^2}$	4(c)
Radiated power	$P_{rad} = 40\pi^2 \left(\frac{dl}{\lambda}\right)^2 I_0^2$	P_{rad} = total radiated power (W)	4(c)
★ **Radiation resistance**	$R_{rad} = \frac{2P_{rad}}{I_0^2} = 80\pi^2 \left(\frac{dl}{\lambda}\right)^2$	R_{rad} = radiation resistance (Ω)	4(c)
★ **Effective height**	$h_e = \frac{V_{oc}}{E} = \frac{1}{I_0} \int I(z)\, dz$	V_{oc} = open-circuit voltage (V) E = incident field (V/m)	6(d)
★ **Effective area**	$A_e = \frac{P_{received}}{P_{inc}} = \frac{\lambda^2}{4\pi} G$	P_{inc} = incident power density (W/m^2) G = antenna gain	6(d)
Area from height	$A_e = \frac{\eta\, h_e^2}{4R_{rad}}$	for a matched load	6(d)
Gain in dB	$G_{dB} = 10\log_{10} G,\ G = 10^{G_{dB}/10}$	G_{dB} = gain in decibels (dBi)	5(c)
★ **EIRP**	$\mathrm{EIRP} = P_t G_t$	P_t = transmitter power (W) G_t = transmit antenna gain	5(c)

Table 5: Hertzian dipole and antennas

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6. Receivers and AM:

Title	Equation	Notation	Ref.
Mixer product	$\cos\omega_{RF}t\cos\omega_{LO}t = \frac{1}{2}\cos(\omega_{RF}-\omega_{LO})t + \frac{1}{2}\cos(\omega_{RF}+\omega_{LO})t$	ω_{RF}, ω_{LO} = signal and local-oscillator frequencies (rad/s)	5(a)
★ **Intermediate frequency**	$f_{IF} = f_{RF} - f_{LO}$	low-side injection, $f_{LO} < f_{RF}$	5(a)
Image frequency	$f_{image} = f_{LO} - f_{IF}$	$2f_{IF}$ away from the wanted signal	5(a)
AM signal	$s(t) = A_c[1+\mu\cos\omega_m t]\cos\omega_c t$	A_c = carrier amplitude μ = modulation index	5(b)
★ **AM total power**	$P_t = P_c\left(1+\frac{\mu^2}{2}\right)$	P_c = carrier power (W)	5(b)

Table 6: Receivers and AM

7. Radiation from moving charges:

Title	Equation	Notation	Ref.
Retarded time	$t_r = t - \frac{R}{c}$	R = distance from source point to field point (m)	6(b)
Retarded potentials	$V = \frac{1}{4\pi\epsilon_0}\int\frac{\rho(t_r)}{R}\,d\tau',\ \mathbf{A} = \frac{\mu_0}{4\pi}\int\frac{\mathbf{J}(t_r)}{R}\,d\tau'$	ρ = charge density (C/m^3) $\mathbf{J}$ = current density (A/m^2)	6(b)
★ **Jefimenko, E**	$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\int\left[\frac{\rho}{R^2}\hat{\mathbf{R}} + \frac{\dot\rho}{cR}\hat{\mathbf{R}} - \frac{\dot{\mathbf{J}}}{c^2R}\right]d\tau'$	dots are $\partial/\partial t$ at t_r; the $1/R$ terms radiate	6(b)
★ **Jefimenko, B**	$\mathbf{B} = \frac{\mu_0}{4\pi}\int\left[\frac{\mathbf{J}}{R^2} + \frac{\dot{\mathbf{J}}}{cR}\right]\times\hat{\mathbf{R}}\,d\tau'$	radiation needs $\dot{\mathbf{J}} \neq 0$: accelerating charges	6(b)

Table 7: Radiation from moving charges

8. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Propagation constant	$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$
Characteristic impedance	$Z_0 = \sqrt{\dfrac{R + j\omega L}{G + j\omega C}}$
Low-loss line	$\alpha \approx \dfrac{R}{2Z_0} + \dfrac{GZ_0}{2},\ \beta \approx \omega\sqrt{LC},\ Z_0 \approx \sqrt{\dfrac{L}{C}}$
Reflection coefficient	$\Gamma = \dfrac{Z_L - Z_0}{Z_L + Z_0}$
Standing wave ratio	$\text{SWR} = \dfrac{\lvert V\rvert_{max}}{\lvert V\rvert_{min}} = \dfrac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert}$
Input impedance	$Z_{in}(l) = \left(\dfrac{Z_L - jZ_0 \tan\beta l}{Z_0 - jZ_L \tan\beta l}\right) Z_0$
Constant-r circle	$\left(u - \dfrac{r}{1+r}\right)^2 + v^2 = \left(\dfrac{1}{1+r}\right)^2$
Constant-x circle	$(u-1)^2 + \left(v - \dfrac{1}{x}\right)^2 = \left(\dfrac{1}{x}\right)^2$
Quarter-wave transformer	$Z_{in} = \dfrac{Z_1^2}{Z_L},\ Z_1 = \sqrt{Z_0 R_L}$
Wave speed	$u = \dfrac{c}{\sqrt{\mu_r \epsilon_r}}$
Intrinsic impedance	$\eta = \eta_0 \sqrt{\dfrac{\mu_r}{\epsilon_r}}$
Cutoff frequency	$f_{c,mn} = \dfrac{u'}{2}\sqrt{\left(\dfrac{m}{a}\right)^2 + \left(\dfrac{n}{b}\right)^2}$
Cavity resonance	$f_{r,mnp} = \dfrac{u'}{2}\sqrt{\left(\dfrac{m}{a}\right)^2 + \left(\dfrac{n}{b}\right)^2 + \left(\dfrac{p}{d}\right)^2}$
Magnetic field	$\mathbf{H} = \dfrac{Idl}{4\pi}\sin\theta\left[\dfrac{j\beta}{r} + \dfrac{1}{r^2}\right] e^{-j\beta r}\,\hat{a}_\phi$
Radial E field	$E_r = \dfrac{\eta Idl}{2\pi}\cos\theta\left[\dfrac{1}{r^2} - \dfrac{j}{\beta r^3}\right] e^{-j\beta r}$
Polar E field	$E_\theta = \dfrac{\eta Idl}{4\pi}\sin\theta\left[\dfrac{j\beta}{r} + \dfrac{1}{r^2} - \dfrac{j}{\beta r^3}\right] e^{-j\beta r}$
Far-field power density	$\mathbf{P}_{avg} = \dfrac{15\pi I_0^2}{r^2}\left(\dfrac{dl}{\lambda}\right)^2 \sin^2\theta\,\hat{a}_r$
Radiation resistance	$R_{rad} = \dfrac{2P_{rad}}{I_0^2} = 80\pi^2\left(\dfrac{dl}{\lambda}\right)^2$
Effective height	$h_e = \dfrac{V_{oc}}{E} = \dfrac{1}{I_0}\int I(z)\,dz$
Effective area	$A_e = \dfrac{P_{received}}{P_{inc}} = \dfrac{\lambda^2}{4\pi}G$

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Title	Equation
EIRP	$\text{EIRP} = P_t G_t$
Intermediate frequency	$f_{IF} = f_{RF} - f_{LO}$
AM total power	$P_t = P_c \left(1 + \frac{\mu^2}{2}\right)$
Jefimenko, E	$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \left[\frac{\rho}{R^2}\hat{\mathbf{R}} + \frac{\dot{\rho}}{cR}\hat{\mathbf{R}} - \frac{\dot{\mathbf{J}}}{c^2 R} \right] d\tau'$
Jefimenko, B	$\mathbf{B} = \frac{\mu_0}{4\pi} \int \left[\frac{\mathbf{J}}{R^2} + \frac{\dot{\mathbf{J}}}{cR} \right] \times \hat{\mathbf{R}}\, d\tau'$

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9. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
R	series resistance (Ω/m)	Telegrapher's equations
L	series inductance (H/m)	Telegrapher's equations
G	shunt conductance (S/m)	Telegrapher's equations
C	shunt capacitance (F/m)	Telegrapher's equations
V_0^+, V_0^-	forward and backward amplitudes (V)	Line voltage
α	attenuation constant (Np/m)	Propagation constant
β	phase constant (rad/m)	Propagation constant
Z_0	characteristic impedance (Ω)	Characteristic impedance
α_c, α_d	conductor and dielectric parts of α (Np/m)	Conductor loss
u_p	phase velocity (m/s)	Phase velocity
Γ	reflection coefficient at the load	Reflection coefficient
Z_L	load impedance (Ω)	Reflection coefficient
l	distance from the load (m); $+j$ in the usual convention	Input impedance
r, x	normalized resistance and reactance	Normalized impedance
u, v	real and imaginary parts of Γ	Constant-r circle
l_s	stub length (m)	Short-circuited stub
Z_1	impedance of the $\lambda/4$ section (Ω)	Quarter-wave transformer
R_L	resistive load (Ω)	Quarter-wave transformer
μ_r, ϵ_r	relative permeability and permittivity	Wave speed
η	ratio of E to H (Ω)	Intrinsic impedance
λ	wavelength (m)	Wavelength
a, b	guide width and height (m)	Cutoff frequency
m, n	mode indices; TM needs $m, n \geq 1$	Cutoff frequency
d	cavity length (m)	Cavity resonance
p	third mode index	Cavity resonance
r, θ	spherical coordinates (m, rad)	Vector potential
$\mathbf{H}$	magnetic field (A/m)	Magnetic field
$\mathbf{P}_{avg}$	power density (W/m^2)	Average Poynting vector
P_{rad}	total radiated power (W)	Radiated power
R_{rad}	radiation resistance (Ω)	Radiation resistance
V_{oc}	open-circuit voltage (V)	Effective height
E	incident field (V/m)	Effective height
P_{inc}	incident power density (W/m^2)	Effective area
G_{dB}	gain in decibels (dBi)	Gain in dB
P_t	transmitter power (W)	EIRP
G_t	transmit antenna gain	EIRP

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Symbol	Meaning	First used in
ω_{RF}, ω_{LO}	signal and local-oscillator frequencies (rad/s)	Mixer product
A_c	carrier amplitude	AM signal
μ	modulation index	AM signal
P_c	carrier power (W)	AM total power
ρ	charge density (C/m^3)	Retarded potentials
$\mathbf{J}$	current density (A/m^2)	Retarded potentials

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